

Causal quantum-channel simulation: memory beyond entropy

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Abstract

What determines the memory needed to simulate a quantum channel repeatedly, when every output must be released before the next input arrives? We exhibit two channels on $\mathbb{C}^{208}$ with the same Choi rank, maximal complementary entropy, flat optimal complementary spectrum, and zero asymptotic purity cost. At their common optimal exchange rate of $5/2$ qubits per use, one has an exact implementation with five memory qubits for arbitrarily many uses; the other requires memory polynomial in the number of uses even at fixed small error, regardless of the available purity. The latter also admits a zero-purity implementation with memory $O(n^{1/3} \log^{4/3} n)$.

More generally, for sublinear memory and vanishing error, exchange and purity have least rates $(H + \kappa)/2$ and κ , where H is the maximal complementary entropy and κ is the closed-device purity cost. We prove that κ vanishes exactly on the closure of finite tracial factorizations and is bounded below by the squared Choi distance to that closure. Within the closure, sequential support tests and reservoir constructions bound memory through finite-approximation profiles. For channels constructed from finite group presentations, these bounds translate directly into quantitative unitary approximation of the group. Houghton’s group supplies the explicit separation above; a channel from Slofstra’s group gives a stronger $n^{1/8-o(1)}$ memory lower bound at logarithmic purity. All operational bounds use complete-experiment error against adaptive observers with quantum references.

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1 Introduction

The question. A unitary channel can be applied again and again by the same device at no cost. A general channel Φ cannot: each use entangles the input with an environment, and a device that is to act as Φ on the next input as well must either carry that environment along or hand it off. Rybár and Ziman called a channel *repeatable* when a single memory device implements it on each individual input [1], and showed that random-unitary channels

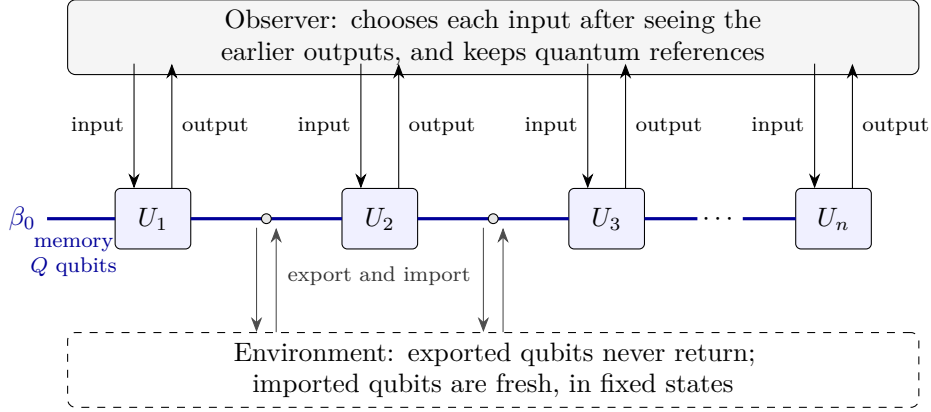


Figure 1: Causal service of a channel for n rounds. The memory (thick line) starts in a fixed state β_0 . In each round a fixed unitary U_t acts on the fresh input and the memory, and the output is released at once. Between rounds some memory qubits are exported for good and replaced by fresh imports. The observer may adapt each input to everything seen so far.

are repeatable while nonunital ones are not. In the closed setting, where nothing may be exchanged with the outside, the bath dimension and initial entropy required per use were characterized in [2].

Here we study the open setting (Figure 1). A device receives n inputs in sequence, applies a fixed public protocol, and releases each output before the next input arrives. Between uses it may export registers permanently and import fresh independent ones. The whole experiment must be within a small error ϵ of n independent uses of Φ , against an arbitrary adaptive observer who chooses each input after seeing the earlier outputs and keeps quantum references (Definition 2.3). We count three resources: the retained memory, the exchange, and the purity the device starts with or imports (see the box below). Exchange at rate $H(\Phi)/2$, half the maximal complementary entropy, is necessary for vanishing error with sublinear memory and purity (Proposition 3.6). The questions are how much purity and how much memory the device then needs, and what decides it.

The resource ledger

- *Error* ϵ : the largest trace distance, over adaptive observers with quantum references, between the whole experiment and n independent uses of Φ .
- *Memory* Q : every retained qubit, including clocks, padding and workspace.
- *Exchange* J : the number of qubits exported, which equals the number imported. Imports happen between rounds; the qubits imported in one step form a register in a fixed state, independent of everything else.
- *Purity*: the initial deficit $P = Q - S(\beta_0)$ plus the imported deficit C , the sum over imports of qubits minus entropy. A purity budget B means $P + C \leq B$.
- $H(\Phi)$ is the maximal complementary entropy. With sublinear memory and purity, exchange at rate $H(\Phi)/2$ is necessary (such services exist only on the closure class, where $\kappa = 0$); with sublinear memory alone, the least rate is $(H(\Phi) + \kappa(\Phi))/2$. There is no free reset, postselection or measurement record. Logarithms are to base two.

The answer in brief. Let $\mathcal{K}_d$ be the closure of the channels on M_d of the form $\rho \mapsto (\text{id} \otimes \tau_m)[U(\rho \otimes I_m)U^*]$, with U unitary and τ_m the normalized trace, and call such channels *exact factors*. At leading order, exchange and purity are set jointly by H and by the closed-device cost κ of [2]: with sublinear memory and vanishing error the least exchange rate is

Channel	Purity and memory for n uses	Where
Any channel	least purity rate at vanishing error $\kappa(\Phi)$, which is 0 exactly on $\mathcal{K}_d$ and at least $\Delta_J(\Phi)^2/(2\ln 2)$ off it; least exchange rate $(H + \kappa)/2$ with sublinear memory; outside $\mathcal{K}_d$, linear memory at rate $H/2$	Thm 3.1, Cor 3.2
Exact factor	memory $O(\log n)$ at exchange $H/2 + s$ and at the rank rate $\frac{1}{2} \log r$; at the optimal rate $H/2$, $O(\log n)$ if the complementary spectrum is flat, and between order $\sqrt{n}$ and $\sqrt{n \log n}$ if not	Thms 3.8, 3.15
Any channel in $\mathcal{K}_d$	memory between the one-round profile at leakage about $(\log n)/n$ and the Choi profile at accuracy $\epsilon/(8dn)$	Thms 3.18, 3.19, 3.21
Gadget channel of a group	memory between the group's approximation profile at relator defect about $(\log n/n)^{1/4}$ and about $(n \log n)^{-1/2}$; if $\Phi_G \in \mathcal{K}_d$, the upper bound at the optimal rate	Thm 3.24
Houghton channel Φ_H	in $\mathcal{K}_{104}$, $H = \log 26$; memory between $(n/\log n)^{\beta_H}$, $\beta_H \approx 1/15.04$, and $n^{1/3} \log^{4/3} n$, the upper bound at the optimal rate	Thm 3.29, Cor 3.37
Slofstra channel Φ_S	memory at least $n^{1/8-o(1)}$, whichever side of $\mathcal{K}_d$ it lies on; at its rank rate at least $n^{1/9-o(1)}$ whatever the purity, and linear if $\Phi_S \notin \mathcal{K}_d$	Thm 3.25

Table 1: Results at a glance, at constant error and a logarithmic purity budget unless stated. H is the maximal complementary entropy and r the Choi rank.

$(H + \kappa)/2$ and the least purity rate is κ (Corollary 3.2). The cost κ vanishes exactly on $\mathcal{K}_d$, so purity decides whether a channel is a limit of exact factors, and on $\mathcal{K}_d$ the exchange threshold is $H/2$. Memory measures what neither H nor κ records: how hard the channel is to approximate.

The sharpest comparison is the pair $\Xi, \hat{\Phi}$ on $\mathbb{C}^{208}$ in Corollary 3.38. Both have Choi rank 32, $H = 5$, a flat optimal complementary spectrum and $\kappa = 0$. At the common optimal exchange rate $5/2$, Ξ is served exactly with five memory qubits and zero purity for any n , whereas $\hat{\Phi}$ needs polynomial memory even at fixed small error and with arbitrary purity. A zero-purity service of $\hat{\Phi}$ with $O(n^{1/3} \log^{4/3} n)$ memory is nevertheless possible. An excess exchange of $O(n^\gamma)$ qubits, for any $\gamma < 1$, still leaves a polynomial memory requirement.

The underlying example is a channel Φ_H on $\mathbb{C}^{104}$ built from Houghton's group (Section 4). It has the same Choi rank 26, entropy $\log 26$, flat optimal complementary spectrum and zero purity cost as the exact factor $\Phi_0 = \text{id}_4 \otimes \Delta_{26}$, where Δ_{26} is complete dephasing. At their optimal exchange rate, Φ_0 needs only $O(\log n)$ memory, while Φ_H needs at least order $(n/\log n)^{1/15.04}$ at logarithmic purity and at most $O(n^{1/3} \log^{4/3} n)$ (Theorem D). Table 1 collects the bounds.

Purity marks the closure. Our first result identifies the purity cost. It is the closed-device cost κ of [2], a smoothed conditional entropy of one use (Section 3.1), and it detects the closure class.

Theorem A (Purity; Theorem 3.1 and Proposition 3.4). *For every channel Φ on M_d the least purity rate $\lim(P + C)/n$ of causal services with vanishing error is $\kappa(\Phi)$. It satisfies $\kappa(\Phi) \geq \Delta_J(\Phi)^2/(2\ln 2)$, where Δ_J is the Choi distance to $\mathcal{K}_d$, and it vanishes exactly on $\mathcal{K}_d$; every service with error ϵ has $P + C \geq n(\Delta_J(\Phi) - \epsilon)_+^2/(2\ln 2)$, at any exchange and with any memory. With sublinear memory the least exchange rate is $(H(\Phi) + \kappa(\Phi))/2$, so outside $\mathcal{K}_d$ memory is linear at the entropy rate $H/2$ (Corollary 3.2). But κ is not continuous at $\mathcal{K}_d$:*

there are qubit channels with $\Delta_J \rightarrow 0$ and $\kappa = 1$.

Inside $\mathcal{K}_d$, zero-purity services exist at every fixed error and horizon at the rank rate (Theorem 3.21). Outside it the purity cost is linear, even for channels realized exactly by a tracial von Neumann algebra: the negative answer to Connes' embedding problem [28], through a theorem of Haagerup and Musat [27], gives a generally factorizable channel outside $\mathcal{K}_d$ (Corollary 3.5). So purity separates finite approximability from its failure. It cannot see the structure inside $\mathcal{K}_d$, where $\kappa = 0$; memory can.

Memory inside the closure. Exact factors need little memory. At every exchange rate above the optimum, and at the rank rate $\frac{1}{2} \log r$ with r the Choi rank, memory $O(\log(n/\epsilon))$ suffices (Theorems 3.8 and 3.15). At the optimal rate itself the answer depends on one spectrum: when the complementary output of an optimal input is flat, the optimal rate is the rank rate and memory stays logarithmic; when it is not, memory of order $\sqrt{n}$ is necessary at small fixed error, and order $\sqrt{n \log n}$ suffices (Corollary 3.16).

For limits of exact factors the memory is bracketed by how large a finite realization must be to approximate one use to accuracy about $1/n$.

Theorem B (The bracket; Theorems 3.18, 3.19 and 3.21). *Let Φ be a channel on M_d and $0 < \epsilon < 1/2$.*

- (a) *Every causal service of Φ for n rounds with error ϵ and purity budget B has, at any exchange,*

$$Q \geq \log D_{\Phi}^{\text{ent}}\left(\frac{\epsilon}{1-\epsilon}, t_n\right), \quad t_n = \frac{B + \epsilon + h_2(\epsilon) + \epsilon \log n}{n(1-\epsilon)}.$$

- (b) *If $\Phi \in \mathcal{K}_d$ and $m = D_{\Phi}^J(\epsilon/(8dn))$, there are causal services with error ϵ , $C = 0$ and $P \leq \lceil \log(n+1) \rceil + 1$ with $Q \leq \log m + \log(2 + \log m) + O_d(\log(n/\epsilon)/s + \log n)$ at exchange $n(H(\Phi)/2 + s)$, and with $Q \leq \log m + \log(2 + \log m) + O_d(\log^2(n/\epsilon))$ at the rank rate $\frac{n}{2} \log r$; the rank-rate service can be chosen with $P = C = 0$.*

Here $D_{\Phi}^{\text{ent}}(u, v)$ is the least bath dimension of one use, with any bath state, that is within diamond distance u of Φ and whose support leakage plus entropy production is at most v , and $D_{\Phi}^J(\delta)$ is the least bath dimension of an exact factor within Choi distance δ (Figure 2).

Memory reads the group. Many channels in $\mathcal{K}_d$ come from groups. Every finite presentation $\mathcal{P}$ of a group G gives a *gadget channel* Φ_G , built from unitary 2×2 blocks that record the multiplication of the relators' prefixes and evaluated in the regular representation (Section 3.4). Its profiles and the approximation profile $\mathbf{h}_{\mathcal{P}}(e, z)$ of the presentation (the least dimension of unitaries with relator defect e and error z on the moments of finitely many words) bound each other up to polynomial changes of scale, and $\Phi_G \in \mathcal{K}_d$ exactly when $\mathbf{h}_{\mathcal{P}}$ is finite everywhere (Theorem 3.23).

Theorem C (Group channels; Proposition 3.22 and Theorem 3.24). *Every gadget channel has maximal complementary entropy $\log r$, with r its Choi rank, so its entropy rate is its rank rate, and this is its optimal exchange rate when $\Phi_G \in \mathcal{K}_d$. At constant error and logarithmic purity, the least memory of n uses of Φ_G is at least $\log \mathbf{h}_{\mathcal{P}}$ at relator defect of order $(\log n/n)^{1/4}$, up to $O(\log n)$, and, if $\Phi_G \in \mathcal{K}_d$, at most $O(\log n)$ times $\log \mathbf{h}_{\mathcal{P}}$ at relator defect of order $(n \log n)^{-1/2}$, plus $O(\log^2 n)$, at the rank rate. At the rank rate the lower bound holds whatever the purity, with the budget replaced by $14Q + 3$.*

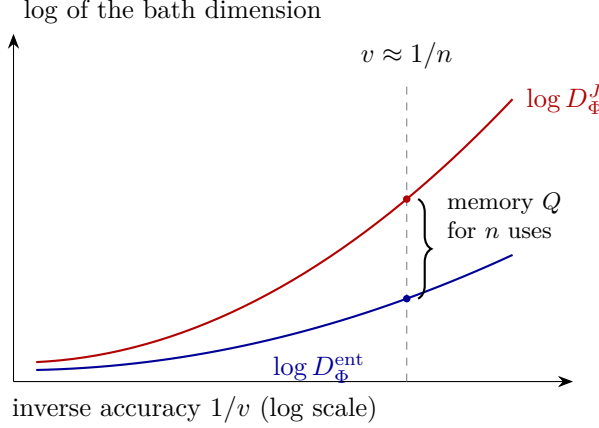


Figure 2: Theorem B in a picture. The memory for n uses lies between two profiles read at accuracy about $1/n$: below, the one-round profile, which allows biased bath states and measures leakage; above, the Choi profile of single exact factors. For Φ_H the one-round profile at distance $1/8$ is between $\exp(c v^{-\beta_H})$ and $\exp(O(v^{-1/3}))$ (Proposition 4.17), and the Choi profile is at most $\exp(O(v^{-1/3} \log^{4/3}(1/v)))$ (Theorem 3.36).

The two bounds read the profile at different scales, roughly e and e^2 . When $\log h_P$ is a power of $1/e$ this fixes the growth exponent of the memory up to a factor of 2; for general profiles it brackets the memory between two readings of the same function. Groups with steep profiles give channels that need polynomial memory.

The Houghton channel. Houghton's group H_3 [19] is finitely presented and amenable, and the companion paper [26] shows that approximate representations of a finite piece of it need stretched-exponential dimension. Its gadget channel Φ_H acts on $\mathbb{C}^{104}$, has Choi rank 26 and $H(\Phi_H) = \log 26$, and lies in $\mathcal{K}_{104}$, so $\kappa(\Phi_H) = 0$.

Theorem D (Polynomial memory in the closure; Theorems 3.29, 3.32 and Corollary 3.37). *Fix $0 < \epsilon \leq 1/16$ and $A \geq 4$, and put $\beta_H = 1/(2 \log 130 + 1) \approx 1/15.04$.*

- (a) *Every causal service of Φ_H for n rounds with error ϵ and $P + C \leq A \log(n + 1)$ has, at any exchange, $Q \geq c_A(n / \log(n + 1))^{\beta_H} - \frac{1}{8}$.*
- (b) *At the optimal exchange $J \leq \frac{n}{2} \log 26$, every causal service with error ϵ has $Q = \Omega(n^{1/(2 \log 130 + 2)})$, whatever its purity.*
- (c) *There is a causal service with error ϵ , $J \leq \frac{n}{2} \log 26$, $P = C = 0$ and $Q = O(n^{1/3} \log^{4/3} n)$.*

This is the comparison with $\Phi_0 = \text{id}_4 \otimes \Delta_{26}$ above: the two channels agree in every invariant that sets exchange or purity, and their memory differs polynomially. Entropy does not price memory.

A Slofstra channel. Slofstra's finitely presented group [4] has a steeper profile, and it is not known whether it is hyperlinear. Its gadget channel, after one Tietze move, needs memory $n^{1/8-o(1)}$ at logarithmic purity, and $n^{1/9-o(1)}$ at its rank rate whatever its purity. Either it lies in $\mathcal{K}_d$, or its purity cost is linear, its memory at the rank rate is linear, and the group is not hyperlinear (Theorem 3.25).

How the proofs go.

- *Purity* (Section 7). Each round of a service, fed a fixed input, is a feasible extension in the definition of κ , and its entropy production is paid from the purity budget. A tracialization lemma turns small entropy production into a nearby finite tracial mixture.
- *One round reads the profile* (Section 6). An observer tests every output against the support of the Choi state, which the ideal passes with certainty; an entropy ledger then finds a round whose leakage and entropy production are both of order $(\log n)/n$.
- *Groups* (Section 12). From a round close to a gadget channel one reads off, block by block, unitaries that satisfy the relators approximately and reproduce the moments; conversely approximate representations assemble into gadgets, and tensor powers with complex conjugates sharpen the moments.
- *Houghton* (Sections 4 to 9). The lower bound combines the one-round extraction with the dimension bounds of the companion paper. The upper bounds come from permutation models on a clock torus that fail at a single corner, and from unitary models that spread that defect thinly, which approximate Φ_H to second order in Choi distance.
- *Services* (Sections 10 and 11). A rolling reservoir with a Haar-random bath mixer serves every exact factor. At the rank rate a pinching bound on Rényi entropies replaces the entropy fluctuations that otherwise fix the buffer.

Related work. Repeated use of one device was introduced by Rybár and Ziman [1]; the closed version of our resource count, with no exchange, is settled in [2], and Theorem 3.1 completes an equivalence left open there. Memory effects across uses [10] and sequential realizations of instruments [11] concern different tasks. Block simulations of $\Phi^{\otimes n}$, by communication with free entanglement in the quantum reverse Shannon theorem [12, 13] or by work in the thermodynamic capacity [14], price other resources and have no causal release; the rank arguments behind our exchange converses parallel the second-order asymptotics of hypothesis testing [20, 21].

Factorizable maps were introduced by Anantharaman-Delaroche [15]. Haagerup and Musat related them to the Connes embedding problem [16, 27], and Musat and Rørdam found factorizable channels that need an infinite-dimensional ancilla and showed that the finite-ancilla class is not closed [8]; see Musat’s survey [17] and, for synchronous correlations, Russell [9]. These results are qualitative; the profiles here quantify the ancilla dimension as a function of accuracy, and the memory of repeated use reads them. Quantitative approximation of groups is surveyed in [6]; the sofic profile is Cornulier’s [3], the hyperlinear profile and its relation to nonlocal games are studied in [5, 4], and hyperlinear approximations of amenable groups come from sofic ones [18]. The diffusive picture behind one of our services is the classical link between amenability and random walks [23].

Organization. Section 2 sets up channels, profiles and the service contract, and Section 3 states all results in full. Section 4 introduces the Houghton channel and its finite models, and states what we use from [26]. Sections 5 to 9 prove the lower bounds, with the purity results in Section 7; Sections 10 and 11 prove the upper bounds; Section 12 treats group channels; Section 13 discusses open problems.

2 Setting

Channels and distances. All channels act on $M_d = M_d(\mathbb{C})$ for a fixed d . For a channel Ψ write $\mathcal{J}(\Psi) = (\Psi \otimes \text{id})(|\Omega\rangle\langle\Omega|)$ for its normalized Choi state, with $|\Omega\rangle = d^{-1/2} \sum_i |ii\rangle$, and put

$d_J(\Psi, \Psi') = \frac{1}{2} \|\mathcal{J}(\Psi) - \mathcal{J}(\Psi')\|_1$ and $d_\diamond(\Psi, \Psi') = \frac{1}{2} \|\Psi - \Psi'\|_\diamond$. Logarithms are base two unless written $\ln$. On M_m we use the normalized trace τ_m and the norms $\|X\|_2 = \tau_m(X^*X)^{1/2}$ and $\|X\|_1 = \tau_m|X|$. We have

$$d_J(\Psi, \Phi) \leq d_\diamond(\Psi, \Phi) \leq d d_J(\Psi, \Phi) : \quad (1)$$

a pure input with any reference has the form $(I \otimes A)|\Omega\rangle$ with $\|A\|_\infty^2 \leq d$, and $\|AXB\|_1 \leq \|A\|_\infty \|X\|_1 \|B\|_\infty$.

Definition 2.1 (Tracial factorizations and profiles). A channel Ψ on M_d has a *finite tracial factorization with bath dimension m* if $\Psi(\rho) = (\text{id} \otimes \tau_m)[U(\rho \otimes I_m)U^*]$ for a unitary U on $\mathbb{C}^d \otimes \mathbb{C}^m$. Let $\mathcal{F}_m$ be the set of finite convex combinations of such channels whose components have bath dimension at most m . For a channel Φ define

$$D_\Phi^J(\delta) = \min\{m : \exists \Psi \text{ with a factorization of bath dimension } m, d_J(\Psi, \Phi) \leq \delta\},$$

$$D_\Phi^{\text{mix}}(\delta) = \min\{m : \exists \Psi \in \mathcal{F}_m, d_J(\Psi, \Phi) \leq \delta\}.$$

Clearly $D^{\text{mix}} \leq D^J$. The *closure class* consists of channels with $D^J(\delta) < \infty$ for every $\delta > 0$.

Definition 2.2 (Support-sensitive profile). Let Π_Φ be the support projection of $\mathcal{J}(\Phi)$ and define the *support leakage* $\ell_\Phi(\Psi) = \text{Tr}[(I - \Pi_\Phi)\mathcal{J}(\Psi)]$. Put

$$\mathfrak{D}_\Phi(u, v) = \min\{m : \exists \Psi \in \mathcal{F}_m, d_\diamond(\Psi, \Phi) \leq u, \ell_\Phi(\Psi) \leq v\}.$$

Since $\text{Tr}[PX] \leq \frac{1}{2}\|X\|_1$ for a projection P and a traceless Hermitian X , the leakage changes by at most d_J under a change of channel, and (1) gives

$$D_\Phi^{\text{mix}}(u) \leq \mathfrak{D}_\Phi(u, v) \leq D_\Phi^{\text{mix}}(\min\{u/d, v\}) \leq D_\Phi^J(\min\{u/d, v\}). \quad (2)$$

For channels of full Choi rank $\Pi_\Phi = I$ and the leakage condition is empty.

For a channel Φ let $H(\Phi) = \max_\rho S(\Phi^c(\rho))$ be its maximal complementary entropy, computed with a minimal complement.

Definition 2.3 (Causal service). A causal service of Φ for n rounds consists of a memory register of Q qubits prepared in a fixed state β_0 , and, for each round t , fixed unitaries on the input and memory, a release of the d -dimensional output, and a replacement step in which a prescribed set of memory qubits is exported permanently and replaced by the same number of imported qubits in a fixed joint state, independent of everything else (the qubits imported in one step may be correlated with each other). The protocol is public. An observer chooses each input, possibly entangled with its own references and adaptively after all earlier outputs. The *error* is the supremum over observers of the half trace distance between the observer's final state in this experiment and in the ideal experiment with n independent uses of Φ . Exported registers are inaccessible to the observer. The resources are: Q ; the exchange J , the total number of exported qubits, which equals the total number of imported qubits since each exported qubit is replaced by an imported one; the initial deficit $P = Q - S(\beta_0)$; and the imported deficit C , the sum over imports of (number of qubits minus entropy). Counting exports and imports separately would double J , and the rate $H/2$ in Proposition 3.6 would become H . A service is *collision-at-call* if each round is one unitary on the fresh input and the retained memory and all imports happen between rounds.

Remark 2.4. The contract counts every qubit that is retained, including clocks, padding and workspace. There is no free reset, postselection or measurement record. Efficient gate synthesis is not required. In Definition 2.3 imports happen between rounds, so after composing the unitaries of a round every service is collision-at-call. Theorem 3.17 also allows imports inside a round. Theorem 3.18 does not: a retained maximally mixed qubit, refreshed between four controlled Pauli twirls, serves the completely depolarizing channel on M_4 exactly with $Q = 1$ and $P = C = 0$, while one unitary with a two-dimensional bath gives Choi rank at most four and so stays at Choi distance at least $3/4$ from it.

3 Results in full

3.1 Purity marks the boundary

Write $\mathcal{K}_d$ for the closure class of channels on M_d (Definition 2.1) and

$$\Delta_J(\Phi) = \inf_m \inf_{\Theta \in \mathcal{F}_m} d_J(\Theta, \Phi)$$

for the Choi distance from Φ to finite tracial mixtures. A channel is *generally factorizable* if $\Phi(\rho) = (\text{id} \otimes \tau_N)[U(\rho \otimes 1)U^*]$ for a finite von Neumann algebra N with a faithful normal tracial state τ_N and a unitary $U \in M_d \otimes N$ [15, 16]. Every channel in $\mathcal{K}_d$ is generally factorizable [27, Theorem 3.6], Φ_H is generally factorizable through the group von Neumann algebra of H_3 , and every generally factorizable channel is unital.

The purity cost of repeated use is the closed-device cost of [2]. For a state $\rho > 0$ on $\mathbb{C}^d$ let $\psi_\rho = (I \otimes \sqrt{d\rho})|\Omega\rangle$ be its purification with a reference Q , whose Q -marginal is ρ^T , and let $\chi_\rho = (\text{id}_Q \otimes \Phi)(\psi_\rho)$ be the joint state of reference and output A . For $\varepsilon > 0$ put

$$k_{\rho,\varepsilon}(\Phi) = \inf \left\{ S(QA | Z)_\sigma : \sigma_{QZ} = \rho^T \otimes \sigma_Z, \frac{1}{2} \|\sigma_{QA} - \chi_\rho\|_1 \leq \varepsilon \right\}, \quad (3)$$

the infimum over all finite systems Z and states σ on QAZ , and

$$\kappa(\Phi) = \sup_{\rho > 0} \lim_{\varepsilon \downarrow 0} k_{\rho,\varepsilon}(\Phi), \quad \kappa_\varepsilon(\Phi) = \sup_{\rho > 0} k_{\rho,\varepsilon}(\Phi)$$

[2, Definition 1.3]. Since $k_{\rho,\varepsilon}$ does not increase with ε , κ_ε increases to κ as $\varepsilon \downarrow 0$. In words, $k_{\rho,\varepsilon}$ is the least conditional entropy of reference and output given a side system that is independent of the reference. For closed devices, which exchange nothing, the least bath-dimension rate is $(H(\Phi) + \kappa(\Phi))/2$ [2, Theorem 2.1].

Theorem 3.1 (Purity marks the closure). *Let Φ be a channel on M_d .*

(a) $0 \leq \kappa(\Phi) \leq \min\{H(\Phi), \log d\}$ and

$$\kappa(\Phi) \geq \frac{\Delta_J(\Phi)^2}{2 \ln 2}.$$

$\Delta_J(\Phi)$ is the Choi distance from Φ to $\mathcal{K}_d$, and $\kappa(\Phi) = 0$ if and only if $\Phi \in \mathcal{K}_d$.

(b) Every causal service of Φ for n rounds with error ϵ satisfies, at any exchange and with any memory,

$$P + C \geq n \kappa_\epsilon(\Phi) \quad \text{and} \quad P + C \geq n \frac{(\Delta_J(\Phi) - \epsilon)_+^2}{2 \ln 2}.$$

(c) The least purity rate at vanishing error is $\kappa(\Phi)$: every sequence of causal services of Φ for n rounds with errors $\epsilon_n \rightarrow 0$ has $\liminf_n (P_n + C_n)/n \geq \kappa(\Phi)$, and for every $\eta > 0$ there are causal services for every n , with errors $\epsilon_n \rightarrow 0$ and no exchange, with $\limsup_n (P_n + C_n)/n \leq \kappa(\Phi) + \eta$.

Inside the closure class zero purity suffices at every fixed horizon and positive error at the rank rate (Theorem 3.21); the general entropy-rate compiler of Theorem 3.19 uses at most a logarithmic initial deficit. Thus purity separates $\mathcal{K}_d$ from its complement, while memory, as the rest of this paper shows, resolves the structure inside $\mathcal{K}_d$. The equivalence in (a) answers a question left open in [2, Section 8], where $\kappa = 0$ on $\mathcal{K}_d$ was proved. The services in (c) come from closed devices and have memory linear in n ; blocking them gives sublinear memory, as the next corollary shows.

Purity and exchange are coupled: with sublinear memory and vanishing error, unavoidable purity consumption raises the least exchange rate from $H/2$ to $(H + \kappa)/2$.

Corollary 3.2 (Exchange and purity together). *Let Φ be a channel on M_d .*

(a) *Every causal service of Φ for n rounds with error ϵ satisfies*

$$2(Q + J) \geq n(H(\Phi) + \kappa_\epsilon(\Phi) - 2\epsilon \log d) - 1.$$

So services with $Q = o(n)$ and vanishing error have $\liminf_n J_n/n \geq (H(\Phi) + \kappa(\Phi))/2$.

(b) *At exchange $J \leq nH(\Phi)/2$, every causal service with error ϵ has*

$$Q \geq \frac{n}{2} \left(\frac{(\Delta_J(\Phi) - \epsilon)_+^2}{2 \ln 2} - 2\epsilon \log d \right) - \frac{1}{2}.$$

So every channel outside $\mathcal{K}_d$ needs memory linear in n at the entropy rate $H/2$ once the error is small enough.

(c) *For every $\eta > 0$ there are causal services of Φ for every n with errors $\epsilon_n \rightarrow 0$, memory $Q_n = o(n)$, $J_n/n \leq (H(\Phi) + \kappa(\Phi))/2 + \eta + o(1)$ and $(P_n + C_n)/n \leq \kappa(\Phi) + \eta + o(1)$.*

So at leading order the exchange and the purity of repeated use are set jointly by H and κ : the least exchange rate with sublinear memory is $(H + \kappa)/2$ and the least purity rate is κ . On $\mathcal{K}_d$, where $\kappa = 0$, the exchange threshold is $H/2$. For a channel whose membership in $\mathcal{K}_d$ is unknown we call $H/2$ its *entropy rate*, and $\frac{1}{2} \log r$ its *rank rate*. The services in (c) have no prescribed error decay and purity linear in n when $\kappa > 0$; finer requirements, such as logarithmic purity at a fixed error, are where approximation enters.

Proposition 3.3 (The range of κ). (a) κ is convex, and $\kappa(\Phi) \leq \log d$.

(b) *For the replacement channel $\mathcal{R}_\sigma(X) = \text{Tr}(X)\sigma$, $\kappa(\mathcal{R}_\sigma) = \log d - S(\sigma)$; so κ takes every value in $[0, \log d]$.*

(c) *If Φ has a minimal Kraus family $K_1, \dots, K_r$ whose products $K_i^* K_j$ are linearly independent, that is, if Φ is an extreme point of the channels on M_d [34], then $\kappa(\Phi) = H(\Phi)$.*

(d) *If $\Theta \in \mathcal{K}_d$ has $\mathcal{J}(\Theta) \geq \lambda I$ with $\lambda > 0$, and $d_J(\Phi, \Theta) \leq \delta$, then*

$$\kappa(\Phi) \leq \min\{\log d, (\log d/\lambda)\delta\}.$$

Part (d) is a Lipschitz bound at interior points of the closure class. At its boundary κ jumps.

Proposition 3.4 (κ is discontinuous at the closure). (a) *For $0 \leq \theta \leq \pi/4$ let*

$$\psi_0 = \cos \theta |0\rangle + \sin \theta |1\rangle, \quad \psi_1 = \sin \theta |0\rangle + \cos \theta |1\rangle,$$

put $c = \sin 2\theta$, and let

$$\Phi_\theta(X) = X_{00} |\psi_0\rangle\langle\psi_0| + X_{11} |\psi_1\rangle\langle\psi_1|.$$

Every Φ_θ has Choi rank 2 and $H(\Phi_\theta) = 1$, and $\Delta_J(\Phi_\theta) = c/2$. For $0 < \theta \leq \pi/4$, $\kappa(\Phi_\theta) = 1$, and every causal service of Φ_θ for n rounds with error ϵ has

$$P + C \geq n \left[1 - w((1 + c^{-1})\sqrt{\epsilon}) \right], \quad w(t) = 4t + (1 + t) h_2\left(\frac{t}{1+t}\right),$$

while one qubit with $P = 1$, $C = J = n - 1$ serves it exactly. The complete dephasing Φ_0 has $\kappa(\Phi_0) = 0$.

- (b) Let $v_1 = (1, 0)$, $v_2 = (0, 1)$, $v_3 = (1, 1)/\sqrt{2}$, $v_4 = (1, e^{i\varphi})/\sqrt{2}$ in $\mathbb{C}^2$, let G_φ be their Gram matrix, and let $\Xi_\varphi(X) = G_\varphi \circ X$ be the Schur multiplier on M_4 . It is a unital channel of Choi rank 2 with $H(\Xi_\varphi) = 1$. The channel Ξ_0 has an exact finite tracial factorization with bath dimension 2, so $\kappa(\Xi_0) = 0$, while $\kappa(\Xi_\varphi) = 1$ for $0 < \varphi < \pi$ and $d_J(\Xi_\varphi, \Xi_0) \rightarrow 0$ as $\varphi \rightarrow 0$.

In particular no bound $\kappa \leq f(\Delta_J)$ with $f(\delta) \rightarrow 0$ as $\delta \rightarrow 0$ holds, for $d = 2$ or for unital channels on M_4 .

The channels Φ_θ with $\theta > 0$ and Ξ_φ with $0 < \varphi < \pi$ are extreme points of the set of channels, and by Haagerup and Musat [16, Corollary 2.3] they are not generally factorizable. Whether κ has a modulus of continuity at $\mathcal{K}_d$ on the generally factorizable channels is open (Section 13). Outside the closure class, generally factorizable channels exist, and for them the purity cost is linear.

Corollary 3.5 (Exact abstract realization, linear purity). *There are $d \geq 3$ and a unital channel Φ on M_d , a Schur multiplier, which is generally factorizable but does not lie in $\mathcal{K}_d$. For such a channel $\Delta_J(\Phi) > 0$; every causal service with error $\epsilon < \Delta_J(\Phi)$ has $P + C \geq n(\Delta_J(\Phi) - \epsilon)^2/(2 \ln 2)$ at every exchange and memory; and the least purity rate at vanishing error is $\kappa(\Phi) \geq \Delta_J(\Phi)^2/(2 \ln 2) > 0$.*

So a channel can be realized exactly by a tracial von Neumann algebra, one use at a time, and still demand purity linear in n from every accurate finite device. The corollary rests on the negative solution of Connes' embedding problem [28]; neither d nor the channel is explicit.

3.2 Exact factors

Proposition 3.6 (Exchange converse). *Every causal service of a channel Φ on M_d for n rounds with error ϵ satisfies*

$$2(Q + J) - P - C \geq n(H(\Phi) - 2\epsilon \log d) - 1.$$

In particular, if $Q + P + C = o(n)$ and $\epsilon \rightarrow 0$, then $J \geq nH(\Phi)/2 - o(n)$.

Proof. Let ρ_* maximize $S(\Phi^c(\rho))$ and let ψ be a purification of ρ_* on the input and a d -dimensional reference. The observer feeds the input halves of n independent copies of ψ and keeps the references; its final system O consists of n outputs and n references, of total dimension d^{2n} . In the ideal experiment its state is $\omega^{\otimes n}$ with $\omega = (\Phi \otimes \text{id})(\psi)$, and $S(\omega) = S(\Phi^c(\rho_*)) = H(\Phi)$ because output, reference and environment of a Stinespring dilation are in a pure state. By the Fannes–Audenaert inequality the actual final state τ_O has $S(\tau_O) \geq nH(\Phi) - 2n\epsilon \log d - 1$.

Purify the initial memory state β_0 by a register F_0 and each imported state ω_t by a register F_t . All operations of the device are unitary, so the final state of O , the retained memory M , the exported registers E and $F = F_0 F_1 \cdots$ is pure, and

$$S(\tau_O) = S(MEF) \leq S(M) + S(E) + S(F_0) + \sum_t S(F_t) \leq Q + J + (Q - P) + (J - C).$$

□

The count uses no structure of Φ ; exported qubits paired with fresh imports carry up to two units of the environment's entropy each, which is the source of the factor one half. A rank count removes the dependence on the error.

Theorem 3.7 (Strong exchange converse). *Let ψ_* purify an input maximizing $S(\Phi^c(\rho))$, let $\omega = (\Phi \otimes \text{id})(\psi_*)$, and let V and μ_4 be the variance and the fourth central moment of $-\log \lambda$ when the eigenvalue λ of ω is drawn with probability λ . Every causal service of Φ for n rounds with error $\epsilon < 1$ satisfies:*

- (a) $2(Q + J) \geq nH(\Phi) - \sqrt{2Vn/(1-\epsilon)} - \log \frac{2}{1-\epsilon}$;
- (b) if $V > 0$, $n \geq \mu_4/V^2$ and $\epsilon + 2^{-\sqrt{nV}/16} < \frac{1}{64}$, then $2(Q + J) > nH(\Phi) + \sqrt{nV}/16$;
- (c) if $\epsilon = 0$, then $2(Q + J) \geq n \log r$, where r is the Choi rank of Φ .

For Φ_H the spectrum of ω is flat on 26 levels, and every service has error at least $1 - 2^{2(Q+J)}/26^n$.

Proof. Use the input $\psi_*^{\otimes n}$. Purify the initial memory and the imports. The observer's final state ρ is a marginal of a pure state whose complement consists of the final memory, the exported qubits and the purifiers, so $\text{rank } \rho \leq 2^Q 2^J 2^Q 2^J$. With Π the support projection of ρ , the test $\{\Pi, I - \Pi\}$ shows that the error is at least $1 - \text{Tr}(\Pi \omega^{\otimes n})$. Let $X = I_n - nH$, where I_n is the sum of n independent copies of $-\log \lambda$, and $\sigma = \sqrt{nV}$. For any threshold a , the eigenvalues of $\omega^{\otimes n}$ above $2^{-(nH+a)}$ carry total weight $\Pr(X < a)$, and at most $\text{rank } \Pi$ eigenvalues below it enter the trace, so $1 - \epsilon \leq \Pr(X < a) + 2^{2(Q+J)-nH-a}$.

(a) If $V = 0$, the spectrum is flat and the support test gives $2(Q + J) \geq nH + \log(1 - \epsilon)$ directly. Otherwise, with $a = -c'\sqrt{n}$, Chebyshev's inequality gives $\Pr(X < a) \leq V/c'^2$, and $c' = \sqrt{2V/(1-\epsilon)}$ gives the bound.

(b) Here $\mathbb{E}X^4 = n\mu_4 + 3n(n-1)V^2 \leq 4\sigma^4$, so Hölder's inequality gives $\mathbb{E}|X| \geq (\mathbb{E}X^2)^{3/2}(\mathbb{E}X^4)^{-1/2} \geq \sigma/2$, hence $\mathbb{E}X_+ \geq \sigma/4$. Since $\mathbb{E}X_+ \leq \sigma/8 + \sigma \Pr(X \geq \sigma/8)^{1/2}$, we get $\Pr(X \geq \sigma/8) \geq 1/64$. With $a = \sigma/8$, the assumption $2(Q + J) \leq nH + \sigma/16$ would give $\epsilon \geq \frac{1}{64} - 2^{-\sigma/16}$.

(c) With maximally entangled inputs the ideal final state is $\mathcal{J}(\Phi)^{\otimes n}$, of rank r^n , and the actual final state again has rank at most $2^{2(Q+J)}$; zero error forces $r^n \leq 2^{2(Q+J)}$.

For Φ_H , Proposition 4.2 gives $\Phi_H^c(\rho_*) = I/26$, so ω has 26 eigenvalues $1/26$ and $\text{Tr}(\Pi \omega^{\otimes n}) \leq \text{rank } \Pi / 26^n$. $\square$

Theorem 3.8 (Rolling reservoir). *Let Φ on M_d have a finite tracial factorization of bath dimension m . For every $n \geq 1$, $0 < \epsilon \leq 1$ and $0 \leq s \leq 1$ there is a causal service with error ϵ , $J \leq n(H(\Phi)/2 + s)$, $C = 0$, $P \leq \lceil \log_2(n+1) \rceil + 1$ and, with $x = \log_2(128n/\epsilon)$ and $x/0 = \infty$,*

$$Q \leq \log m + \log(2 + \log m) + C_d(\min\{\sqrt{nx}, x/s\} + \log(n+1)) \quad \text{if } x < n,$$

and $Q \leq \lceil n \log m \rceil + \lceil \log(n+1) \rceil$ otherwise. The constant C_d depends only on d .

Exact factors are thus cheap at any fixed exchange rate above the optimum. At the optimum itself, and below half the logarithm of the Choi rank, the picture depends on the channel.

Theorem 3.9 (A slack law). *Let ω be as in Theorem 3.7, with positive eigenvalues $\lambda_1, \dots, \lambda_r$, and suppose $H = H(\Phi) < \log r$. Put $Z_i = -\log \lambda_i$, $V = \sum_i \lambda_i (Z_i - H)^2$, $R = \max_i Z_i - \min_i Z_i$, $a = (\ln 2)V/2$, $A = (\ln 2)R^2/4$, $\gamma = \min\{1/(2R), a/(4A), 1/(2\sqrt{A})\}$, $c = a\gamma/2$ and $x_0 = \max\{2, 64R^2/(a\gamma)^2, 8\log(8/3)/(a\gamma), 64R^2\gamma^2\}$. Let a causal service of Φ for n rounds have error ϵ .*

- (a) If $J \leq n(H/2 + \sigma)$ with $0 \leq \sigma \leq c/16$, and $x = \min\{\log(1/\epsilon), n/4\} \geq x_0$, then $Q \geq \min\{c\sqrt{nx}/8, c^2x/(64\sigma)\}$, the second term being absent when $\sigma = 0$.

(b) If $\epsilon \leq 1/4$ and $J \leq n(H/2 + s)$, then $Q \geq \frac{3c}{16}\sqrt{by} - 2sb$ for every integer $1 \leq N \leq n$ with $b = \lfloor n/N \rfloor \geq y = \log(\delta_0 N/(2\epsilon)) \geq x_0$, where $\delta_0 = \min\{\frac{1}{2}, c\sqrt{x_0}/(8(2\sqrt{2V} + 2))\}$. Consequently, for fixed $0 < \eta < 1$ there are $s_1, n_1 > 0$ depending on Φ and η such that $Q \geq (\eta c^2/1300) \log(n/\epsilon)/s$ whenever $n \geq n_1$, $n^{-(1-\eta)/2} \leq s \leq s_1$ and $\log(1/\epsilon) \leq n^{\eta/3}$.

The rank r of ω is the Choi rank (the maximizing complementary output is faithful), so the hypothesis says that ω is not flat. For an exact factor, Theorem 3.8 gives $O_\Phi(\min\{\sqrt{nx}, x/s\} + \log n)$ with $x = \log(128n/\epsilon)$, so the two bounds agree up to constants at polynomially small error and over the range of slack in (b). Only at fixed error and zero slack does a gap between $\sqrt{n}$ and $\sqrt{n \log n}$ remain.

Theorem 3.10 (Logarithmic memory below the rank rate). *Let the normalized Choi state of Φ have rank r and least nonzero eigenvalue $\lambda < 1/r$, and put $\mu = \log(1/(r\lambda))$. Fix $0 < g \leq \frac{1}{2} \log r$ and $0 \leq \epsilon < 1$. Every causal service of Φ for n rounds with error ϵ and $J \leq n(\frac{1}{2} \log r - g)$ satisfies*

$$Q \geq \frac{g}{2\mu} \log n - O_{g,r,\lambda,\epsilon}(\log \log(n+2)).$$

If Φ has an exact finite tracial factorization and $H(\Phi) < \log r$, then at every fixed exchange rate strictly between $H(\Phi)/2$ and $\frac{1}{2} \log r$ the least memory at fixed error $0 < \epsilon < 1$ is $\Theta(\log n)$.

Corollary 3.11 (The exchange threshold for sublogarithmic memory). *Let Φ have Choi rank r . At any fixed error bound $\epsilon < 1$, every sequence of causal services with $Q_n = o(\log n)$ satisfies $\liminf_{n \rightarrow \infty} J_n/n \geq \frac{1}{2} \log r$.*

Proof. For a nonflat Choi state, any subsequence with exchange rate bounded a fixed positive amount below $\frac{1}{2} \log r$ contradicts Theorem 3.10. For a flat Choi state, fresh Bell inputs and the support test used in Theorem 3.7 give $2(Q_n + J_n) \geq n \log r + \log(1 - \epsilon)$ directly. The rank-one case follows from $J_n \geq 0$. $\square$

Proposition 3.12 (Dephasing by an involution). *Let R be a Hermitian unitary on $\mathbb{C}^d$ and $0 < p < 1$. For every n the channel $\rho \mapsto p\rho + (1-p)R\rho R$ has a causal service with error 0, $Q = 1$, $P = C = 0$ and $J = \lfloor (n-1)/2 \rfloor$. For $R = Z$ on M_2 the Choi rank is 2, so by Theorems 3.7 and 3.10 the rate $\frac{1}{2}$ is the least exchange rate at which bounded memory suffices, at zero error and at every fixed error below one; for $p = \frac{1}{2}$ it is the optimal rate $H/2$.*

Proposition 3.13 (Equal mixtures of orthonormal unitaries). *Let $U_0, \dots, U_{r-1}$ be unitaries on $\mathbb{C}^d$ with $\tau_d(U_a^* U_b) = \delta_{ab}$ and $r = 2^q$, and let $\Phi(\rho) = \frac{1}{r} \sum_a U_a \rho U_a^*$. Then $H(\Phi) = q$, and for every n there is a causal service with error 0, $Q = q$, $P = C = 0$ and $J = q \lfloor (n-1)/2 \rfloor$, at the optimal rate $H/2$. Complete dephasing on $\mathbb{C}^{2^q}$ and the completely depolarizing qubit channel are examples.*

Remark 3.14. Biased dephasing $\rho \mapsto \frac{1}{4}\rho + \frac{3}{4}Z\rho Z$ has the exact factorization $U = I \otimes |0\rangle\langle 0| + Z \otimes (I_4 - |0\rangle\langle 0|)$ with a four-dimensional bath, so $D^J \leq 4$ at every accuracy. Its Choi state has eigenvalues $\frac{1}{4}, \frac{3}{4}$, so $r = 2$, $H = h_2(1/4) \approx 0.811$ and $V = \frac{3}{16} \log^2 3$. At $J \leq nH/2$ Theorem 3.9 forces memory of order $\sqrt{n}$, at a fixed rate strictly between $H/2$ and $\frac{1}{2}$ Theorem 3.10 gives $\Theta(\log n)$, and at rate $\frac{1}{2}$ Proposition 3.12 gives one qubit. Bounded profiles thus coexist with memory of order $\sqrt{n}$, $\log n$ or 1, according to the exchange rate.

At the rank rate $\frac{1}{2} \log r$ the buffer of Theorem 3.8 shrinks to a logarithm, and for exact factors (more generally, channels in $\mathcal{K}_d$) whose optimal complementary spectrum is flat that rate is the optimal one.

Theorem 3.15 (Exact factors at the rank rate). *Let Φ on M_d have a finite tracial factorization of bath dimension m and Choi rank $r \geq 2$. For every $n \geq 1$ and $0 < \epsilon \leq 1$, with $x = \log(128n/\epsilon)$, there is a causal service of Φ with error ϵ ,*

$$J = \left\lfloor \frac{n}{2} \log r \right\rfloor, \quad P = C = 0,$$

and $Q \leq \log m + \log(2 + \log m) + O_d(x) + \lceil \log(n+1) \rceil$. If $H(\Phi) = \log r$, no history of the construction ever exceeds its cost threshold.

Corollary 3.16 (Flat and not flat at the optimal rate). *Let Φ be an exact factor, and let ω be as in Theorem 3.7.*

- (a) *If ω is flat, the optimal rate $H(\Phi)/2$ equals the rank rate, and Φ has causal services at the optimal rate with $C = 0$, $P = 0$ and $Q = O_\Phi(\log(2n/\epsilon))$.*
- (b) *If ω is not flat, every causal service at $J \leq nH(\Phi)/2$ with fixed error $\epsilon \leq 2^{-x_0}$ has $Q = \Omega_\Phi(\sqrt{n})$ (Theorem 3.9), while Theorem 3.8 gives $O_\Phi(\sqrt{n \log(n/\epsilon)})$.*

So at the optimal rate a flat exact factor needs only logarithmic memory, and a non-flat one needs at least order $\sqrt{n}$ and at most order $\sqrt{n \log n}$. Above the optimum every exact factor needs $\Theta(\log n)$ at fixed rates strictly between $H/2$ and $\frac{1}{2} \log r$ (Theorem 3.10) and $O(\log(n/\epsilon))$ at the rank rate; Propositions 3.12 and 3.13 show that bounded memory can suffice there.

3.3 Service and approximation profiles

Theorem 3.17 (One round bounds the mixture profile). *Every causal service of a channel Φ for n rounds with memory Q , error ϵ and $P + C \leq n\zeta$ satisfies*

$$D_\Phi^{\text{mix}}(\epsilon + K\sqrt{\zeta}) \leq 2^Q, \quad K = (4 + \frac{1}{\sqrt{2}})\sqrt{\ln 2}.$$

For collision-at-call services, K may be replaced by $\sqrt{2 \ln 2}$.

Theorem 3.18 (Sequential extraction). *Let a causal service of a channel Φ on M_d for n rounds, with imports between rounds as in Definition 2.3, have memory Q , error $\epsilon < 1/2$ and $P + C \leq B$. Put*

$$t_n = \frac{B + \epsilon + h_2(\epsilon) + \epsilon \log n}{n(1 - \epsilon)}, \quad h_n = \sqrt{2 \ln 2 t_n}.$$

Then, at any exchange,

$$\mathfrak{D}_\Phi\left(\frac{\epsilon}{1 - \epsilon} + d h_n, t_n + h_n\right) \leq 2^Q.$$

For fixed ϵ and $B = O(\log n)$ the leakage parameter is $t_n + h_n = O(\sqrt{\log n/n})$. More precisely, some round t defines a channel $\Psi_t(X) = \text{Tr}_M U_t(X \otimes \rho_t) U_t^*$ with bath dimension 2^Q , $d_\diamond(\Psi_t, \Phi) \leq \epsilon/(1 - \epsilon)$, support leakage q_t and entropy production $a_t = S(T_t(\rho_t)) - S(\rho_t)$ with $q_t + a_t \leq t_n$, where $T_t(\rho) = \text{Tr}_d U_t(\tau_d \otimes \rho) U_t^*$.

Theorem 3.19 (From profiles to services). *Let Φ on M_d lie in the closure class, and let $0 < \epsilon \leq 1/2$, $0 \leq s \leq 1$ and $x = \log(256n/\epsilon) < n$. There is a causal service of Φ for n rounds with error ϵ , $J \leq n(H(\Phi)/2 + s)$, $C = 0$, $P \leq \lceil \log(n+1) \rceil + 1$ and*

$$Q \leq \log m + \log(2 + \log m) + C_d(\min\{\sqrt{nx}, x/s\} + \log(n+1)), \quad m = D_\Phi^J\left(\frac{\epsilon}{8dn}\right),$$

with C_d depending only on d .

Together with Theorem 3.18, this brackets the memory of every closure channel between its support profile at leakage of order $\sqrt{(B + \log n)/n}$ and its Choi profile at accuracy of order ϵ/n , up to the buffers. For Φ_H this general bound, fed with the corner models, already gives memory $O(\sqrt{n/\epsilon} \log(n/\epsilon))$ at the optimal rate (Corollary 4.14); the construction of Section 10 gives the same order in n with the better dependence $\sqrt{\log(1/\epsilon)}$ on the error, and the Choi profile of Theorem 3.36 with the compiler of Theorem 3.21 lowers the order to $(n/\epsilon)^{1/3} \log^{4/3}(n/\epsilon)$ (Corollary 3.37).

Remark 3.20 (The profile read by one round). Let $D_{\Phi}^{\text{ent}}(u, v)$ be the least M for which some unitary U on $\mathbb{C}^d \otimes \mathbb{C}^M$ and bath state ρ give a channel $\Psi(X) = \text{Tr}_M U(X \otimes \rho) U^*$ with

$$d_{\diamond}(\Psi, \Phi) \leq u, \quad \ell_{\Phi}(\Psi) + S(T(\rho)) - S(\rho) \leq v,$$

where $T(\rho) = \text{Tr}_d U(\tau_d \otimes \rho) U^*$ as in Theorem 3.18. That theorem gives $D_{\Phi}^{\text{ent}}(\epsilon/(1-\epsilon), t_n) \leq 2^Q$, with no square root. The square root enters only when the round is converted into a finite tracial mixture:

$$\mathfrak{D}_{\Phi}(u + d\sqrt{2 \ln 2 v}, v + \sqrt{2 \ln 2 v}) \leq D_{\Phi}^{\text{ent}}(u, v) \leq d^4 \mathfrak{D}_{\Phi}(u, v).$$

The first inequality is Lemma 6.1 together with the fact that leakage moves by at most d_J . For the second, write a mixture in $\mathcal{F}_m$ with at most d^4 components (Carathéodory) and realize it as one round whose bath is the direct sum of the component baths, in the state $\bigoplus_i p_i \tau_{m_i}$; tracial channels are unital, so its entropy production is zero. Hence, if $\log \mathfrak{D}_{\Phi}(u_0, v) \geq \gamma \log(1/v) - O(1)$ for some $u_0 > \epsilon/(1-\epsilon)$ and $\gamma > 0$, and $\log D_{\Phi}^J(\delta) = O(\log(1/\delta))$, then Theorems 3.18 and 3.19 give memory $\Theta(\log n)$ at purity budget $O(\log n)$ and fixed exchange slack: the accuracy scale matters only for profiles that grow faster than every power.

The compiler of Theorem 3.19 runs at exchange $n(H(\Phi)/2 + s)$ for every $0 \leq s \leq 1$; it absorbs the excess entropy of the approximant in spare flat slots of the memory, and at $s = 0$ its buffer is of order $\sqrt{n \log(n/\epsilon)}$. At the rank rate no excess has to be absorbed and the buffer is polylogarithmic.

Theorem 3.21 (A compiler at the rank rate). *Let Φ on M_d have Choi rank $r \geq 2$, let $n \geq 1$ and $0 < \epsilon \leq 1$, and put $x = \log(256n/\epsilon)$. Let Ψ have a finite tracial factorization of bath dimension m with $d_J(\Psi, \Phi) \leq \epsilon/(8dn)$; for $\Phi \in \mathcal{K}_d$ one may take $m = D_{\Phi}^J(\epsilon/(8dn))$. There is a causal service of Φ with error ϵ ,*

$$J = \left\lfloor \frac{n}{2} \log r \right\rfloor, \quad P = C = 0, \quad Q \leq \log m + \log(2 + \log m) + O_d(x^2).$$

The constants depend only on d .

When $\Phi \in \mathcal{K}_d$ and $H(\Phi) = \log r$ this is the optimal rate itself, where Theorem 3.19 pays a buffer of order $\sqrt{n \log(n/\epsilon)}$. The price of running at the rank rate for an approximant is the quadratic term x^2 , which comes from the rare Kraus directions of Ψ outside the support of $\mathcal{J}(\Phi)$.

3.4 Group channels

The construction of Φ_H in Section 4 applies to every finite presentation $\mathcal{P} = \langle S \mid R \rangle$ of a group G whose relators $r_i = \sigma_{i,1} \cdots \sigma_{i,\ell_i}$ have length $\ell_i \geq 2$. Its symbols are the signed generators and the prefixes $p_{i,l} = \sigma_{i,1} \cdots \sigma_{i,l}$ with $2 \leq l \leq \ell_i - 1$; its triangles are $(p_{i,l-1}, \sigma_{i,l}, p_{i,l})$ for $2 \leq l \leq \ell_i$, with $p_{i,1} = \sigma_{i,1}$, and (s, s^{-1}, e) for $s \in S$. With $p = |S|$ and $k = |R|$ the system dimension is $d = 1 + 4p + 3 \sum_i \ell_i - 4k$; for Johnson's presentation of H_3 this is 104. Evaluating the gadget U_{λ} in the regular representation with the canonical trace gives the *gadget channel*

Φ_G on M_d , which is generally factorizable. The nonzero entries of the gadget sit at a list $\mathcal{T}$ of slots a , each with a coefficient $c_a \in \{\pm 1, \pm 2^{-1/2}\}$ and a word w_a (the empty word, a signed generator, a prefix, or the concatenation xy of a triangle), and $|\mathcal{T}| \leq 2d - 1$.

Proposition 3.22 (Gadget entropy). *Let r be the number of distinct elements of G carried by the entries of the gadget. Then Φ_G has Choi rank r and $H(\Phi_G) = \log r$, and some input has flat complementary output I_r/r . So the entropy rate $H(\Phi_G)/2$ of every gadget channel is its rank rate, which is its optimal exchange rate when $\Phi_G \in \mathcal{K}_d$, and its probe spectrum in Theorem 3.7 is flat of rank r .*

Proof. Group the entries by element: $U_\lambda = \sum_g A_g \otimes \lambda(g)$, where the A_g have disjoint supports and are therefore independent. Since $\tau(\lambda(g)\lambda(h)^*) = \delta_{gh}$, $\mathcal{J}(\Phi_G) = \frac{1}{d} \sum_g |A_g\rangle\langle A_g|$ has rank r . Every element carried is e or the element of a symbol, since the product slot of a triangle carries a prefix or e . So every g has an anchor: a diagonal entry i_g with coefficient 1, the identity entry or a symbol entry, whose column contains no other entry of the gadget. For $\rho_* = \frac{1}{r} \sum_g |i_g\rangle\langle i_g|$ this gives $\Phi_G^c(\rho_*)_{gh} = \text{Tr}(A_g \rho_* A_h^*) = \delta_{gh}/r$, and $\log r$ is the largest entropy of a complement of dimension r . The probe output of ρ_* has the spectrum of $\Phi_G^c(\rho_*)$. $\square$

For a unitary assignment $\phi : S \rightarrow U(M)$, extended to words with $\phi(s^{-1}) = \phi(s)^*$, put

$$e(\phi) = \max_{r \in R} \|\phi(r) - I\|_2, \quad z(\phi) = \max_{a,b \in \mathcal{T}} |\tau_M(\phi(w_a)\phi(w_b)^*) - \mathbf{1}[w_a = w_b \text{ in } G]|.$$

The *approximation profile* $\mathbf{h}_{\mathcal{P}}(e, z)$ is the least M for which some ϕ has $e(\phi) \leq e$ and $z(\phi) \leq z$, or ∞ ; $\mathbf{h}_{\mathcal{P}}^\oplus$ is the same with M_M replaced by finite direct sums of matrix algebras of size at most M with any faithful tracial state. The moments read by z are those of the finitely many gadget words, a finite piece of G in the sense of Cornulier's chunks. Let $A_{\mathcal{P}} \geq 1$ be the largest area of a word $w_a w_b^{-1}$ with $w_a = w_b$ in G , and $L = \max\{2, \ell_1, \dots, \ell_k\}$. Put

$$h_d = 1 + 2\sqrt{d}, \quad b_d = 2\sqrt{d}(1 + \sqrt{2})h_d, \quad c_{\mathcal{P}} = h_d + 2Lb_d,$$

and $C_{\mathcal{P}} = 4d\sqrt{2\ln 2} + 4dc_{\mathcal{P}}(1 + \sqrt{2\ln 2})^{1/2}$.

Theorem 3.23 (Channel profiles are group profiles). *Let Φ_G on M_d be the gadget channel of $\mathcal{P}$.*

- (a) *For all $e, z \geq 0$, $D_{\Phi_G}^J(dz) \leq \mathbf{h}_{\mathcal{P}}(e, z)$ and $\mathfrak{D}_{\Phi_G}(d^2z, A_{\mathcal{P}}^2e^2) \leq \mathbf{h}_{\mathcal{P}}^\oplus(e, z)$.*
- (b) *If $\Psi \in \mathcal{F}_M$ has $d_\diamond(\Psi, \Phi_G) \leq u$ and $\ell_{\Phi_G}(\Psi) \leq v$, then $\mathbf{h}_{\mathcal{P}}^\oplus(2Lb_d\sqrt{v}, 4u + 4dc_{\mathcal{P}}\sqrt{v}) \leq M$. For a single factorization, $\mathbf{h}_{\mathcal{P}}(2Lb_d\sqrt{\delta}, 4d(\delta + c_{\mathcal{P}}\sqrt{\delta})) \leq D_{\Phi_G}^J(\delta)$.*
- (c) *For $0 < v \leq 1$ and $u \geq 0$, $\mathbf{h}_{\mathcal{P}}^\oplus(C_{\mathcal{P}}v^{1/4}, 4u + C_{\mathcal{P}}v^{1/4}) \leq D_{\Phi_G}^{\text{ent}}(u, v)$.*
- (d) *For $0 < \eta \leq 1$, $\mathbf{h}_{\mathcal{P}}(\sqrt{e^2 + 4\eta}, z + \eta) \leq \lceil 3N_{\mathcal{P}}M/\eta \rceil$, where $M = \mathbf{h}_{\mathcal{P}}^\oplus(e, z)$ and $N_{\mathcal{P}} = 2|\mathcal{T}|^2 + |R| + 1$.*
- (e) *For $z_0 \in (0, 1)$, $0 < \delta \leq 1$, $k = \lceil \log(2d/\delta)/(2\log(1/z_0)) \rceil \geq 1$ and $e = (\delta/(2dkA_{\mathcal{P}}^2))^{1/2}$, $\log D_{\Phi_G}^J(\delta) \leq 2k \log \mathbf{h}_{\mathcal{P}}(e, z_0)$.*
- (f) *$\Phi_G \in \mathcal{K}_d$ if and only if $\mathbf{h}_{\mathcal{P}}(e, z) < \infty$ for all $e, z > 0$, and if and only if the same holds for $\mathbf{h}_{\mathcal{P}}^\oplus$. If G is hyperlinear, then $\Phi_G \in \mathcal{K}_d$.*

So the static profiles of a gadget channel and the approximation profile of its presentation bound each other up to polynomial changes of scale: a square root from leakage to relator defect, a further square root through a round with biased memory, and tensor-conjugate amplification (e) from a fixed moment tolerance to accuracy δ . Combined with Theorems 3.18 and 3.19 this places the memory of a gadget channel between two readings of its group's profile.

Theorem 3.24 (Memory reads the group). *Let Φ_G be the gadget channel of $\mathcal{P}$, let $0 < \epsilon \leq 1/16$, and for a purity budget B let t_n be as in Theorem 3.18.*

- (a) *For all n with $t_n \leq 1$ and $C_{\mathcal{P}} t_n^{1/4} \leq 7/30$, every causal service of Φ_G for n rounds with error ϵ and $P + C \leq B$ has, at any exchange,*

$$Q \geq \log h_{\mathcal{P}}^{\oplus}(C_{\mathcal{P}} t_n^{1/4}, \frac{1}{2}) \geq \log h_{\mathcal{P}}(\sqrt{2} C_{\mathcal{P}} t_n^{1/4}, \frac{3}{4}) - \log \left(N_{\mathcal{P}} \left(\frac{8}{C_{\mathcal{P}}^2 \sqrt{t_n}} + 1 \right) \right) - 1,$$

the second inequality once $C_{\mathcal{P}}^2 \sqrt{t_n} \leq 1$. For $B = A \log(n+1)$ the subtracted term is $O_{\mathcal{P}, \epsilon, A}(\log n)$.

- (b) *Suppose $\Phi_G \in \mathcal{K}_d$, and let $z_0 \in (0, 1)$, $0 < s \leq 1$ and $x = \log(256n/\epsilon) < n$. Put*

$$k_n = \left\lceil \frac{\log(16d^2 n/\epsilon)}{2 \log(1/z_0)} \right\rceil, \quad e_n = \left(\frac{\epsilon}{16d^2 n k_n A_{\mathcal{P}}^2} \right)^{1/2}.$$

There is a causal service with error ϵ , $J \leq n(H(\Phi_G)/2 + s)$, $C = 0$, $P \leq \lceil \log(n+1) \rceil + 1$ and

$$Q \leq 2k_n \log h_{\mathcal{P}}(e_n, z_0) + \log(2 + 2k_n \log h_{\mathcal{P}}(e_n, z_0)) + C_d \left(\frac{x}{s} + \log(n+1) \right).$$

- (c) *Under the hypotheses of (b), without the condition $x < n$, there is also a causal service at the optimal exchange $J \leq nH(\Phi_G)/2$, with $C = 0$, $P = 0$ and the same bound on Q with $C_d(x/s + \log(n+1))$ replaced by $O_d(\log^2(2n/\epsilon))$.*

- (d) *At the rank rate $J \leq nH(\Phi_G)/2$, every causal service with error ϵ has $P + C \leq 14Q + 3$, so (a) holds with $B = 14Q + 3$, whatever the purity budget.*

At constant error and logarithmic purity the lower bound reads the relator defect at $(\log n/n)^{1/4}$, and the upper bound reads it at $(n \log n)^{-1/2}$, roughly the square, with a factor of order $\log n$. When $\log h_{\mathcal{P}}(e, z)$ is a power of $1/e$, this determines the growth exponent of the memory up to a factor of 2; for general profiles the two readings of the same function can be further apart. The loss has two sources: the two square roots on the channel side in Theorem 3.23(c), and the accuracy $\epsilon/(8dn)$ at which the compiler needs its approximant. Closing the gap needs a lower bound that reads the profile at defect of order $n^{-1/2}$, or a service that reads it at a coarser defect, or both.

The correspondence turns any group with a steep profile into a channel that needs polynomial memory. Slofstra's group [4, Theorem 1.2] is finitely presented and has a word w with $\text{hlp}(w; \delta, \epsilon) \geq C' \exp[C(\delta/\epsilon)^\alpha]$ for every $\alpha < \frac{1}{2}$, where $\text{hlp}(w; \delta, \epsilon)$ is the least dimension of an ϵ -representation that moves w by at least δ in normalized Hilbert–Schmidt norm. It is not known whether this group is hyperlinear. Let $\mathcal{P}$ be its presentation, let $\mathcal{P}_S = \langle S \cup \{j\} \mid R \cup \{j^{-1}w\} \rangle$, a Tietze transformation that makes w a symbol of the gadget, and let Φ_S on M_{d_S} be its gadget channel.

Theorem 3.25 (A Slofstra channel). *Let $0 < \epsilon \leq 1/16$ and $A \geq 0$. There are $c > 0$ and n_0 , depending on $\mathcal{P}_S$, ϵ and A , such that every causal service of Φ_S for $n \geq n_0$ rounds with error ϵ and $P + C \leq A \log(n+1)$ has, at any exchange,*

$$Q \geq c \frac{n^{1/8}}{(\log n)^{13+1/8}}.$$

At its rank rate $J \leq nH(\Phi_S)/2$, every causal service with error ϵ has $Q \geq n^{1/9-o(1)}$, whatever its purity budget. Moreover exactly one of the following holds.

- (i) $\Phi_S \in \mathcal{K}_d$, and such services exist for every large n (Theorem 3.24(c), even with $P = C = 0$); the rank rate is then the optimal exchange rate.
- (ii) $\Phi_S \notin \mathcal{K}_d$. Then Φ_S is generally factorizable with $\Delta = \Delta_J(\Phi_S) > 0$, and Slofstra's group is not hyperlinear. Every service with error $\epsilon < \Delta$ has $P + C \geq n(\Delta - \epsilon)^2 / (2 \ln 2)$, so services with logarithmic purity exist only for finitely many n ; and every service at the rank rate with error $\epsilon < \min\{\Delta, 1/16\}$ has

$$Q \geq \frac{n(\Delta - \epsilon)^2}{28 \ln 2} - \frac{3}{14}.$$

The memory bounds of the first display hold in both branches; branch (ii) makes the memory at the rank rate linear. Its exponent $\frac{1}{8}$ exceeds β_H , but it concerns Φ_S , whose presentation is produced by an embedding theorem and is not enumerated, so neither d_S nor the constants are explicit. A proof of branch (ii) would show that Slofstra's group is not hyperlinear. No conclusion about that unresolved branch is assumed here.

3.5 The Houghton channel

Theorem 3.26 (Static profile). *The channel Φ_H of Section 4 satisfies, for $0 < \delta \leq 1/2$,*

$$\log D_{\Phi_H}^J(\delta) \leq \frac{55}{\delta} \log \frac{54}{\delta},$$

and, for every $\delta > 0$,

$$\log D_{\Phi_H}^{\text{mix}}(\delta) \geq c_H \delta^{-\beta_H} - \frac{1}{16}, \quad \beta_H = \frac{1}{2 \log 130 + 1} \approx \frac{1}{15.04}.$$

Here $c_H = T_H^{-2\beta_H}/16$ with $T_H = 750 \cdot 30 \cdot 3^{\log 130} \cdot 68\sqrt{624} \approx 8.567 \cdot 10^{10}$; numerically $c_H > 2.2 \cdot 10^{-3}$, and the bound is nontrivial for $\delta < T_H^{-2} \approx 1.4 \cdot 10^{-22}$. In particular Φ_H lies in the closure class but has no exact finite tracial factorization, nor an exact finite mixture of them.

Theorem 3.27 (Support profile of Φ_H). *Let $d = 104$. For every $v > 0$,*

$$\log \mathfrak{D}_{\Phi_H}(1/8, v) \geq c_H v^{-\beta_H} - \frac{1}{16},$$

and more precisely $\log \mathfrak{D}_{\Phi_H}(1/8, v) \geq \frac{1}{16} \lfloor (T_H^2 v)^{-\beta_H} \rfloor$ for $v \leq T_H^{-2}$. No finite mixture of finite tracial channels with zero support leakage lies within diamond distance $1/8$ of Φ_H . Conversely $\log \mathfrak{D}_{\Phi_H}(1/8, v) \leq 8M + 8 + 2 \log M$ with $M = \max\{48, \lceil (0.3/v)^{1/3} \rceil\}$ (Proposition 4.17), so the exponent of the support profile lies between β_H and $1/3$; approximants by permutations, as in Proposition 4.12, give the exponent $1/2$.

Theorem 3.28 (One round with a memory state that is not flat). *Let $\Psi(X) = \text{Tr}_M U(X \otimes \rho) U^*$ with U unitary on $\mathbb{C}^{104} \otimes \mathbb{C}^M$ and ρ a state, and suppose $d_\diamond(\Psi, \Phi_H) \leq 1/8$. Let ℓ be its support leakage and $a = S(T(\rho)) - S(\rho)$ its entropy production, where $T(\rho) = \text{Tr}_{104} U(\tau_{104} \otimes \rho) U^*$. Then*

$$\log \text{rank } \rho \geq S(\rho) \geq c_*(\ell + a)^{-\beta_H} - \frac{1}{8},$$

with $c_* = (16K_H^{2\beta_H})^{-1} > 2.5 \cdot 10^{-4}$ and $K_H = 10^8 \cdot 30 \cdot 3^{\log 130} \cdot 2^{17}$. In particular no such channel has $\ell + a = 0$.

Theorem 3.29 (Constant-error lower bound). *Every causal service of Φ_H for n rounds with error $\epsilon \leq 1/16$ and $P + C \leq B$ has, at any exchange,*

$$Q \geq c_a \left(\frac{n}{B + \log(n+1)} \right)^{\beta_H} - \frac{1}{8}, \quad c_a = c_*(16/15)^{-\beta_H} > 2.5 \cdot 10^{-4}.$$

The constants are far from optimal: the bound is positive only when $n/(B + \log(n+1)) > (8c_a)^{-1/\beta_H} \approx 2.8 \cdot 10^{40}$. For $A \geq 0$ we write $c_A = c_a(A+2)^{-\beta_H}$: if $P + C \leq A \log(n+1) + 1$, then $B + \log(n+1) \leq (A+2) \log(n+1)$ and the theorem gives $Q \geq c_A(n/\log(n+1))^{\beta_H} - \frac{1}{8}$. What Theorem 3.29 establishes is the exponent.

Theorem 3.30 (A sublinear service at the optimal rate). *Let $n \geq 1$ and $0 < \epsilon < 1$. Put $c = 1 - 2^{-1/4}$ and $N = 2^{\lceil \log n \rceil}$, let t be the least even integer with $t \geq 4$ and $2^t \geq 4/\epsilon$, and put*

$$R = \left\lceil \frac{4}{c} \sqrt{2tN} \right\rceil, \quad M = 2R + 123.$$

There is a causal service of Φ_H with error at most ϵ and

$$J \leq \frac{n}{2} \log 26, \quad C = 0, \quad P < 2 \log M + 2 \lceil \log(n+1) \rceil + 2,$$

and $Q = O(\sqrt{n \log(2/\epsilon)} \log(n/\epsilon) + \log(2/\epsilon) \log(n+1))$, which is $O_\epsilon(\sqrt{n} \log(n+1))$ at fixed error. The register sizes are explicit (Section 10); at fixed error the bath register of $\lceil \log(M^2(2M)!) \rceil$ qubits dominates.

Corollary 3.31 (Two-sided separation at constant error). *Fix $0 < \epsilon \leq 1/16$ and a purity budget $P + C \leq A \log(n+1)$, with $A \geq 4$ so that the constructions below fit and n large. Every channel with an exact finite tracial factorization has a causal service with $Q = O(\log n)$ at exchange $J \leq n(H/2 + s)$ for any fixed $s > 0$. For Φ_H the least memory satisfies*

$$c_A \left(\frac{n}{\log n} \right)^{\beta_H} \leq Q_{\min} \leq O(\sqrt{n} \log n),$$

where the lower bound holds at any exchange and the upper bound is attained with $C = 0$ at the optimal exchange $J \leq \frac{n}{2} \log 26$.

At the rank rate of a channel with flat probe spectrum a device cannot buy memory with purity. The next bound is a purity budget in terms of memory, at every fixed error.

Theorem 3.32 (Memory bounds purity at the rank rate). *Let Φ have an input ρ_* maximizing $S(\Phi^c(\rho))$ whose probe output $\omega = (\Phi \otimes \text{id})(\psi_*)$, for a purification ψ_* of ρ_* , has flat spectrum of rank r . Every causal service of Φ for n rounds with error $\epsilon \leq 1/16$ and excess exchange $E_x = (J - \frac{n}{2} \log r)_+$ satisfies*

$$P + C \leq 14Q + 4E_x + 2 \log \frac{16}{7}.$$

In particular $P + C \leq 14Q + 2 \log \frac{16}{7}$ at $J \leq \frac{n}{2} \log r$.

For Φ_H the probe spectrum is flat on 26 levels (Theorem 3.7), so Theorem 3.29 applies with a budget $B = 14Q + 3$ and gives a memory bound at the optimal rate that needs no assumption on the purity (Corollary 3.33(c)).

Corollary 3.33 (Purity and memory). *(a) Every causal service of Φ_H with error $\epsilon \leq 1/16$ and memory Q has, at any exchange, $P + C \geq n(c_a/(Q + \frac{1}{8}))^{1/\beta_H} - \log(n+1)$; bounded memory needs a purity budget linear in n .*

(b) With pure imports, Φ_H has a causal service with error 0, $Q = 5$, $J = 5(n-1)$ and $P + C = 5n$.

(c) Every causal service of Φ_H with error $\epsilon \leq 1/16$ and $J \leq \frac{n}{2} \log 26$ satisfies $(14Q + 3 + \log(n+1))(Q + \frac{1}{8})^{1/\beta_H} \geq c_a^{1/\beta_H} n$, so $Q = \Omega(n^{\beta_H/(1+\beta_H)}) = \Omega(n^{1/(2 \log 130+2)})$ whatever its purity budget.

(d) More generally, every causal service of Φ_H with error $\epsilon \leq 1/16$ and excess exchange $E_x = (J - \frac{n}{2} \log 26)_+$ satisfies, whatever its purity budget,

$$(14Q + 4E_x + 3 + \log(n+1))(Q + \frac{1}{8})^{1/\beta_H} \geq c_a^{1/\beta_H} n.$$

So bounded memory needs excess exchange linear in n , and if $E_x = O(n^\gamma)$ with $0 \leq \gamma < 1$, then $Q = \Omega(n^{\min\{\beta_H/(1+\beta_H), \beta_H(1-\gamma)\}})$; for instance an excess of $O(\sqrt{n})$ exchanged qubits leaves $Q = \Omega(n^{\beta_H/2})$.

Proof. (a) rearranges Theorem 3.29. (b) Extend the Stinespring isometry $|\psi\rangle \mapsto \sum_g K_g |\psi\rangle |g\rangle$ of Proposition 4.2 to a unitary on the input and five qubits in the pure state $|0^5\rangle$, apply it every round, and replace the five qubits by fresh pure ones after every round but the last. (d) Theorem 3.32 gives $P + C \leq 14Q + 4E_x + 3$; insert this into Theorem 3.29. If $E_x \leq Cn^\gamma$, then either $Q \geq n^\gamma$, and the first factor is $O(Q + \log n)$, so $Q = \Omega(n^{\beta_H/(1+\beta_H)})$; or $Q < n^\gamma$, the first factor is $O(n^\gamma + \log n)$, and $Q = \Omega(n^{\beta_H(1-\gamma)})$. If $E_x = o(n)$ and Q were bounded, the left side would be $o(n)$. Part (c) is the case $E_x = 0$: Theorem 3.32 gives $P + C \leq 14Q + 2\log(16/7) < 14Q + 3$ at this exchange; insert this into Theorem 3.29. $\square$

Theorem 3.34 (Vanishing error). *Every causal service of Φ_H with error $\epsilon \leq 1/8$ and $P + C \leq n\zeta$ has $Q \geq c_*(\epsilon + \zeta)^{-\beta_H} - \frac{1}{8}$ at any exchange.*

Corollary 3.35. *Fix $s \in (0, 1]$ and $a > 0$, and let $\epsilon_n = n^{-a}$. Every channel with an exact finite tracial factorization has a causal service with error ϵ_n , deficit $O(\log n)$, exchange $n(H/2 + s)$ and memory $O(\log n)$. Every causal service of Φ_H with error ϵ_n and deficit $O(\log n)$ has memory*

$$Q = \Omega\left(\left(\frac{n}{\log n}\right)^{\beta_H}\right),$$

and Φ_H has causal services with error ϵ_n , $C = 0$, $P = O(\log n)$, the optimal exchange $J \leq \frac{n}{2} \log 26$, and $Q = O_a(\sqrt{n}(\log n)^{3/2})$.

The smoothed corner models of Section 4.4 approximate Φ_H in Choi distance to second order in their defect, which lowers the exponent of the Choi profile from $\frac{1}{2}$ to $\frac{1}{3}$.

Theorem 3.36 (Choi profile of Φ_H). *For $0 < \delta \leq 1/2$,*

$$\log D_{\Phi_H}^J(\delta) = O\left(\delta^{-1/3} \log^{4/3} \frac{1}{\delta}\right).$$

Corollary 3.37 (Memory $n^{1/3}$ at the optimal rate). *For $n \geq 1$ and $0 < \epsilon \leq 1$, Φ_H has a causal service with error ϵ ,*

$$J \leq \frac{n}{2} \log 26, \quad P = C = 0, \quad Q = O\left(\left(\frac{n}{\epsilon}\right)^{1/3} \log^{4/3} \frac{2n}{\epsilon}\right).$$

At fixed error $\epsilon \leq 1/16$ and purity budget $O(\log n)$ the least memory at the optimal rate therefore lies between $c(n/\log n)^{\beta_H}$ and $O(n^{1/3} \log^{4/3} n)$.

Theorem 3.30 remains the better bound when the error is small: its memory grows like $\sqrt{\log(1/\epsilon)}$ in the error, and the corollary's like $\epsilon^{-1/3}$; at $\epsilon = n^{-a}$ the corollary is better exactly when $a \leq \frac{1}{2}$, at $a = \frac{1}{2}$ by a factor of order $(\log n)^{1/6}$. A lower bound read solely from the one-round profile cannot exceed order $n^{1/3}$ at a fixed error (Proposition 4.18), so the exponent $\frac{1}{3}$ is the natural target for a lower bound.

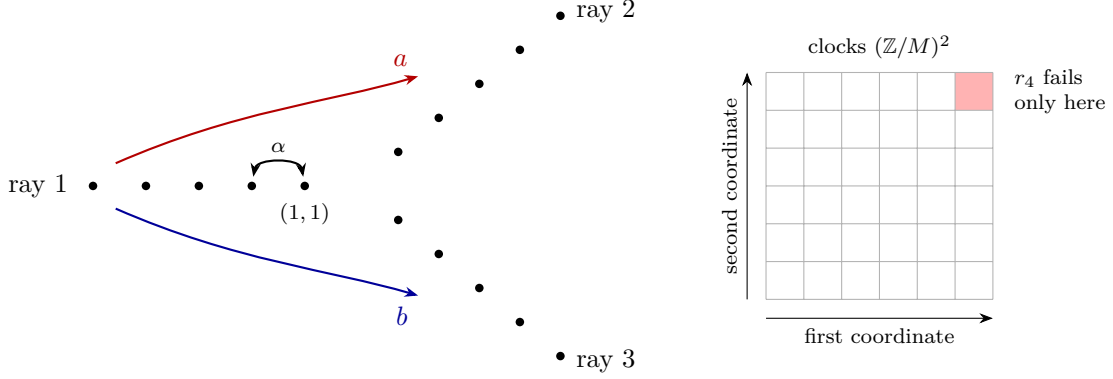


Figure 3: Left: Houghton's group H_3 acts on three rays of points. The generator a translates along rays 1 and 2: points of ray 1 move toward the centre and on into ray 2. The generator b does the same with rays 1 and 3, and α swaps the first two points of ray 1. Right: a corner model (Section 4.3) follows the quotient $\mathbb{Z}^2$ on a clock torus $(\mathbb{Z}/M)^2$ and colours the points. The relation $r_4 = aba^{-1}b^{-1}\alpha^{-1}$ fails only over one corner clock, a fraction M^{-2} of the model, whose size is $\exp(O(M))$.

The comparison with $\text{id}_4 \otimes \Delta_{26}$ in the introduction contrasts polynomial with logarithmic memory. Doubling the dimension gives constant against polynomial memory, with the same invariants matched. Let $Z_D = \text{diag}(e^{2\pi ij/D})_{j=0}^{D-1}$, and put

$$\Xi(X) = \frac{1}{32} \sum_{a=0}^{31} Z_{208}^a X Z_{208}^{-a}, \quad \Theta_6(X) = \frac{1}{6} \sum_{a=0}^5 Z_{104}^a X Z_{104}^{-a},$$

and, writing $X \in M_{208} = M_2 \otimes M_{104}$ in 2×2 blocks X_{ij} of a tag qubit,

$$\hat{\Phi}(X) = |0\rangle\langle 0| \otimes \Phi_H(X_{00}) + |1\rangle\langle 1| \otimes \Theta_6(X_{11}).$$

Corollary 3.38 (Constant against polynomial memory). *The channels Ξ and $\hat{\Phi}$ on M_{208} both have Choi rank 32, maximal complementary entropy 5 with a flat optimal complementary output, and lie in $\mathcal{K}_{208}$, so both have $\kappa = 0$ and optimal exchange rate $\frac{5}{2}$. At exchange $J \leq \frac{5}{2}n$:*

- (a) Ξ has a causal service with error 0, $Q = 5$ and $P = C = 0$;
- (b) every causal service of $\hat{\Phi}$ with error $\epsilon \leq 1/16$ has $Q = \Omega(n^{1/(2\log 130+2)})$, whatever its purity, and $Q \geq c_A(n/\log(n+1))^{\beta_H} - \frac{9}{8}$ if $P + C \leq A \log(n+1)$;
- (c) $\hat{\Phi}$ has causal services with error ϵ , $P = C = 0$ and $Q = O((n/\epsilon)^{1/3} \log^{4/3}(2n/\epsilon))$.

Above that exchange, with excess $E_x = (J - \frac{5}{2}n)_+$, every service of $\hat{\Phi}$ with error $\epsilon \leq 1/16$ satisfies the bound of Corollary 3.33(d) with $Q+1$ in place of Q ; so the separation survives an excess exchange of $O(n^\gamma)$ for every $\gamma < 1$.

4 Houghton's group and the channel

The group. Let $Y = \{(i, j) : i \in \{1, 2, 3\}, j \geq 1\}$ be three rays. Permutations act on the right and gh means g followed by h . Let a send $(1, 1) \mapsto (2, 1)$, $(1, j) \mapsto (1, j-1)$ for $j \geq 2$ and $(2, j) \mapsto (2, j+1)$, fixing ray 3; let b be defined in the same way with rays 2 and 3

exchanged; let α be the transposition of $(1, 1)$ and $(1, 2)$. These generate H_3 , the group of permutations of Y that are eventually translations on each ray. They satisfy

$$r_1 = \alpha^2, \quad r_2 = (\alpha a \alpha a^{-1})^3, \quad r_3 = [\alpha, a^2 \alpha a^{-2}], \quad r_4 = a b a^{-1} b^{-1} \alpha^{-1}, \quad r_5 = a \alpha a^{-1} b \alpha^{-1} b^{-1}, \quad (4)$$

with $[x, y] = xyx^{-1}y^{-1}$; each identity can be checked directly on the finitely many points where the word moves anything. By a theorem of Johnson (see [7, Theorem 2.8]) these relators present H_3 . The lower bound below uses only that $r_1, \dots, r_5$ hold in H_3 , the far-commutator areas of Section 4.2, and that the three-cycle $\gamma = \alpha(a\alpha a^{-1})$ on $(1, 1), (1, 2), (1, 3)$ is nontrivial. The short exact sequence $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ shows that H_3 is elementary amenable.

The channel. Introduce formal symbols for the six letters $a^{\pm 1}, b^{\pm 1}, \alpha^{\pm 1}$ and for every proper prefix of length at least two of each relator; there are 33 symbols. Each relator of length l contributes $l - 1$ multiplication triangles (x, y, z) meaning $xy = z$ along its prefixes, and three further triangles record $ss^{-1} = e$ for $s \in \{a, b, \alpha\}$; there are 35 triangles. For unitaries $\lambda(w)$ assigned to the symbols, set

$$U_\lambda = \text{diag}(1, (\lambda(w))_w) \oplus \bigoplus_{(x, y, z)} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & \lambda(y) \\ \lambda(x) & -\lambda(x)\lambda(y) \end{pmatrix},$$

an operator on $\mathbb{C}^{104} \otimes \mathcal{H}$ ($104 = 1 + 33 + 2 \cdot 35$). Each block is unitary for arbitrary unitaries $\lambda(x), \lambda(y)$, so U_λ is unitary. The channel Φ_H is defined with λ the right regular representation of H_3 and the bath in the canonical trace of the group von Neumann algebra: $\Phi_H(\rho) = (\text{id} \otimes \tau)[U_\lambda(\rho \otimes 1)U_\lambda^*]$. Because the relators hold, the lower right entry $-\lambda(x)\lambda(y)$ of each triangle block equals $-\lambda(z)$.

Proposition 4.1 (Finite approximants). *For every $M \geq 1$ there is a finite tracial factorization of bath dimension $M^2(2M)!$ whose channel is within d_J -distance $\min(1, 26/M)$ of Φ_H . Hence $\log D_{\Phi_H}^J(\delta) \leq (55/\delta) \log(54/\delta)$ for $0 < \delta \leq 1/2$.*

Proof. Let $\Omega = \{(1, j) : 1 \leq j \leq 2M\}$ and $F_M = \{\sigma a^m b^n : \sigma \in \text{Sym}(\Omega), 0 \leq m, n < M\}$. The translation part determines (m, n) and then σ , so $|F_M| = M^2(2M)!$. With $t(p) = a^m b^n$ and the cocycle $c(p, s) = t(p) s t(p + \pi(s))^{-1}$ one computes $c(p, b) = e$, $c(p, a) = a^m \varsigma_n a^{-m}$ with $\varsigma_n = b^n a b^{-n} a^{-1}$ the cycle $(1, 1) \rightarrow (1, 2) \rightarrow \dots \rightarrow (1, n+1) \rightarrow (1, 1)$, and $c(p, \alpha)$ the transposition of $(1, m+n+1)$ and $(1, m+n+2)$. All lie in $\text{Sym}(\Omega)$ when the translation coordinates stay in $[0, M)^2$. Hence $F_M \alpha = F_M$, and each of $a^{\pm 1}, b^{\pm 1}$ moves exactly a fraction $1/M$ of F_M out of F_M . Extend the partial bijections $x \mapsto xs^{-1}$ inside F_M to permutations P_s of F_M and use them as $\lambda(s)$; for a prefix symbol use the product of the corresponding P_s . No relation is imposed on the P_s , and U_P is unitary by construction.

The normalized Choi state of the finite channel is the average over $x \in F_M$ of states $\mathcal{J}_x$, each the (normalized) Gram matrix of the entry vectors at x . Call x good if every entry word, read letter by letter from x , stays inside F_M ; then the entry vectors at x are distinct basis vectors exactly when the words are distinct in H_3 , so $\mathcal{J}_x = \mathcal{J}(\Phi_H)$. The five relator paths contain 0, 6, 8, 4, 4 translation letters and the four single-letter symbols add four more; each translation step expels at most a fraction $1/M$ of the surviving points, because the preceding partial word map is injective. So at most a fraction $26/M$ of points is bad, and $\mathcal{J}(\Phi_M) = (1-t)\mathcal{J}(\Phi_H) + t\mathcal{J}_{\text{bad}}$ with $t \leq \min(1, 26/M)$. Choosing $M = \lceil 26/\delta \rceil \leq 27/\delta$ and $\log(M^2(2M)!) \leq (2M+2) \log(2M)$ gives the bound. $\square$

$\pi(g)$	labels $g = h a^u b^v$ and their permutations h of the points $(1, 1), \dots, (1, 4)$, written $1, \dots, 4$
$(0, 0)$	e : id; $\alpha = \alpha^{-1} = p_{4,4}$: (1 2); $p_{2,4}$: (1 3 2); $p_{2,5}$: (1 3); $p_{2,8}$: (1 2 3); $p_{2,9} = p_{5,3}$: (2 3); $p_{3,6}$: (1 2)(3 4); $p_{3,7}$: (3 4)
$(1, 0)$	$a = p_{2,11} = p_{3,11}$: id; $p_{2,2} = p_{3,2}$: (1 2); $p_{2,3}$: (1 3 2); $p_{2,6}$: (1 3); $p_{2,7}$: (1 2 3); $p_{2,10} = p_{5,2}$: (2 3); $p_{3,5}$: (1 2)(3 4); $p_{3,8}$: (3 4)
$(2, 0)$	$p_{3,3}$: (1 2); $p_{3,4}$: (1 2)(3 4); $p_{3,9}$: (3 4); $p_{3,10}$: id
$(0, 1)$	$b = p_{5,5}$: id; $p_{4,3}$: (1 2); $p_{5,4}$: (2 3)
$(1, 1)$	$p_{4,2}$: id
$(-1, 0), (0, -1)$	a^{-1} : id; b^{-1} : id

Table 2: The 26 elements of H_3 carried by the entries of U_λ , grouped by translation. Here $p_{i,l}$ is the prefix of length l of r_i , a permutation is written in cycle notation, and $g = h a^u b^v$ means h followed by $a^u b^v$.

Proposition 4.2 (Choi rank and entropy). *The entries of U_λ carry exactly 26 distinct elements of H_3 . Grouping entries by element, $U_\lambda = \sum_{g \in \mathcal{G}} K_g \otimes \lambda(g)$ with $|\mathcal{G}| = 26$ and $K_g \in M_{104}$ of pairwise disjoint supports, so*

$$\Phi_H(\rho) = \sum_{g \in \mathcal{G}} K_g \rho K_g^*, \quad \mathcal{J}(\Phi_H) = \frac{1}{d} \sum_{g \in \mathcal{G}} |K_g\rangle\langle K_g|,$$

the Choi rank of Φ_H is 26, and $H(\Phi_H) = \log 26$.

Proof. The entry labels are the identity, the 33 symbols, and the lower-right entries of the triangles, which equal $-\lambda(x)\lambda(y) = -\lambda(z)$ and whose elements are therefore among the symbols or the identity. Evaluating the 34 labels as permutations of the rays shows exactly 26 distinct elements, listed in Table 2. Writing $p_{i,l}$ for the prefix of length l of r_i , the eight coincidences are $p_{2,2} = p_{3,2}$ (the same word $\alpha\alpha$) and seven consequences of the relators: $p_{2,11} = p_{3,11} = a$, $p_{5,5} = b$, $\alpha^{-1} = p_{4,4} = \alpha$, $p_{2,9} = p_{5,3} = \alpha\alpha\alpha^{-1}$ and $p_{2,10} = p_{5,2} = \alpha\alpha$. Every label is $h a^u b^v$ with $(u, v) = \pi(g)$ and h a permutation of the points $(1, 1), \dots, (1, 4)$, and labels with the same translation have different h . Disjoint supports make the K_g orthogonal, hence linearly independent, and $\tau(\lambda(g)\lambda(h)^*) = \delta_{gh}$ gives the displayed formulas. Every g has a singleton diagonal position i_g whose column contains only the entry of K_g . For $\rho = \frac{1}{26} \sum_g |i_g\rangle\langle i_g|$ the complementary output

$$\Phi_H^c(\rho)_{gh} = \text{Tr}(K_g \rho K_h^*) = \delta_{gh}/26$$

is maximally mixed, and $\log 26$ is the largest entropy of a complement of dimension 26. $\square$

4.1 A symmetry of H_3 and exact translations in a box

Let $\pi : H_3 \rightarrow \mathbb{Z}^2$ be the quotient map with $\pi(a) = (1, 0)$, $\pi(b) = (0, 1)$, $\pi(\alpha) = 0$. An element g with $\pi(g) = (u, v)$ is eventually a translation by $(-u - v, u, v)$ on the three rays.

Let θ be the automorphism of H_3 given by conjugation with the relabelling $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ of the rays. A direct check on the rays gives

$$\begin{array}{c|ccc} & a & b & \alpha \\ \hline \theta & a^{-1}b & a^{-1} & a^{-2}\alpha a^2 \\ \theta^2 & b^{-1} & b^{-1}a & b^{-2}\alpha b^2 \end{array} \quad (5)$$

and $\theta^3 = \text{id}$. Since θ permutes the translation vector $(-u - v, u, v)$ cyclically, and its entries sum to zero,

$$\sum_{j=0}^2 \pi(\theta^j(g)) = 0 \quad (g \in H_3). \quad (6)$$

Every entry label of U_λ has a word of length at most 12, each θ^j multiplies word length by at most 5 by (5), and $\|\pi(\theta^j(g))\|_\infty \leq 2$ for every label: $\pi(\theta(g)) = (-u - v, u)$ and $\pi(\theta^2(g)) = (v, -u - v)$ when $\pi(g) = (u, v)$, and Table 2 lists the possible $\pi(g)$. The θ -twisted unitaries

$$U_j^\infty = \sum_{g \in \mathcal{G}} K_g \otimes \lambda(\theta^j(g)), \quad j = 0, 1, 2,$$

arise from the same block construction with labels $\lambda(\theta^j(w))$; their lower-right entries are $-\lambda(\theta^j(x))\lambda(\theta^j(y)) = -\lambda(\theta^j(z))$, and their tracial channel is Φ_H because θ^j is injective.

Lemma 4.3 (Exact translations in the box). *Let F_M be as in the proof of Proposition 4.1. If $f \in F_M$, $s \in \{a^{\pm 1}, b^{\pm 1}, \alpha^{\pm 1}\}$ and both $\pi(f)$ and $\pi(fs)$ lie in $[0, M]^2$, then $fs \in F_M$. Consequently a word of length L is evaluated exactly, letter by letter inside F_M , from every f whose quotient position has distance at least L from the boundary of the box.*

Proof. Write $f = \sigma t(p)$ with $t(p) = a^u b^v$. For a positive letter, $fs = \sigma c(p, s) t(p + \pi(s))$, and the cocycles $c(p, a) = a^u \varsigma_v a^{-u}$, $c(p, b) = e$ and $c(p, \alpha)$ are permutations of $(1, 1), \dots, (1, u + v + 2)$, which lie in Ω when both positions are in the box. For an inverse letter, $t(p)s^{-1} = c(p - \pi(s), s)^{-1} t(p - \pi(s))$. The property was also checked by machine for $M = 2, 3$. $\square$

The support profile asks only for constant diamond accuracy, and a coarser register suffices for it.

Proposition 4.4 (Coloring approximants). *Let $0 < v \leq 2^{-17}$, $M = \lceil 104/v \rceil$ and $q = 2^{20}$. There is a finite tracial factorization of bath dimension at most $M^2 q^{2M}$ whose channel is within half-diamond distance $1/8$ of Φ_H and has support leakage at most v . Hence $\log \mathfrak{D}_{\Phi_H}(1/8, v) \leq 40M + 2 \log M$.*

Proof. Split the $2M$ points of Ω into q classes of sizes $k_1, \dots, k_q$ differing by at most one, let G be the product of the symmetric groups of the classes, and let the bath basis be the set $G \backslash F_M$ of cosets Gx , $x \in F_M$; it has $M^2(2M)! / \prod_j k_j! \leq M^2 q^{2M}$ elements. Left multiplication by G commutes with right multiplication, so the partial bijections $x \mapsto xs^{-1}$ of Proposition 4.1 descend to partial bijections of the cosets. Extend them to permutations, use them and their products for the prefix symbols in place of λ , and keep the bath flat. Every entry of U_λ carries a word of length at most 12, so the entries a, b of the gadget satisfy $R_a R_b^* = R_w$ for a word w of length at most 24. Call a coset good if its clock position has distance at least 24 from the boundary of the box; the bad fraction is $\beta \leq 96/M \leq v$. By Lemma 4.3, every word of length at most 24 acts on good cosets through right multiplication by the element of H_3 it represents.

Accuracy. If w represents 1, then $|\tau(R_w) - 1| \leq \beta$. If w represents an element $g \neq 1$ and moves the clock position, it fixes no good coset. Otherwise g acts on a good coset $G\sigma t(p)$ as $\sigma \mapsto \sigma c$ for a nonidentity permutation c of Ω , and fixes the coset only if $\sigma c \sigma^{-1} \in G$, which requires two points of a nontrivial cycle of c to lie in the same class after applying σ . For a uniformly distributed coset at a fixed clock this has probability at most $\sum_j k_j(k_j - 1)/(2M(2M - 1)) \leq 2/q$. Hence $|\tau(R_w)| \leq \beta + 2/q$. The normalized Choi state is $\frac{1}{d} \sum_{a,b} c_a \bar{c}_b \tau(R_a R_b^*) |r_a s_a\rangle \langle r_b s_b|$ over the 174 nonzero entries $c_a |r_a\rangle \langle s_a|$ of the gadget, whose coefficients satisfy $\sum_a |c_a|^2 = d = 104$, while that of Φ_H has $\delta_{g_a g_b}$ in place of the traces. The

trace-norm difference is therefore at most $\frac{1}{d}(\sum_a |c_a|)^2(\beta + 2/q) \leq 174(\beta + 2/q)$, and by (1) the half-diamond distance is at most $9048(\beta + 2/q) \leq 9048(2^{-17} + 2^{-19}) < 1/8$.

Leakage. The flat bath is the uniform mixture of the basis cosets. For a good coset x , the Kraus operators with bath input x are $\sum_g [y = xg^{-1}] K_g$ for the matrices K_g of Proposition 4.2, because entries carrying the same element of H_3 act identically on x ; so the Choi state of that input lies in the support of $\mathcal{J}(\Phi_H)$. The leakage is at most $\beta \leq v$. $\square$

At fixed q this construction does not approach Φ_H in Choi distance, so the Choi profile keeps the bound of Proposition 4.1.

4.2 What we use from the group paper

The lower bounds rest on four results about approximate representations of Johnson's presentation, proved in the companion paper [26]; we state them here without proof. Let $P = \langle a, b, \alpha \mid r_1, \dots, r_5 \rangle$, put $\sigma_j = a^j \alpha a^{-j}$, and let $A^*(M)$ be the largest area in P of a commutator $[\sigma_p^{\pm 1}, \sigma_q^{\pm 1}]$ with $0 \leq p, q \leq M$ and $|p - q| \geq 2$. These commutators are trivial in H_3 because the transpositions have disjoint supports.

Theorem 4.5 (Reweighted far-commutator area). *$A^*(M) \leq 30 M^{\log 130}$ for every $M \geq 2$.*

Proposition 4.6 (Quadratic lower bound). *For odd $k \geq 9$, $\text{Area}(C_k) \geq (k - 7)(k - 5)/8$, where $C_k = [\alpha, a^k \alpha a^{-k}]$. Hence $A^*(R) \geq R^2/16$ for $R \geq 23$.*

The area bound is proved by a doubling self-similarity of H_3 , with finite relations certified by machine; [26] also gives the certified bound $A^*(M) \leq M^{11}$. Write $k(u, v) = \|uv - vu\|_2$ and $G(t) = h_2(\min\{\frac{2}{3}t, \frac{2}{3}\}) + \min\{\frac{2}{3}t, \frac{2}{3}\}$.

Lemma 4.7 (Projection products). *Let $P_1, \dots, P_m$ be orthogonal projections on a Hilbert space and $\|x\| = 1$. Then $1 - \text{Re}\langle x, P_m \cdots P_1 x \rangle \leq 2 \sum_j \|(I - P_j)x\|^2$.*

Lemma 4.8 (Exact S_3 rounding). *If $A, B \in U(D)$ satisfy $\|A^2 - I\|_2, \|B^2 - I\|_2, \|(AB)^3 - I\|_2 \leq e$, there is a unitary representation u of $S_3 = \langle s, t \mid s^2, t^2, (st)^3 \rangle$ on $\mathbb{C}^D$ with $\|u(s) - A\|_2 \leq e$, $\|u(t) - B\|_2 \leq 8e$ and $\|u(st) - AB\|_2 \leq 9e$, and each normal form $1, s, t, st, ts, sts$ evaluated in u is within $10e$ of the same word in A, B .*

Theorem 4.9 (Phase-robust sector bound). *Let $c_i, x_i \in U(D)$, $1 \leq i \leq N$, satisfy $c_i^3 = x_i^2 = I$ and $x_i c_i x_i = c_i^{-1}$, with no condition on the commutators of x_i and x_j . Put $\Delta_i = \|c_i - I\|_2$ and $f(g) = c_1^{g_1} \cdots c_N^{g_N}$ for $g \in \mathbb{Z}_3^N$, let $g^{(i)}$ be g with its i -th coordinate negated, and, for uniform g ,*

$$\hat{a}_i^2 = \mathbb{E}_g \|f(g + e_i) - f(g)c_i\|_2^2, \quad \hat{b}_i^2 = \mathbb{E}_g \|x_i f(g^{(i)}) - f(g)x_i\|_2^2.$$

Then $\hat{a}_i^2 \leq \frac{4}{3} \sum_{j>i} k(c_i, c_j)^2$, $\hat{b}_i^2 \leq \frac{4}{3} \sum_{j \neq i} k(x_i, c_j)^2$, and

$$\log D \geq \sum_{i=1}^N \left[\frac{(\Delta_i^2 - \hat{a}_i^2)_+}{3} - \frac{\hat{b}_i^2}{2 \ln 2} - G(\hat{a}_i^2) \right].$$

In particular $\Delta_i \geq 49/100$ and $\hat{a}_i, \hat{b}_i \leq 1/100$ for all i give $\log D \geq N/16$.

Call $\phi : \{a, b, \alpha\} \rightarrow U(D)$ an e -representation if $\|\phi(r_i) - I\|_2 \leq e$ for $i = 1, \dots, 5$, where ϕ is extended to the free group. If w is null in P with area A , then $\|\phi(w) - I\|_2 \leq Ae$.

Proposition 4.10. *Let ϕ be an e -representation with $\|\phi(\gamma) - I\|_2 \geq 1/2$, and let $N \geq 1$ satisfy $750\sqrt{N} A_N e \leq 1$, where $A_N = \max\{1, A^*(3N)\}$. Then $\log D \geq N/16$.*

4.3 Corner models

The approximants above fail on a layer of about M clock positions out of M^2 . A torus of clock positions with a single bad corner does better. Fix integers $M \geq 2$ and $s \geq 2$, put $\ell = 2M + 2$ and $L = 3\ell$, and let $\Omega_{M,s} = (\mathbb{Z}/M\mathbb{Z})^2 \times [s]^L$. For $2 \leq r \leq L$ let $c_r = (1 \ r \ r-1 \ \dots \ 2) \in \text{Sym}(L)$ be the reverse cycle and $\tau = (1 \ 2)$; products of permutations are read from left to right, and a permutation π of the coordinates acts on colorings by $(x \cdot \pi)(t) = x(t\pi^{-1})$. For $0 \leq i, j < M$ put $A_{i,j} = c_{\ell-i-j}$ and, for $j < M-1$, $B_{i,j} = c_{2\ell-j}$. On the top row put $B_{i,M-1} = D_i$, where

$$D_0 = c_{2\ell-M+1}, \quad D_{i+1} = c_{\ell-i-(M-1)}^{-1} \tau D_i c_{\ell-i} \quad (0 \leq i < M-1). \quad (7)$$

Define permutations of $\Omega_{M,s}$, with clock coordinates modulo M , by

$$(i, j, x) \cdot a = (i+1, j, x \cdot A_{i,j}), \quad (i, j, x) \cdot b = (i, j+1, x \cdot B_{i,j}), \quad (i, j, x) \cdot \alpha = (i, j, x \cdot \tau).$$

Proposition 4.11 (Corner models [26]). *In these permutations r_1, r_2, r_3, r_5 hold exactly and r_4 fails only at the clock $(M-1, M-1)$, where it acts as the product $R_M = (1 \ M+4) \prod_{t=5}^{M+3} (t \ t+M)$ of M disjoint transpositions of the colours; so every relator moves at most a fraction M^{-2} of $\Omega_{M,s}$, and γ moves a fraction $1 - s^{-2}$. At every clock the colour permutations of $\alpha\alpha\alpha^{-1}$, $\alpha^2\alpha\alpha^{-2}$ and $b\alpha b^{-1}$ are $(2 \ 3)$, $(3 \ 4)$ and $(2 \ 3)$. A word of length at most t that represents a nonidentity element of H_3 fixes at most a fraction $s^{-1} + 4t/M$ of $\Omega_{M,s}$ when $M > 2t$.*

The proof is in [26]; it also records the identities $c_r \tau c_r^{-1} = (2 \ 3)$ and $c_r (2 \ 3) c_r^{-1} = (3 \ 4)$ for $r \geq 4$ used below.

Proposition 4.12 (Corner channels). *For $s = 2^{20}$ and $M \geq 2^{24}$ the gadget built from these permutations is a finite tracial factorization of bath dimension $M^2 s^{6M+6}$ within half-diamond distance $1/16$ of Φ_H , with support leakage at most M^{-2} . Hence, for every $v > 0$,*

$$\log \mathfrak{D}_{\Phi_H}(1/8, v) \leq 120M + 120 + 2 \log M, \quad M = \max\{2^{24}, \lceil v^{-1/2} \rceil\}.$$

Proof. Put $P(g)|\omega\rangle = |\omega g^{-1}\rangle$ and use these permutation unitaries in the gadget of Section 4; every block stays unitary, so this is a finite tracial factorization of bath dimension $M^2 s^{6M+6}$. Two literal entry words that carry the same element of H_3 are related by one relator: the full relators equal e , and the other coincidences are $p_{2,11} = p_{3,11} = a$, $p_{5,5} = b$, $\alpha^{-1} = p_{4,4} = \alpha$, $p_{2,9} = a\alpha a^{-1}$ and $p_{2,10} = a\alpha$, where $p_{i,l}$ is the prefix of length l of r_i (for instance $p_{2,11} a^{-1} = r_2$ and $p_{4,4} \alpha^{-1} = r_4$ freely). Since r_1, r_2, r_3, r_5 are exact, only the identifications through r_4 can fail, on at most a fraction M^{-2} of the states; so $1 - \tau(P(w_\nu)^* P(w_\mu)) \leq M^{-2}$ whenever two entry words w_μ, w_ν carry the same element. Write the entries of the gadget as $c_\mu P(w_\mu)$, put $a_\mu = |c_\mu|^2/104$ and $p_g = \sum_{w_\mu=g} a_\mu$. The support projection of $\mathcal{J}(\Phi_H)$ is $\sum_g v_g v_g^*/p_g$ with $v_g = 104^{-1/2} \sum_{w_\mu=g} c_\mu |\mu\rangle$, and a direct computation gives the leakage of the resulting channel Ψ

$$\ell_{\Phi_H}(\Psi) = \sum_g \frac{1}{p_g} \sum_{w_\mu=w_\nu=g} a_\mu a_\nu [1 - \tau(P(w_\nu)^* P(w_\mu))] \leq M^{-2} \sum_g p_g = M^{-2}.$$

For entries carrying different elements the comparison word has length at most 24, so its normalized trace is at most $\eta = s^{-1} + 96/M$; for equal elements the error is at most $M^{-2} \leq \eta$. As in the proof of Proposition 4.4 the half-diamond distance is at most 9048η , which for $s = 2^{20}$ and $M \geq 2^{24}$ is at most $9048 \cdot 112 \cdot 2^{-24} < 1/16$. The bath dimension has $\log(M^2 s^{6M+6}) = 120M + 120 + 2 \log M$, and for $v > 0$ the choice $M = \max\{2^{24}, \lceil v^{-1/2} \rceil\}$ gives leakage at most v . $\square$

Proposition 4.13 (Sharper corner models). *Let $M \geq 4$ and $s \geq 2$, and drop the last $2M + 2$ colour coordinates, which no generator moves. The gadget built from the resulting permutations of $(\mathbb{Z}/M\mathbb{Z})^2 \times [s]^{4M+4}$ is a finite tracial factorization $\Psi_{M,s}$ with*

$$\ell_{\Phi_H}(\Psi_{M,s}) = \varkappa_0 \frac{1 - s^{-M}}{M^2}, \quad \varkappa_0 = \frac{122}{2925}, \quad d_{\diamond}(\Psi_{M,s}, \Phi_H) \leq \frac{7}{2s} + \frac{2}{M^2}.$$

Consequently, for $0 < \delta \leq 1/4$, $\log D_{\Phi_H}^J(\delta) \leq 2 \log M + (4M + 4) \log \lceil 7/\delta \rceil$ with $M = \lceil 2/\sqrt{\delta} \rceil$, which is $O(\delta^{-1/2} \log(1/\delta))$; and $\log \mathfrak{D}_{\Phi_H}(1/8, v) \leq 24M + 24 + 2 \log M$ with $M = \max\{6, \lceil (\varkappa_0/v)^{1/2} \rceil\}$.

Proof. Normal forms. By Proposition 4.11, at every clock α , $a\alpha a^{-1}$, $a^2\alpha a^{-2}$ and $b\alpha b^{-1}$ act on the colours as $s_0 = (1\ 2)$, $s_1 = (2\ 3)$, $s_2 = (3\ 4)$ and s_1 without moving the clock, and α^{-1} acts as α . In every literal entry word in the letters $a^{\pm 1}, \alpha^{\pm 1}$, each $\alpha^{\pm 1}$ occurs at running a -exponent 0, 1 or 2, so moving the powers of a to the right writes the word, as a permutation of the bath, as $h a^u$ with $h \in \langle s_0, s_1, s_2 \rangle \cong S_4$. The other literal words are $b^{\pm 1}$, bb^{-1} , ab , $a\alpha a^{-1}b = s_1b$, $a\alpha a^{-1}b\alpha^{-1} = b$ and $r_5 = 1$, and the three words

$$aba^{-1} = r_4 \alpha b, \quad aba^{-1}b^{-1} = r_4 \alpha, \quad aba^{-1}b^{-1}\alpha^{-1} = r_4$$

(identities in the free group), where r_4 acts as R_M at the clock $(M-1, M-1)$ and trivially elsewhere. The permutations s_0, s_1, s_2 are Coxeter generators of S_4 , as are the transpositions α , $a\alpha a^{-1}$, $a^2\alpha a^{-2}$ of $(1, 1), \dots, (1, 4)$ in H_3 . So, apart from these three words, two literal words carrying the same element of H_3 act identically on the bath, and two carrying different elements with the same translation act as ht and $h't$ with $h \neq h'$ (Table 2).

Two exceptional clocks. An entry acts by $P(w)|\omega\rangle = |\omega w^{-1}\rangle$. For the three words above, ωw^{-1} differs from the value of the normal form only when the path of w^{-1} meets r_4^{-1} at the clock $(M-1, M-1)$: for the last two words this happens when ω is at $(M-1, M-1)$, and for aba^{-1} , whose inverse first applies b^{-1} , when ω is at $(M-1, 0)$. At every other clock c each entry acts as its normal form. Let τ_c be the uniform state on the colourings at c . For entries with normal forms ht and $h't'$, $\tau_c(P(w')^*P(w))$ is 0 if $t \neq t'$, because the two words end at different clocks: the translations in Table 2 differ by vectors with coordinates at most 3 and 2 in absolute value, and $M \geq 4$. If $t = t'$, it is the fraction of colourings fixed by a permutation conjugate to $h^{-1}h'$, which is $s^{-\iota(h^{-1}h')}$, with ι the number of moved points minus the number of nontrivial cycles. So at every clock outside the two exceptional ones the channel is $\Phi_s(X) = \sum_{g,h} G_{gh} K_g X K_h^*$, where $G_{gg} = 1$, $G_{gh} = s^{-\iota} \leq 1/s$ for distinct labels with the same translation, and $G_{gh} = 0$ otherwise. Each clock gives a channel, because the gadget is unitary for any assignment, so $\Psi_{M,s} = (1 - 2/M^2)\Phi_s + (2/M^2)\Xi$ for a channel Ξ .

Distance. The map $V_0 = \sum_g K_g \otimes |g\rangle$ is an isometry, and for a Hermitian matrix H , $\sum_{g,h} H_{gh} K_g X K_h^* = \text{Tr}_2[(I \otimes H^T) V_0 X V_0^*]$. With a reference system, the trace norm of such a weighted partial trace is at most $\|H\|_{\text{op}}$ times that of its argument, so

$$\left\| X \mapsto \sum_{g,h} H_{gh} K_g X K_h^* \right\|_{\diamond} \leq \|H\|_{\text{op}}. \quad (8)$$

The matrix $G - I$ is symmetric, with nonnegative entries at most $1/s$ and at most seven of them in each row, since no translation class has more than eight labels. So $\|G - I\|_{\text{op}} \leq 7/s$, $d_{\diamond}(\Phi_s, \Phi_H) \leq 7/(2s)$, and $d_{\diamond}(\Psi_{M,s}, \Phi_H) \leq 7/(2s) + 2/M^2$.

Leakage. In the leakage formula of the proof of Proposition 4.12, only the classes of e and α contain literal words that act differently. In squared-coefficient weight the class of e has $45/2$, of which $1/2$ is the entry $aba^{-1}b^{-1}\alpha^{-1}$ (the last triangle of r_4); the class of α has $25/2$, of which 2 belongs to $aba^{-1}b^{-1}$. The word aba^{-1} is alone in its class and causes no leakage.

For each pair of one of these two words and a correctly acting word of its class, $1 - \tau(P^*P')$ is the fraction $(1 - s^{-M})/M^2$ of states moved by R_M , and all other pairs contribute zero. Hence

$$\ell_{\Phi_H}(\Psi_{M,s}) = \frac{1}{104} \left(\frac{2 \cdot \frac{1}{2} \cdot 22}{45/2} + \frac{2 \cdot 2 \cdot \frac{21}{2}}{25/2} \right) \frac{1 - s^{-M}}{M^2} = \frac{122}{2925} \cdot \frac{1 - s^{-M}}{M^2}.$$

Profiles. For $0 < \delta \leq 1/4$ take $s = \lceil 7/\delta \rceil$ and $M = \lceil 2/\sqrt{\delta} \rceil \geq 4$: the half-diamond distance is at most $\delta/2 + \delta/2$, so by (1) the Choi distance is at most δ , with bath dimension $M^2 s^{4M+4}$. For the support profile take $s = 64$ and $M \geq 6$: then $7/128 + 2/M^2 < 1/8$ and the leakage is at most $\varkappa_0/M^2 \leq v$. $\square$

Corollary 4.14 (A service from the corner models). *Let $0 < \epsilon \leq 1/2$ and $\log(256n/\epsilon) < n$. There is a causal service of Φ_H with error ϵ , exchange $J \leq nH(\Phi_H)/2$, $C = 0$, $P \leq \lceil \log(n+1) \rceil + 1$ and memory $Q = O(\sqrt{n/\epsilon} \log(n/\epsilon))$.*

Proof. Apply Theorem 3.19 with exchange slack 0, so that its buffer is $C_d(\sqrt{nx} + \log(n+1))$ with $x = \log(256n/\epsilon)$, and with $m = D_{\Phi_H}^J(\epsilon/(8dn))$, $d = 104$. By Proposition 4.13, $\log m = O(\sqrt{n/\epsilon} \log(n/\epsilon))$. $\square$

At fixed error this is the order of Theorem 3.30, reached through the general compiler and the static profile alone.

4.4 Smoothed corner models

In a permutation model a relator that fails costs the fraction of points it moves. In a unitary model it costs a sum of squared angles, so a single defect can be spread thin. Doing this for the corner models gives unitary approximants of Φ_H whose leakage is of order M^{-3} , not M^{-2} , at the same size. Throughout, products are read from left to right, the left factor acting first, as in Section 4.3, and $e_1 = (1, 0)$, $e_2 = (0, 1)$.

One-particle models. Use the reduced colour coordinates $1, \dots, L$ with $L = 4M + 4$, and read each link permutation A_c , B_c and τ of Section 4.3 as a permutation matrix on $\mathbb{C}^L$. A one-particle model replaces some of these link matrices by unitaries. For a word w and a clock c the holonomy $W_c(w) \in U(L)$ is the product of the link matrices along the path of w from c , with inverse matrices for inverse letters. For a two-element set $S = \{x, y\}$ let Q_S be the projector onto $(e_x - e_y)/\sqrt{2}$; then $\exp(i\pi Q_S)$ is the transposition $(x\ y)$, and $\exp(i\vartheta Q_S)$ is a phase rotation in the plane of S .

Fermionic quantization. For $W \in U(L)$ let $\Lambda(W)$ act on the exterior algebra $\Lambda(\mathbb{C}^L)$, of dimension 2^L , by $v_1 \wedge \dots \wedge v_m \mapsto Wv_1 \wedge \dots \wedge Wv_m$. It is multiplicative on all of $U(L)$, and $\text{Tr } \Lambda(W) = \det(I + W)$, since in degree m its eigenvalues are the products of m eigenvalues of W . For $k \geq 1$ put $\Lambda_k = \Lambda^{\otimes k}$, so that $2^{-kL} \text{Tr } \Lambda_k(W) = \det((I + W)/2)^k$. The quantized model has bath $\mathcal{H} = \mathbb{C}^{M^2} \otimes \Lambda(\mathbb{C}^L)^{\otimes k}$ with its normalized trace τ . A letter moves the clock c to $c + \pi(s)$ and applies Λ_k of its link matrix at c , and $\rho(s)$ is the adjoint of this operator, as for $P(g)$ in Proposition 4.11, so ρ is a representation of the free group on a, b, α . A word whose translation is nonzero modulo M has $\tau(\rho(w)) = 0$, and a word that closes has

$$\|\rho(w) - I\|_2^2 = \mathbb{E}_c \left[2 - 2 \text{Re} \det((I + W_c(w))/2)^k \right],$$

with $\mathbb{E}_c$ the average over clocks. For a three-cycle σ of coordinates $\det((I + \sigma)/2) = \frac{1}{4}$.

Lemma 4.15 (Rotating the links). *In a corner model, replace every link A_c by $U_c A_c$ and every link B_c by $V_c B_c$, where U_c commutes with $(2\ 3)$ and $(3\ 4)$, and V_c commutes with $(2\ 3)$. Then at every clock the holonomies of α , $\alpha\alpha\alpha^{-1}$, $\alpha^2\alpha\alpha^{-2}$ and $b\alpha b^{-1}$ are $(1\ 2)$, $(2\ 3)$, $(3\ 4)$ and $(2\ 3)$, so r_1, r_2, r_3, r_5 have identity holonomy everywhere and γ acts as a three-cycle.*

A plaquette q with base v_0 has the vertices $v_0, v_1 = v_0 + e_1, v_2 = v_0 + e_2, v_0 + e_1 + e_2$; its bottom and top edges are the a -links at v_0 and v_2 , and its left and right edges the b -links at v_0 and v_1 . Its r_4 holonomy is $W_q = A_{v_0} B_{v_1} A_{v_2}^{-1} B_{v_0}^{-1} \tau$. A *region* consists of a set R of plaquettes, a source $q_0 \in R$, a sign $s_R = \pm 1$, and a two-element label $\text{lab}(v) \subset \{1, \dots, L\}$ at every vertex of every plaquette of R . An edge is *interior* if it lies on two plaquettes of R .

Lemma 4.16 (Spreading a defect). *Take a permutation model satisfying the local identities $A_c \tau A_c^{-1} = (2\ 3)$, $A_c (2\ 3) A_c^{-1} = (3\ 4)$ and $B_c \tau B_c^{-1} = (2\ 3)$ of Lemma 4.15 at every clock. Suppose W_q is the identity except at finitely many sources, where it is the product of the transpositions given by the base labels of the regions with that source. Let finitely many regions satisfy:*

- (R1) *every edge of a non-source plaquette, from v to w , carries $\text{lab}(v)$ onto $\text{lab}(w)$ under its link permutation;*
- (R2) *every base label avoids $\{1, 2\}$, and the label at the initial vertex of every interior edge avoids $\{2, 3, 4\}$;*
- (R3) *each region is connected through its interior edges, and the only interior edge of its source is the bottom edge;*
- (R4) *labels of different regions are disjoint wherever both are defined, and disjoint from any other modification of the link.*

Let $F_{R,q}$ be real numbers with $\sum_{q \in R} F_{R,q} = s_R \pi$ for each region. Then there are potentials φ_R on the interior edges of R such that the model with $U_c = \prod_R \exp(i\varphi_R(a\text{-edge at } c)) Q_{\text{lab}_R(c)}$ and $V_c = \prod_R \exp(i\varphi_R(b\text{-edge at } c)) Q_{\text{lab}_R(c)}$ still has r_1, r_2, r_3, r_5 exact, and its r_4 holonomy at every plaquette q is $\prod_{R \ni q} \exp(iF_{R,q} Q_{\text{lab}_R(v_0)})$, with v_0 the base of q .

Lemmas 4.15 and 4.16 are proved in [26, Section 6.2], where the same construction gives unitary models of Johnson's presentation.

Proposition 4.17 (Smoothed corner models). *For every $M \geq 48$ there is a finite tracial factorization with bath dimension $M^2 2^{8M+8}$ whose channel is within half-diamond distance 0.106 of Φ_H and has support leakage at most $0.3 M^{-3}$. Its bath is flat, so its entropy production is zero. Consequently, for every $v > 0$, with $M = \max\{48, \lceil (0.3/v)^{1/3} \rceil\}$,*

$$\log \mathfrak{D}_{\Phi_H}(1/8, v) \leq 8M + 8 + 2 \log M \quad \text{and} \quad \log D_{\Phi_H}^{\text{ent}}(1/8, v) \leq 8M + 8 + 2 \log M.$$

Proof. The unitary model. We recall the construction of [26, Section 6.2]. Let $c^* = (M-1, M-1)$, where the r_4 holonomy is R_M (Proposition 4.11). Replace A_{c^*} by $\sigma_1 A_{c^*}$ and $B_{(0, M-2)}$ by $\sigma_2 B_{(0, M-2)}$, with $\sigma_1 = (1\ M+4)$ and $\sigma_2 = (5\ M+5)$. Then r_1, r_2, r_3, r_5 stay exact, the r_4 holonomy at c^* becomes $\prod_{t=5}^{M+3} (t\ t+M)$, at $(0, M-2)$ it becomes σ_2 , and elsewhere it is the identity.

The regions. For $5 \leq t \leq M+3$ let R_t have source c^* and sign $(-1)^{t-5}$, and consist of c^* , the plaquettes $(M-1, j)$ with $1 \leq j \leq M-2$, and the plaquettes (i, j) with $0 \leq i \leq M-3$, $0 \leq j \leq M-2$ and $i+j \leq t+M-8$; label the vertex (i, j) by $\{t+M-1-i-j, t+2M-1-j\}$ for $0 \leq i \leq M-2$ and $(M-1, j)$ by $\{t+M-1-j, t+2M-1-j\}$. Let T have source $(0, M-2)$ and sign $(-1)^{M-1}$, and consist of $(0, M-2)$ and the plaquettes (i, j) with $i, j \geq 0$ and $i+j \leq M-3$, with labels $\{M+3-i-j, 2M+3-j\}$. The source labels are the defect pairs $\{t, t+M\}$ at c^* and $\{5, M+5\}$ at $(0, M-2)$.

These regions satisfy (R1)–(R4) [26, Section 6.2]. Moreover $R_t \subset R_{t+1}$, and every region has at least $m_M = (M-1)(M-2)/2 + 1$ plaquettes.

The defect. Take $F_{R,q} = s_R \pi / |R|$ on each region and $k = 2$. By Lemma 4.16 the r_4 holonomy at q is a product of phase rotations by angles $F_{R,q}$ in disjoint planes, so $\det((I +$

$W'_q/2) = \prod_R (1 + e^{iF_{R,q}})/2 = e^{i\Theta_q} C_q$ with $\Theta_q = \frac{1}{2} \sum_R F_{R,q}$ and $C_q = \prod_R \cos(F_{R,q}/2) \in [0, 1]$. Using $1 - C^k \leq k(1 - C)$, $1 - \prod_R C_R \leq \sum_R (1 - C_R)$ and $1 - \cos x \leq x^2/2$,

$$2 - 2\operatorname{Re}(e^{ik\Theta_q} C_q^k) = 2(1 - C_q^k) + 2C_q^k(1 - \cos k\Theta_q) \leq \frac{k}{4} \sum_R F_{R,q}^2 + k^2 \Theta_q^2.$$

Summed over the plaquettes, $\sum_q \sum_R F_{R,q}^2 = \sum_R \pi^2/|R| \leq M\pi^2/m_M$. The bulk regions containing q are those with t at least some t_q ; their signs alternate and their magnitudes $\pi/|R_t|$ decrease, so they sum to at most π/m_M in absolute value, and T adds at most π/m_M . Hence $|\Theta_q| \leq \pi/m_M$, and averaging over the M^2 clocks,

$$M^3 e_4^2 := M^3 \|\rho(r_4) - I\|_2^2 \leq \frac{\pi^2 M^2}{2m_M} + \frac{4\pi^2 M^3}{m_M^2} \leq \frac{\pi^2 M^2}{(M-1)(M-2)} + \frac{16\pi^2 M^3}{(M-1)^2(M-2)^2},$$

which decreases in M and is below 14.26 at $M = 48$. So $e_4^2 \leq 14.26 M^{-3}$ for $M \geq 48$.

The channel. Build the gadget from ρ with $k = 2$; its bath has dimension $M^2 2^{2L} = M^2 2^{8M+8}$. By Lemma 4.15 the normal forms in the proof of Proposition 4.13 hold for ρ : every literal entry word u carrying a label g has $\rho(u) = \rho(\hat{g})$, where $\hat{g} = ht$ is the normal form of g , except aba^{-1} , $aba^{-1}b^{-1}$ and $aba^{-1}b^{-1}\alpha^{-1}$, for which $\|\rho(u) - \rho(\hat{g})\|_2 = e_4$. For labels g, g' with normal forms $\hat{g} = ht$ and $\hat{g}' = h't'$, $G_{gg'} = \tau(\rho(\hat{g}')^* \rho(\hat{g}))$ is 0 if $t \neq t'$; otherwise the holonomy is conjugate to $h'^{-1}h$, a permutation of the coordinates $1, \dots, 4$, so $G_{gg'}$ is $\det((I_4 + h'^{-1}h)/2)^k$ up to conjugation: 1 if $h = h'$, 4^{-k} for a three-cycle, and 0 for every other element of S_4 , each of which has the eigenvalue -1 . By Table 2, in each translation class of eight labels the three-cycle relation splits by parity into two copies of the complete graph on four vertices minus one edge, whose largest eigenvalue is $(1 + \sqrt{17})/2$, and the other classes give smaller graphs. So $\|G - I\|_{\text{op}} = 4^{-k}(1 + \sqrt{17})/2$.

Write $U_\rho = W + R$ with $W = \sum_g K_g \otimes \rho(\hat{g})$. For operators A, B on $\mathbb{C}^{104} \otimes \mathcal{H}$ put $V_A \psi = (A \otimes I)(\psi \otimes \Omega)$, with $\Omega \in \mathcal{H} \otimes \mathcal{H}$ a purification of τ ; then $X \mapsto (\text{id} \otimes \tau)[A(X \otimes I)B^*]$ has diamond norm at most $\|V_A\| \|V_B\|$, and $\|V_A\|$ is at most its Hilbert–Schmidt norm. The remainder R lives on the entries carrying the three exceptional words, of squared-coefficient weight $2 + 2 + \frac{1}{2}$, so $\|V_R\|^2 \leq \frac{9}{2} e_4^2$, and $\|V_W\| \leq 1 + \|V_R\|$ because V_{U_ρ} is an isometry. The channel of W is $X \mapsto \sum_{g,g'} G_{gg'} K_g X K_{g'}^*$, within diamond norm $\|G - I\|_{\text{op}}$ of Φ_H by (8). Therefore

$$d_\diamond(\Psi, \Phi_H) \leq \frac{1}{2} \|G - I\|_{\text{op}} + \|V_R\|(1 + \|V_R\|) + \frac{1}{2} \|V_R\|^2 < 0.0801 + 0.0247 + 0.0003 < 0.106$$

for $M \geq 48$, since $\|V_R\|^2 \leq \frac{9}{2} \cdot 14.26 \cdot 48^{-3} < 5.81 \cdot 10^{-4}$. In the leakage formula of the proof of Proposition 4.12, with $1 - \operatorname{Re} \tau(\rho(u')^* \rho(u)) = \frac{1}{2} \|\rho(u) - \rho(u')\|_2^2$ in place of $1 - \tau(P^* P')$, only the pairs counted in the proof of Proposition 4.13 contribute, each with $\frac{1}{2} e_4^2$. So the leakage is $\frac{40}{2} e_4^2 = \frac{61}{2925} e_4^2 \leq \frac{61}{2925} \cdot 14.26 M^{-3} < 0.3 M^{-3}$.

Profiles. The bath is flat, so the round has entropy production zero, and the choice of M makes the leakage at most v and the distance below $1/8$. The logarithm of the bath dimension is $8M + 8 + 2 \log M$. $\square$

So the one-round profile of Φ_H at distance $1/8$ grows at most like $\exp(O(v^{-1/3}))$, and Theorem 3.28 cannot hold with any exponent above $1/3$ in place of β_H . Sequential extraction reads the profile at the stricter distance $\epsilon/(1 - \epsilon)$, so the ceiling has to hold at every fixed distance; it does, with the fermionic power chosen from the distance.

Proposition 4.18 (One-round profile at every fixed distance). *For every $u_0 > 0$ there is $C(u_0)$ such that $\log D_{\Phi_H}^{\text{ent}}(u_0, v) \leq C(u_0) v^{-1/3}$ and $\log \mathfrak{D}_{\Phi_H}(u_0, v) \leq C(u_0) v^{-1/3}$ for $0 < v \leq 1$.*

Proof. Run the construction of the proof of Proposition 4.17 with fermionic power k and $M \geq 6$. Its estimates hold for every k : $\|G - I\|_{\text{op}} = 4^{-k}(1 + \sqrt{17})/2$, $\|V_R\|^2 \leq \frac{9}{2}e_4^2$, the leakage is $\frac{z_0}{2}e_4^2$, and the defect computation gives $e_4^2 \leq \frac{k\pi^2}{4Mm_M} + \frac{k^2\pi^2}{m_M^2} \leq \frac{k\pi^2}{M^3} + \frac{16k^2\pi^2}{M^4}$, since $m_M \geq M^2/4$. Choose k with $4^{-k}(1 + \sqrt{17})/4 \leq u_0/2$, and then M so large that $\|V_R\|(1 + \|V_R\|) + \frac{1}{2}\|V_R\|^2 \leq u_0/2$ and the leakage is at most v ; this needs M of order $(k/v)^{1/3}$. The bath is flat, so the entropy production is zero, and its dimension is $M^2 2^{k(4M+4)}$. $\square$

A memory lower bound obtained solely from this two-parameter one-round profile, as in Theorem 3.29, therefore cannot exceed order $(n/(B + \log n))^{1/3}$ at a fixed error. This does not exclude stronger arguments that use more information about the extracted round, or several rounds; and $C(u_0)$ grows as $u_0 \rightarrow 0$. Our permutation models have no such smoothing: in them the fraction of points moved by a relator adds up over a van Kampen diagram [26], and the corner models, with exponent $1/2$, are the best permutation approximants we know. Whether better permutation models exist is open.

4.5 The Choi profile

The proof of Proposition 4.17 bounds the distance to Φ_H through the remainder V_R , which is of first order in the defect e_4 ; this suffices at constant diamond accuracy. In Choi distance the defect enters only at second order. Fix $k \geq 1$ and an even $M \geq 6$; the construction of the proof of Proposition 4.17, and Lemmas 4.15 and 4.16, need only $M \geq 6$. Let $\rho = \rho_{M,k}$ be the resulting representation with fermionic power k , $\Psi_{M,k}$ its gadget channel, of bath dimension $M^2 2^{k(4M+4)}$, and $z = \tau(\rho(r_4))$. Matrix products below are written with later letters on the left. For words u, v ,

$$\tau(\rho(v)^* \rho(u)) = M^{-2} \sum_c \mathbf{1}[\text{end}_u(c) = \text{end}_v(c)] \det \left(\frac{I + W_c(v)W_c(u)^*}{2} \right)^k, \quad (9)$$

where $\text{end}_u(c)$ is the clock at which the path of u from c ends: on the fibre over c the operator acts by $\Lambda_k(W_c(v)W_c(u)^*)$ when the ends agree. Index the Choi matrices by the 174 entries of the gadget as in the proof of Proposition 4.12, and let Γ , Γ^c and Γ^H be 104 times the Choi matrices of $\Psi_{M,k}$, of the channel $\Theta_G : X \mapsto \sum_{g,g'} G_{gg'} K_g X K_{g'}^*$ of the proof of Proposition 4.17, and of Φ_H . Let EXC be the seven entries that carry the words aba^{-1} , $aba^{-1}b^{-1}$ and $aba^{-1}b^{-1}\alpha^{-1}$.

Lemma 4.19 (Structure). $\Gamma_{\mu\nu} = \Gamma_{\mu\nu}^c$ unless exactly one of μ, ν lies in EXC. For $\mu \in \text{EXC}$ and $\nu \notin \text{EXC}$, $\Gamma_{\mu\nu} = z \Gamma_{\mu\nu}^c$ and $\Gamma_{\nu\mu} = \bar{z} \Gamma_{\nu\mu}^c$.

Proof. Holonomies. A non-exceptional entry word u with normal form $\hat{u} = h t$ has $W_c(u) = W_c(t)P_h$ at every clock, where P_h permutes the coordinates $1, \dots, 4$: by Lemma 4.15 the conjugates $\alpha, a\alpha a^{-1}, a^2\alpha a^{-2}$ act as (1 2), (2 3), (3 4), and for words containing b the relator r_5 is exact. An exceptional word is $u = r_4 \hat{u}$ in the free group (proof of Proposition 4.13), so $W_c(u) = W_c(\hat{u})W'_c$ with W'_c the r_4 holonomy at the plaquette with base c , and the ends agree. By Lemma 4.16 W'_c is a product of rotations in planes whose coordinates are at least 5, so $W'_c = I_4 \oplus R_c$; put $\zeta_c = \det((I + R_c)/2)$. Since $\rho(s)$ is the adjoint of the transport operator, $z = M^{-2} \sum_c \bar{\zeta}_c^k$.

Canonical entries. For labels with normal forms $h t$ and $h' t'$ and $t \neq t'$, the ends differ at every clock, since $M \geq 6$ exceeds the translation differences, and both entries vanish. For $t = t'$, $W_c(\hat{u})W_c(\hat{u})^*$ is similar to $P_{h'h^{-1}} \oplus I$, and $\det((I_4 + P)/2)^k$ is 1 for the identity, 4^{-k} for a three-cycle and 0 otherwise; this is Γ^c .

Exceptional entries. Let μ be exceptional and ν not. Then $W_c(w_\nu)W_c(w_\mu)^* = W_c(\hat{w}_\nu)W'_c{}^*W_c(\hat{w}_\mu)^*$, which is similar to

$$W_c(\hat{w}_\mu)^*W_c(\hat{w}_\nu)W'_c{}^* = (P \oplus I)(I_4 \oplus R_c^*) = P \oplus R_c^*,$$

whose determinant factorizes; by (9) the entry is the canonical one times $M^{-2} \sum_c \bar{\zeta}_c^k$. If both are exceptional the factors W'_c cancel. The transposed entries are complex conjugates. $\square$

Lemma 4.20 (Second order). *For even $M \geq 6$, with $m_M = (M-1)(M-2)/2 + 1$,*

$$|1 - z| \leq \frac{k\pi^2}{8Mm_M} + \frac{k^2\pi^2}{2m_M^2}.$$

Proof. In the notation of the proof of Proposition 4.17, $\zeta_q = e^{i\Theta_q}C_q$ with $\Theta_q = \frac{1}{2} \sum_{R \ni q} F_{R,q}$ and $C_q = \prod_{R \ni q} \cos(F_{R,q}/2) \in [0, 1]$. The net flux vanishes for even M : $\sum_q 2\Theta_q = \pi \sum_R s_R$, the $M-1$ bulk signs $(-1)^{t-5}$ sum to 1, and $s_T = (-1)^{M-1} = -1$. Hence, since $|1 - z| = |1 - \bar{z}|$,

$$|1 - z| = \left| M^{-2} \sum_q (1 + ik\Theta_q - \zeta_q^k) \right| \leq M^{-2} \sum_q \left[\frac{k^2\Theta_q^2}{2} + \frac{k}{8} \sum_{R \ni q} F_{R,q}^2 \right],$$

by $|e^{ix} - 1 - ix| \leq x^2/2$, $1 - C^k \leq k(1 - C)$, $1 - \prod_R c_R \leq \sum_R (1 - c_R)$ and $1 - \cos x \leq x^2/2$. As in that proof $|\Theta_q| \leq \pi/m_M$ and $\sum_q \sum_R F_{R,q}^2 \leq M\pi^2/m_M$. $\square$

For odd M the net flux is π and z differs from 1 at first order; this is why M is even.

Lemma 4.21 (Choi distance). $d_J(\Psi_{M,k}, \Phi_H) \leq \frac{41}{32}4^{-k} + \Upsilon_k|1 - z|$, where $104\Upsilon_k = \sqrt{21/2} + 2\sqrt{21/4} + \sqrt{11} + 4^{-k}(2 + 2\sqrt{2} + \sqrt{6} + 2\sqrt{3})$.

Proof. We have $d_J(\Psi_{M,k}, \Phi_H) \leq \frac{1}{208} \|\Gamma - \Gamma^c\|_1 + \frac{1}{208} \|\Gamma^c - \Gamma^H\|_1$. The second term is at most $d_\diamond(\Theta_G, \Phi_H) \leq \frac{1}{2} \|G - I\|_{\text{op}}$ by (8), and $\frac{1}{2} \|G - I\|_{\text{op}} = 4^{-k}(1 + \sqrt{17})/4 \leq \frac{41}{32}4^{-k}$ (proof of Proposition 4.17). By Lemma 4.19, $\Gamma - \Gamma^c = (z - 1)K + (\bar{z} - 1)K^*$ with K the part of Γ^c with rows in EXC and columns outside it, so $\|\Gamma - \Gamma^c\|_1 \leq 2|1 - z| \|K\|_1$. The trace norm is at most the sum of the Euclidean norms of the rows. Write $K = K_1 + 4^{-k}K_3$ for its same-label and three-cycle parts. With the class weights of the proof of Proposition 4.13, the squared row norms of K_1 are $\frac{21}{2}, \frac{21}{4}, \frac{21}{4}$ for the three entries of $aba^{-1}b^{-1}$, 11 for the entry of r_4 and 0 for those of aba^{-1} , which is alone in its class. Those of K_3 are 6, 3, 3 (partners $p_{2,5}$ and $p_{2,9}$), 2 (partners $p_{2,4}, p_{2,8}$) and 2, 1, 1 (partner $p_{5,4}$). So $\|K\|_1 \leq 104\Upsilon_k$. $\square$

Proof of Theorem 3.36. Let $0 < \delta \leq \frac{1}{2}$ and $k = \lceil \log_4(41/(16\delta)) \rceil$, so $\frac{41}{32}4^{-k} \leq \delta/2$ and $k \geq 2$. For $k \geq 2$, $\Upsilon_k \leq \Upsilon_2 < 0.1136 < \frac{1}{8}$, and for $M \geq 6$, $m_M \geq M^2/4$, so Lemma 4.20 gives $|1 - z| \leq k\pi^2/(2M^3) + 8k^2\pi^2/M^4$. Every even $M \geq \max\{6, (k\pi^2/(2\delta))^{1/3}, (8k^2\pi^2/\delta)^{1/4}\}$ makes each term at most 2δ , so $\Upsilon_k|1 - z| \leq \delta/2$ and $d_J(\Psi_{M,k}, \Phi_H) \leq \delta$ by Lemma 4.21. The least such M is $O(k^{1/3}\delta^{-1/3} + k^{1/2}\delta^{-1/4})$, and the bath dimension is $M^2 2^{k(4M+4)}$, so $\log D_{\Phi_H}^J(\delta) \leq k(4M + 4) + 2 \log M = O(\delta^{-1/3} \log^{4/3}(1/\delta))$. $\square$

For example, at $\delta = 10^{-9}$ the choice is $k = 16$ and $M = 2120$, with $\log D_{\Phi_H}^J(\delta) \leq 135766$.

5 Extraction from a mixture

Lemma 5.1 (Direct extraction). *Let $d = 104$, and let $\Psi \in \mathcal{F}_m$ satisfy $d_\diamond(\Psi, \Phi_H) \leq 1/8$ and $\ell_{\Phi_H}(\Psi) \leq v \leq (144^2 d)^{-1}$. Some component of Ψ , of bath dimension at most m , carries unitaries V_a, V_b, V_α that define an e -representation ϕ with $e = 68\sqrt{6d}\sqrt{v} \approx 1698.6\sqrt{v}$ and $\|\phi(\gamma) - I\|_2 \geq 1/2$.*

Proof. Write $\Psi = \sum_i w_i \Psi_i$ with factorization unitaries $U^{(i)}$ in $d \times d$ blocks of $m_i \times m_i$ matrices, and let $v_i = \ell_{\Phi_H}(\Psi_i)$, so $\sum_i w_i v_i \leq v$. By Proposition 4.2, $\mathcal{J}(\Phi_H) = \frac{1}{d} \sum_g |K_g\rangle\langle K_g|$ with disjointly supported K_g and $\sum_g \|K_g\|_F^2 = d$.

Rounding the table, column by column. Fix a component, drop the index i , let $W = \sum_g K_g \otimes A_g$ be the orthogonal projection of U onto $\text{span}\{K_g\} \otimes M_m$, and put $R = U - W$. Measuring each $m \times m$ block in its normalized norm, the squared norms of the blocks of R sum to dv_i . For each g choose a singleton diagonal position j_g , whose column of U_λ contains only the entry of K_g , and let δ_g be the squared norm of column j_g of R ; these columns are distinct, so $\sum_g \delta_g \leq dv_i$. The off-diagonal blocks of column j_g of U belong to R , so column unitarity gives $\tau(I - U_{j_g j_g}^* U_{j_g j_g}) \leq \delta_g - \|U_{j_g j_g} - A_g\|_2^2$. A unitary polar extension V_g of $U_{j_g j_g}$ therefore satisfies $\|U_{j_g j_g} - V_g\|_2^2 \leq \delta_g - \|U_{j_g j_g} - A_g\|_2^2$, since $(1 - \sigma)^2 \leq 1 - \sigma^2$ on $[0, 1]$, and by the Cauchy–Schwarz inequality $\|A_g - V_g\|_2^2 \leq 2\delta_g$. Replacing U by $U(I \otimes V_e^*)$ changes neither the channel nor the leakage, so we may assume $V_e = I$.

Triangles. For a multiplication triangle $t = (x, y, z)$ let r_t^2 be the squared norm of R in the two system columns of its block. The two columns of U in that block are orthonormal isometries, and the two columns of $\tilde{T} = 2^{-1/2} \begin{pmatrix} I & V_y \\ V_x & -V_z \end{pmatrix}$ are isometries with cross-Gram term $\frac{1}{2}(V_y - V_x^* V_z)$. Comparing the two pairs of columns, with the Cauchy–Schwarz inequality for the two column errors and the bound on $\|A_g - V_g\|_2$, gives

$$\|V_x V_y - V_z\|_2 \leq 2\sqrt{2} \left(r_t + \sqrt{\delta_e + \delta_x + \delta_y + \delta_z} \right),$$

with repeated labels counted with multiplicity. The triangle columns are disjoint from the chosen singleton columns, and no label occurs more than twice in a triangle. Since $(r + \sqrt{S})^2 \leq 3r^2 + \frac{3}{2}S$ and $r_t^2 + \sum_g \delta_g \leq dv_i$, every triangle satisfies

$$\|V_x V_y - V_z\|_2 \leq 2\sqrt{6dv_i} =: E_i.$$

Define ϕ on the generators by V_a, V_b, V_α , with inverse letters evaluated as adjoints. A relator is assembled from its prefix and inverse triangles, 1, 14, 17, 7, 8 of them for $r_1, \dots, r_5$, so its defect is at most $17E_i$.

Keeping the three-cycle displaced. Let p and q be the singleton diagonal positions of α and of $s_1 = \alpha\alpha\alpha^{-1}$, a prefix of r_5 . For the target, $[\Phi_H(|p\rangle\langle q|)]_{pq} = \tau(\lambda(\alpha)\lambda(s_1)^*) = 0$ because $\alpha s_1 = \gamma \neq 1$. The input $(|p\rangle|0\rangle + |q\rangle|1\rangle)/\sqrt{2}$ with a qubit reference, and the bound $|X_{kl}| \leq \frac{1}{2}\|X\|_1$ for traceless Hermitian X , give $|\sum_i w_i \tau(U_{pp}^{(i)} U_{qq}^{(i)*})| \leq 1/4$. Since $|\tau(U_{pp} U_{qq}^*) - \tau(V_\alpha V_{s_1}^*)| \leq 2\sqrt{dv_i}$ and $\sum_i w_i \sqrt{v_i} \leq \sqrt{v}$,

$$\sum_i w_i \|V_\alpha^{(i)} - V_{s_1}^{(i)}\|_2^2 \geq \frac{3}{2} - 4\sqrt{dv}.$$

Components with $v_i > 4v$ have total weight at most $1/4$ and contribute at most 1, so some component has $v_i \leq 4v$ and $\|V_\alpha - V_{s_1}\|_2^2 \geq \frac{2}{3} - \frac{16}{3}\sqrt{dv} \geq \frac{17}{27} > (\frac{19}{24})^2$. The prefix triangles $(a, \alpha, a\alpha)$ and $(a\alpha, a^{-1}, s_1)$ of r_5 and one inverse triangle give $\|V_a V_\alpha V_a^* - V_{s_1}\|_2 \leq 3E_i$, and the triangle of r_1 gives $\|V_\alpha - V_\alpha^*\|_2 \leq E_i$. For this component $E_i \leq 4\sqrt{6dv} \leq \sqrt{6}/36$, so

$$\|\phi(\gamma) - I\|_2 = \|V_a V_\alpha V_a^* - V_\alpha^*\|_2 \geq \frac{19}{24} - 4E_i \geq \frac{19}{24} - \frac{5}{18} > \frac{1}{2},$$

and its relator defects are at most $17E_i \leq 68\sqrt{6dv}$. $\square$

Leakage controls the relations of the recovered table, and constant diamond accuracy keeps the three-cycle displaced. Neither suffices alone: the trivial one-dimensional representation of H_3 gives a unitary $\sum_g K_g$ with zero leakage in which γ acts trivially.

Proof of the lower bounds in Theorems 3.26 and 3.27. Let $\Psi \in \mathcal{F}_m$ have $d_\diamond(\Psi, \Phi_H) \leq 1/8$ and leakage at most $v \leq T_H^{-2}$, which is far below $(144^2 d)^{-1}$, and put $N = \lfloor (T_H^2 v)^{-\beta_H} \rfloor \geq 1$. Lemma 5.1 gives an e -representation with $e = 68\sqrt{6d}\sqrt{v}$ and $\|\phi(\gamma) - I\|_2 \geq 1/2$ on a bath of dimension at most m . With $A^*(3N) \leq 30(3N)^{\log 130}$ (Theorem 4.5) the condition of Proposition 4.10 reads $T_H N^{1/(2\beta_H)} \sqrt{v} \leq 1$, which holds by the choice of N . Hence $\log m \geq N/16 \geq c_H v^{-\beta_H} - 1/16$. For $v > T_H^{-2}$ the right side is negative. With $v = 0$ every N is allowed, so no mixture with zero leakage lies within diamond distance $1/8$. If $d_J(\Psi, \Phi_H) \leq \delta \leq 1/(8d)$, the leakage is at most δ and $d_\diamond(\Psi, \Phi_H) \leq d\delta \leq 1/8$ by (1), and for larger δ the bound is negative; this gives the bound for D^{mix} . The upper bound in Theorem 3.27 is Proposition 4.12. (Propositions 4.4 and 4.1 give the weaker bounds $O(1/v)$ and $(55/v) \log(54/v)$; the approximant of the latter at Choi accuracy v has leakage at most v and diamond distance at most $dv \leq 1/8$.) $\square$

6 From service to profile

Proof of Theorem 3.17. Entropy ledger. Feed maximally mixed inputs and keep their purifications. The complete-experiment guarantee implies $d_J(\Lambda_t, \Phi) \leq \epsilon$ for the effective channel Λ_t of each round t , whose bath is the unconditional memory marginal η_t . Let $F(\eta) = \log \dim - S(\eta)$ be the memory deficit. Put all imports of round t in a register A of dimension a in product state χ , with $c_t = \log a - S(\chi)$, and let U dilate the round on system, memory and A . With $\rho = U(\tau_d \otimes \eta_t \otimes \chi)U^*$ and $\sigma = \tau_d \otimes \eta_{t+1} \otimes \tau_a$ one has exactly $D(\rho\|\sigma) = \ell_t \ln 2$ with $\ell_t = F(\eta_t) + c_t - F(\eta_{t+1})$. Hence $\ell_t \geq 0$ and $\sum_t \ell_t \leq P + C$.

Transport and repair. Let V be a unitary on $\mathbb{C}^d \otimes \mathbb{C}^m$ with bath spectrum λ (decreasing) and a comparison spectrum μ , and let $T_{ij} = d^{-1} \sum_{a,b} |V_{(a,i),(b,j)}|^2$ in the respective eigenbases. T is doubly stochastic. Put $R = \sum_{ij} T_{ij} |\lambda_j - \mu_i|$. Sorted matching gives $\sum_i |\lambda_i - \mu_i| \leq R$, so after a bath-only rotation $\sum_{ij} T_{ij} |\lambda_j - \lambda_i| \leq 2R$. For the spectral projection P_s of the bath above level s let $e_s = \|(I - I \otimes P_s)V(I \otimes P_s)\|_{HS}^2$; then $\int e_s ds/d = \sum T_{ij} (\lambda_j - \lambda_i)_+ \leq R$. On a layer of rank r write $A = PVP = VH$ in polar form, $B = V\sqrt{I - H^2}$ and $W_\pm = A \pm iB$. Both $W_\pm$ are unitary, the original channel on the flat layer is $\Gamma_A + \Gamma_L$ with $L = (I - P)VP$, the mixture of $W_\pm$ is $\Gamma_A + \Gamma_B$, and both remainders are positive with trace $\|L\|_{HS}^2/(dr) = \|B\|_{HS}^2/(dr)$. Writing $\eta = \sum_j w_j P_j/j$ as a mixture of flat layers, the repaired channel lies in $\mathcal{F}_m$ at distance at most R . By Cauchy–Schwarz, $R \leq \sqrt{h(4-h)} \leq 2\sqrt{h}$ with $h = \sum T_{ij} (\sqrt{\lambda_j} - \sqrt{\mu_i})^2$.

Arbitrary import timing. Replace the imports by maximally mixed states. Exchanging k memory qubits for flat ones is the map $X \mapsto \text{Tr}_k(X) \otimes \tau_{2^k} = 4^{-k} \sum_c W_c X W_c^*$, with W_c the Weyl operators on those qubits, so the resulting comparison channel N_t is a mixture $\sum_c p_c \text{Ad}_{V_c}$ of unitaries V_c on the actual system and the Q memory qubits; this is why the repaired channel below lies in $\mathcal{F}_{2Q}$. By data processing and Pinsker’s inequality the replacement changes the round by at most $\sqrt{c_t \ln 2/2}$. With $h(x, y) = \|\sqrt{x} - \sqrt{y}\|_{HS}^2 = 2 - 2 \text{Tr} \sqrt{x} \sqrt{y} \leq -2 \ln \text{Tr} \sqrt{x} \sqrt{y} \leq D_{\text{nat}}(x\|y)$ (the middle quantity is the Rényi divergence of order $\frac{1}{2}$) and the triangle inequality for $\sqrt{h}$, $h_t := h(\tilde{\rho}, \sigma) \leq \ln 2 (\sqrt{c_t} + \sqrt{\ell_t})^2$. For Kraus operators K of N_t the sum $\sum_K \frac{1}{d} \|K(I \otimes \sqrt{\eta_t}) - (I \otimes \sqrt{\eta_{t+1}})K\|_{HS}^2$ does not depend on the Kraus decomposition, and for the Kraus operators of the dilation of the round by U with the flat register it equals h_t ; for the Weyl decomposition this is the identity

$$h_t = \sum_c p_c \frac{1}{d} \|V_c(I \otimes \sqrt{\eta_t}) - (I \otimes \sqrt{\eta_{t+1}})V_c\|_{HS}^2.$$

Applying the repair to each V_c with comparison state η_{t+1} and Jensen’s inequality gives a channel in $\mathcal{F}_{2Q}$ within $2\sqrt{\ln 2} \sqrt{\ell_t} + (2 + 2^{-1/2})\sqrt{\ln 2} \sqrt{c_t}$ of Λ_t . Averaging over t and choosing

the best round proves the claim. In the collision-at-call case the import step is absent and the bath entropy gain $k_t \geq 0$ of one collision directly gives $R \leq \sqrt{2k_t \ln 2}$. $\square$

6.1 Local tracialization

Lemma 6.1 (Small entropy production gives a nearby flat mixture). *Let U be a unitary on $\mathbb{C}^d \otimes \mathbb{C}^M$, ρ a bath state, $\Psi(X) = \text{Tr}_M U(X \otimes \rho) U^*$ and $T(\rho) = \text{Tr}_d U(\tau_d \otimes \rho) U^*$. Then $a = S(T(\rho)) - S(\rho) \geq 0$, and some $\Theta \in \mathcal{F}_M$ satisfies $d_J(\Psi, \Theta) \leq \sqrt{2 \ln 2 a}$.*

Proof. Diagonalize $\rho = \text{diag}(\lambda_1, \dots, \lambda_M)$ and $T(\rho) = \text{diag}(\mu_1, \dots, \mu_M)$, both decreasing, write the blocks of U in these bases as $U_{ji} \in M_d$, and put $t_{ji} = \frac{1}{d} \text{Tr}(U_{ji}^* U_{ji})$. The matrix t is doubly stochastic and $\mu_j = \sum_i t_{ji} \lambda_i$. The distributions $p_{ji} = t_{ji} \lambda_i$ and $q_{ji} = t_{ji} \mu_j$ satisfy $D(p||q) = S(\mu) - S(\lambda) = a$, so Pinsker's inequality gives $\sum_{ji} t_{ji} |\lambda_i - \mu_j| \leq \sqrt{2 \ln 2 a}$. Integrating threshold indicators shows $\sum_i |\lambda_i - \mu_i| \leq \sum_{ji} t_{ji} |\lambda_i - \mu_j|$.

Write $\rho = \sum_k w_k P_k / k$ with P_k the top- k input projection and $w_k = k(\lambda_k - \lambda_{k+1})$, and let Q_k be the top- k output projection. The compression $A_k = (I \otimes Q_k) U (I \otimes P_k)$ is a contraction between dk -dimensional spaces; with a unitary polar extension $A_k = V_k H_k$, the unitaries $W_{k,\pm} = V_k (H_k \pm i \sqrt{I - H_k^2})$ satisfy $\frac{1}{2} \sum_{\pm} W_{k,\pm} X W_{k,\pm}^* = A_k X A_k^* + C_k X C_k^*$ with $C_k = V_k \sqrt{I - H_k^2}$. For the input bath P_k / k the original channel and this flat mixture share the part from A_k , and their positive remainders have equal trace $e_k = \frac{1}{k} \sum_{i \leq k < j} t_{ji}$, so their distance is at most e_k . Mixing with weights w_k gives $\Theta \in \mathcal{F}_M$ with

$$d_J(\Psi, \Theta) \leq \sum_k w_k e_k = \sum_{i < j} t_{ji} (\lambda_i - \lambda_j) = \frac{1}{2} \sum_{ij} t_{ji} |\lambda_i - \lambda_j| \leq \sum_{ij} t_{ji} |\lambda_i - \mu_j|,$$

where the middle equality uses that the signed sum $\sum_{ij} t_{ji} (\lambda_i - \lambda_j)$ vanishes by double stochasticity. $\square$

6.2 Sequential extraction

Proof of Theorem 3.18. A test with perfect completeness. In every round the observer submits half of a fresh maximally entangled pair, measures $\{\Pi_\Phi, I - \Pi_\Phi\}$ on output and reference, keeps the outcome, discards the pair and continues. Let p_t be the probability that the tests of rounds $1, \dots, t-1$ all pass, q_t the probability of failing round t given that, and $F \in \{1, \dots, n, \infty\}$ the first failure, with ∞ for none. The ideal experiment never fails, so $f = \Pr(F \leq n) \leq \epsilon$. Hence $p_t \geq 1 - \epsilon$ and $\sum_t p_t q_t = f \leq \epsilon$.

Entropy ledger. Let ρ_t be the memory state given that the first $t-1$ tests passed; for the bookkeeping, a branch that fails is frozen. Since the input marginal is τ_d , the round takes ρ_t to $T_t(\rho_t)$, raising the entropy by a_t . Recording the outcome lowers the average conditional entropy by at most $h_2(q_t)$. Exporting k qubits lowers the entropy by at most k , and importing k qubits in a state of entropy $k - c$ raises it by $k - c$, so an exchange lowers it by at most the imported deficit c . The average entropy starts at $Q - P$ and ends at most at Q , so

$$\sum_t p_t a_t \leq P + C + \sum_t p_t h_2(q_t) \leq B + H(F) \leq B + h_2(\epsilon) + \epsilon \log n,$$

where $\sum_t p_t h_2(q_t) = H(F)$ by the chain rule and $H(F) \leq h_2(f) + f \log n$. Therefore $\sum_t p_t (q_t + a_t) \leq B + \epsilon + h_2(\epsilon) + \epsilon \log n$, and since $\sum_t p_t \geq n(1 - \epsilon)$, some round has $q_t + a_t \leq t_n$.

The extracted channel. Each round's unitary acts only on the input and the Q memory qubits, and the memory state ρ_t conditioned on the earlier passes does not depend on the current input, since the earlier test registers are discarded; so the round defines a channel Ψ_t

with bath of dimension 2^Q . Its support leakage is exactly q_t . For an arbitrary current input with reference, the actual state is p_t times a normalized pass-block state σ plus a failure block, while the ideal state ω lies in the pass block; closeness of the complete experiments gives $\frac{1}{2}\|p_t\sigma - \omega\|_1 + \frac{1-p_t}{2} \leq \epsilon$, hence $p_t\frac{1}{2}\|\sigma - \omega\|_1 \leq \epsilon$ and $d_\diamond(\Psi_t, \Phi) \leq \epsilon/(1-\epsilon)$. Lemma 6.1 gives $\Theta \in \mathcal{F}_{2^Q}$ with $d_J(\Theta, \Psi_t) \leq h_n$; by (1) and the leakage bound after Definition 2.2, $d_\diamond(\Theta, \Phi) \leq \epsilon/(1-\epsilon) + dh_n$ and $\ell_\Phi(\Theta) \leq q_t + h_n \leq t_n + h_n$. $\square$

The device performs no test; the tests and the conditioning on success belong to one particular observer and to the proof.

7 Purity

7.1 A round is an extension

Lemma 7.1 (Entropy ledger at a fixed input). *Let a causal service of Φ for n rounds have error ϵ , let $\rho > 0$, and let an observer feed the input half of a fresh copy of ψ_ρ in every round and keep the references. Let ρ_t be the memory state entering round t , U_t the unitary of that round, and*

$$a_t = S(\text{Tr}_d U_t(\rho \otimes \rho_t) U_t^*) - S(\rho_t)$$

its entropy production at input ρ . Then $\sum_{t=1}^n a_t \leq P + C$, and $a_t \geq k_{\rho, \epsilon}(\Phi)$ for every t .

Proof. Ledger. After round t the memory state is $\text{Tr}_d U_t(\rho \otimes \rho_t) U_t^*$, since the input marginal is ρ . Exporting j_t qubits lowers its entropy by at most j_t , and importing j_t qubits in a product state of entropy $j_t - c_t$, independent of everything else, raises it by exactly $j_t - c_t$. So $S(\rho_{t+1}) \geq S(\rho_t) + a_t - c_t$. Summing, with $S(\rho_1) = Q - P$ and entropy at most Q at the end, gives $\sum_t a_t \leq P + C$.

Extension. Purify the initial memory and the imports. The observer does not act, so before round t the fresh pair ψ_ρ on the reference Q_t and the input is in product with a pure state of the memory M and the remaining registers F (earlier references and outputs, exported qubits, purifiers). The round acts on the input and M only. Afterwards $\sigma_{Q_t F} = \rho^T \otimes \sigma_F$ exactly, because neither register is touched; the marginal $\sigma_{Q_t A_t}$ of reference and output is kept unchanged to the end, and it is χ_ρ in the ideal experiment, so $\frac{1}{2}\|\sigma_{Q_t A_t} - \chi_\rho\|_1 \leq \epsilon$; and since the global state is pure, $S(Q_t A_t | F) = S(M_{\text{out}}) - S(F) = S(\text{Tr}_d U_t(\rho \otimes \rho_t) U_t^*) - S(\rho_t) = a_t$. So σ is feasible in (3) with $Z = F$, and $a_t \geq k_{\rho, \epsilon}(\Phi)$. $\square$

Proof of Theorem 3.1. (b) Lemma 7.1 gives $P+C \geq \sum_t a_t \geq nk_{\rho, \epsilon}(\Phi)$ for every $\rho > 0$; take the supremum. For the second bound take $\rho = \tau_d$, so that ψ_ρ is maximally entangled. The round- t channel $\Psi_t(X) = \text{Tr}_M U_t(X \otimes \rho_t) U_t^*$ has as Choi state the marginal $\sigma_{Q_t A_t}$, so $d_J(\Psi_t, \Phi) \leq \epsilon$, and its entropy production is a_t . Lemma 6.1 gives $\Theta_t \in \mathcal{F}_{2^Q}$ with $d_J(\Psi_t, \Theta_t) \leq \sqrt{2 \ln 2 a_t}$, so $\Delta_J(\Phi) \leq \epsilon + \sqrt{2 \ln 2 a_t}$ and $a_t \geq (\Delta_J(\Phi) - \epsilon)_+^2 / (2 \ln 2)$. Sum over t .

(a) *Range.* If $\sigma_{QZ} = \rho^T \otimes \sigma_Z$ then $S(QA | Z) = S(Q | Z) + S(A | QZ) \geq S(\rho) - \log d$, which is 0 for $\rho = \tau_d$; so $\kappa \geq 0$. With Z trivial and $\sigma_{QA} = \chi_\rho$, $k_{\rho, \epsilon} \leq S(\chi_\rho) = S(\Phi^c(\rho)) \leq H(\Phi)$, since reference, output and environment of a Stinespring dilation are in a pure state. The bound $\kappa \leq \log d$ is Proposition 3.3(a).

The closure class. A mixture with rational weights n_k/N of factorizations U_k with bath dimensions m_k is itself a factorization, with unitary $\bigoplus_k U_k \otimes I_{n_k L / m_k}$ on a bath of dimension NL , L the least common multiple of the m_k , in its flat state. Approximating the weights, every element of $\bigcup_m \mathcal{F}_m$ is a d_J -limit of single factorizations. So $\mathcal{K}_d$, which is closed, is the d_J -closure of $\bigcup_m \mathcal{F}_m$, and $\Delta_J(\Phi) = \inf_{\Theta \in \mathcal{K}_d} d_J(\Theta, \Phi)$.

$\kappa = 0$ on $\mathcal{K}_d$. Let $\Phi \in \mathcal{K}_d$, $\rho > 0$ and $\epsilon > 0$. By (1) there is a factorization Ψ with bath dimension m and $d_\diamond(\Psi, \Phi) \leq \epsilon$. Feed ψ_ρ into it with the flat bath purified by F . The

resulting state has $\sigma_{QF} = \rho^T \otimes \sigma_F$, $\frac{1}{2}\|\sigma_{QA} - \chi_\rho\|_1 \leq \epsilon$ and $S(QA | F) = S(B_{\text{out}}) - S(\tau_m) \leq \log m - \log m = 0$. So $k_{\rho, \epsilon}(\Phi) \leq 0$ for all ρ and ϵ , and $\kappa(\Phi) = 0$. This is the direction proved in [2].

The quantitative converse. Take any feasible σ_{QAZ} in (3) for $\rho = \tau_d$, and put $D = \max\{\text{rank } \sigma_{QAZ}, \text{rank } \sigma_Z\}$. Purify σ_{QAZ} with a memory M of dimension D , and purify σ_Z with another copy of M . The latter purification tensored with a Bell state on QA_{in} and the former purification have the same QZ marginal $\tau_d \otimes \sigma_Z$. Their purifying spaces both have dimension dD , so a unitary on $A_{\text{in}}M$ maps one to the other. With bath state β from the purification of σ_Z , this unitary realizes a channel Ψ with Choi state σ_{QA} . Its bath entropy gain at τ_d is $a = S(\sigma_{QAZ}) - S(\sigma_Z) = S(QA | Z)_\sigma \geq 0$. Lemma 6.1 gives $\Delta_J(\Phi) \leq \epsilon + \sqrt{2 \ln 2} a$. Taking the infimum over feasible extensions proves the finite-error bound

$$k_{\tau_d, \epsilon}(\Phi) \geq \frac{(\Delta_J(\Phi) - \epsilon)_+^2}{2 \ln 2}.$$

Letting $\epsilon \downarrow 0$ gives the assertion for κ . Thus $\kappa(\Phi) = 0$ implies $\Delta_J(\Phi) = 0$ and $\Phi \in \mathcal{K}_d$.

(c) The lower bound follows from (b), since $\kappa_{\epsilon_n}(\Phi) \rightarrow \kappa(\Phi)$. For the upper bound, [2, Theorem 2.1] gives, for every $\eta > 0$, closed devices for every horizon n with errors tending to zero, bath dimensions R_n and bath states ω_n such that $(\log R_n - S(\omega_n))/n \leq \kappa(\Phi) + \eta + o(1)$. A closed device applies one fixed unitary to the input and its bath in every round and exchanges nothing, and its error is the complete-experiment error against adaptive observers with references. Embedding the bath in $\lceil \log R_n \rceil$ qubits, with the added dimensions unoccupied, gives a causal service with $J = C = 0$ and $P \leq \log R_n - S(\omega_n) + 1$. $\square$

Proof of Corollary 3.2. The bounds of Proposition 3.6 and Theorem 3.1(b) concern the resources of one device, so they hold together.

(a) Proposition 3.6 gives $2(Q + J) \geq n(H - 2\epsilon \log d) - 1 + P + C$, and Theorem 3.1(b) gives $P + C \geq n\kappa_\epsilon$. If $Q = o(n)$ and $\epsilon_n \rightarrow 0$, divide by $2n$ and use $\kappa_{\epsilon_n} \rightarrow \kappa$.

(b) Insert instead $P + C \geq n(\Delta_J - \epsilon)_+^2 / (2 \ln 2)$ from Theorem 3.1(b), and $J \leq nH/2$.

(c) By [2, Theorem 2.1] the pair $((H + \kappa)/2, (H - \kappa)/2)$ lies in the closure of the achievable rate pairs, so for $\eta > 0$ there are closed devices for every horizon L , with errors $\delta_L \rightarrow 0$, bath dimensions R_L and bath states ω_L , such that $\log R_L/L \leq (H + \kappa)/2 + \eta + o(1)$ and $(\log R_L - S(\omega_L))/L \leq \kappa + \eta + o(1)$. Put $s_n = \sup_{L \geq \sqrt{n}} \delta_L$, which tends to zero, $g(n) = \min\{\sqrt{n}, s_n^{-1/2}\}$, which tends to infinity, and $L_n = \lceil n/g(n) \rceil \geq \sqrt{n}$. Serve n rounds in consecutive blocks of L_n rounds, the last block taking the remainder, so that every block has length between L_n and $2L_n$. Serve each block by the closed device for its length. The memory has $Q_n = \max_L \lceil \log R_L \rceil$ qubits over the two block lengths used, the unused ones maximally mixed; the first block's bath is the initial memory. Before each block after the first, export the $\lceil \log R_L \rceil$ qubits that will hold that block's bath (L its length) and import its fresh bath ω_L embedded in them. Fresh baths are independent of everything before, so a hybrid argument over the at most $g(n)$ blocks bounds the error of the whole experiment by $g(n)s_n \leq s_n^{1/2} \rightarrow 0$. The memory is $O(L_n) = O(n/g(n)) = o(n)$. The step before a block of length L exchanges $\lceil \log R_L \rceil \leq \log R_L + 1$ qubits, and each block, including the first, costs at most $\log R_L - S(\omega_L) + 1$ in purity, so J_n/n and $(P_n + C_n)/n$ obey the stated bounds. $\square$

7.2 The range of κ

Lemma 7.2 (Support-injective channels). *Let Φ on M_d have a minimal Kraus family $K_1, \dots, K_r$ whose products $K_i^* K_j$ are linearly independent, let $\rho > 0$, and let P_ρ be the support projection of χ_ρ . The partial trace over A is injective on the operators supported on*

P_ρ , so it has a linear left inverse $L_\rho : M_d \rightarrow P_\rho(M_d \otimes M_d)P_\rho$; let C_ρ be the least diamond norm of such a left inverse. Then every feasible σ in (3) satisfies

$$S(QA | Z)_\sigma \geq S(\chi_\rho) - w_d((1 + C_\rho)\sqrt{\epsilon}), \quad w_d(t) = 4t \log d + (1 + t) h_2\left(\frac{t}{1+t}\right).$$

Proof. Injectivity. Write $\chi_\rho = \sum_i |\phi_i\rangle\langle\phi_i|$ with $|\phi_i\rangle = (I \otimes K_i)|\psi_\rho\rangle$. Since ρ is invertible and the K_i are independent, the ϕ_i are independent, P_ρ has rank r , and the operators supported on P_ρ are spanned by the $|\phi_i\rangle\langle\phi_j|$. Their partial traces are $(\sqrt{\rho} K_j^* K_i \sqrt{\rho})^T$, which are independent. So left inverses exist, and $C_\rho < \infty$: expanding in matrix units, $\|L\|_\diamond \leq \sum_{a,b} \|L(E_{ab})\|_1$ for every linear map L .

Projection. Since $\text{Tr}[(I - P_\rho)\chi_\rho] = 0$ and $\text{Tr}[EY] \leq \frac{1}{2}\|Y\|_1$ for a projection E and a traceless Hermitian Y , $t_0 = \text{Tr}[(I - P_\rho)\sigma_{QA}] \leq \epsilon$. The compression $\sigma' = (P_\rho \otimes I)\sigma(P_\rho \otimes I)$ satisfies $\|\sigma - \sigma'\|_1 \leq \sqrt{t_0(4 - 3t_0)} \leq 2\sqrt{\epsilon}$: purify σ and split the purification into its components in the ranges of P_ρ and $I - P_\rho$; the rank-two difference of the two projections has trace norm $\sqrt{t_0(4 - 3t_0)}$, and partial traces contract.

Inversion. The operator $Y = \sigma' - \chi_\rho \otimes \sigma_Z$ is supported on $P_\rho \otimes I$, and $(\text{Tr}_A \otimes \text{id})(Y) = \text{Tr}_A \sigma' - \rho^T \otimes \sigma_Z$, because $\text{Tr}_A \chi_\rho = \rho^T$. By the exact product constraint $\text{Tr}_A \sigma = \rho^T \otimes \sigma_Z$, this has trace norm at most $\|\sigma' - \sigma\|_1 \leq 2\sqrt{\epsilon}$. Applying $L_\rho \otimes \text{id}$, $\|Y\|_1 \leq 2C_\rho\sqrt{\epsilon}$, so $\frac{1}{2}\|\sigma - \chi_\rho \otimes \sigma_Z\|_1 \leq t := (1 + C_\rho)\sqrt{\epsilon}$.

Continuity. By Winter's continuity bound for the conditional entropy [33, Lemma 2], applied with the conditioned system QA of dimension d^2 , $S(QA | Z)_\sigma \geq S(QA | Z)_{\chi_\rho \otimes \sigma_Z} - w_d(t) = S(\chi_\rho) - w_d(t)$ for $t \leq 1$; for $t > 1$ the claim holds because $S(QA | Z) \geq -2 \log d$. $\square$

Proof of Proposition 3.3. (a) *Convexity.* Let $\Phi = \sum_j p_j \Phi_j$, fix ρ and ϵ , and take feasible σ^j for Φ_j on QAZ , with a common Z . The state $\sigma = \sum_j p_j \sigma^j \otimes |j\rangle\langle j|_L$ is feasible for Φ with side system ZL , since χ_ρ is affine in the channel, and $S(QA | ZL)_\sigma = \sum_j p_j S(QA | Z)_{\sigma^j}$. So $k_{\rho,\epsilon}(\Phi) \leq \sum_j p_j k_{\rho,\epsilon}(\Phi_j)$; let $\epsilon \downarrow 0$ and take the supremum over ρ . *The bound $\log d$.* By Choi's theorem [34] an extreme channel has linearly independent products $K_i^* K_j$, hence Choi rank at most d , and every channel is a finite convex combination of extreme channels. A channel of Choi rank $r \leq d$ has $\kappa \leq H \leq \log r \leq \log d$, and convexity finishes the proof.

(b) *Upper bound.* Let the bath $\mathbb{C}^d$ be in the state σ , purified by F , and let the unitary be the swap, which realizes $\mathcal{R}_\sigma$ exactly. Feeding ψ_ρ leaves Q entangled with the bath and AF in a pure state, so $\sigma_{QF} = \rho^T \otimes \sigma_F$ and $S(QA | F) = S(\rho) - S(\sigma) \leq \log d - S(\sigma)$. *Lower bound.* For feasible σ , $S(QA | Z) = S(Q | Z) + S(A | QZ) \geq S(\rho) - S(\sigma_A)$, and $\frac{1}{2}\|\sigma_A - \mathcal{R}_\sigma(\rho)\|_1 \leq \epsilon$, so the Fannes–Audenaert inequality gives $k_{\rho,\epsilon} \geq S(\rho) - S(\sigma) - \epsilon \log(d - 1) - h_2(\epsilon)$. Take $\rho = \tau_d$ and let $\epsilon \downarrow 0$.

(c) Lemma 7.2 gives $\lim_{\epsilon \downarrow 0} k_{\rho,\epsilon}(\Phi) \geq S(\chi_\rho) = S(\Phi^c(\rho))$ for every $\rho > 0$. Taking the supremum over full-rank ρ , which are dense, gives $\kappa \geq H$; the reverse inequality is part of Theorem 3.1(a). The order of limits matters: C_ρ may diverge as ρ approaches a state that maximizes $S(\Phi^c(\rho))$ on the boundary.

(d) The operator $Y = \mathcal{J}(\Phi) - \mathcal{J}(\Theta)$ is Hermitian and traceless with $\|Y\|_1 \leq 2\delta$, so $\|Y\|_{\text{op}} \leq \delta$. Put $p = \delta/\lambda$; if $p \geq 1$ the claim is (a). Otherwise $\mathcal{J}(\Phi) - (1 - p)\mathcal{J}(\Theta) = Y + p\mathcal{J}(\Theta) \geq 0$ has trace p and reference marginal $p\tau_d$, so it is p times the Choi state of a channel Λ , and $\Phi = (1 - p)\Theta + p\Lambda$. By convexity, $\kappa(\Theta) = 0$ and (a), $\kappa(\Phi) \leq p \log d$. $\square$

7.3 Discontinuity

Proof of Proposition 3.4. (a) The Kraus operators are $K_i = |\psi_i\rangle\langle i|$, with $K_0^* K_0 = |0\rangle\langle 0|$, $K_1^* K_1 = |1\rangle\langle 1|$ and $K_0^* K_1 = c|0\rangle\langle 1|$; so the Choi rank is 2, and the products are independent exactly when $c > 0$. The complementary channel is the Schur multiplier by $(\frac{1}{c} \ c; \ c \ 1)$, a unital qubit channel, so $H(\Phi_\theta) = 1$, attained at τ_2 .

Distance. Every channel in $\mathcal{F}_m$ is unital, and $\Psi(\tau_d)$ is the reference-traced Choi state, so $d_J(\Psi, \Phi_\theta) \geq \frac{1}{2} \|\Phi_\theta(\tau_2) - \tau_2\|_1$. Since $\Phi_\theta(\tau_2) = \tau_2 + \frac{c}{2}X$, $\Delta_J(\Phi_\theta) \geq c/2$. Conversely $\Theta_\theta = \cos^2 \theta \Delta_Z + \sin^2 \theta \text{Ad}_X \circ \Delta_Z$, with Δ_Z the complete dephasing, is a mixture of four unitary channels, so $\Theta_\theta \in \mathcal{F}_1$, and $(\Phi_\theta - \Theta_\theta)(Y) = \frac{c}{2} \text{Tr}(Y)X$ gives $d_J(\Phi_\theta, \Theta_\theta) = c/2$.

Purity. Take $\rho = \tau_2$ in Lemma 7.2. With the isometry $W|i\rangle = |i\rangle|\psi_i\rangle$, $\chi_{\tau_2} = W\tau_2W^*$, the operators supported on P_{τ_2} are $W\xi W^*$, and $\text{Tr}_A(W\xi W^*) = D_c(\xi)$, the Schur multiplier by $\begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$. Its inverse is $D_c^{-1} = \frac{1+c^{-1}}{2} \text{id} + \frac{1-c^{-1}}{2} \text{Ad}_Z$, of diamond norm at most c^{-1} , and $Y \mapsto WD_c^{-1}(Y)W^*$ is a left inverse of the same norm, so $C_{\tau_2} \leq c^{-1}$. Since $S(\chi_{\tau_2}) = 1$ and $w_2 = w$, $k_{\tau_2, \epsilon}(\Phi_\theta) \geq 1 - w((1+c^{-1})\sqrt{\epsilon})$. Lemma 7.1 gives the bound on $P+C$, and Proposition 3.3(c) gives $\kappa(\Phi_\theta) = H(\Phi_\theta) = 1$. For the exact service, extend $|i\rangle|0\rangle \mapsto |\psi_i\rangle|i\rangle$ to a unitary on the input and one memory qubit in the state $|0\rangle$, apply it in every round, and replace the memory qubit by a fresh $|0\rangle$ after every round but the last. The complete dephasing $\Phi_0 = \frac{1}{2}(\text{id} + \text{Ad}_Z)$ is an exact factor, so $\kappa(\Phi_0) = 0$.

(b) The Kraus operators are $K_1 = \text{diag}(1, 0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $K_2 = \text{diag}(0, 1, \frac{1}{\sqrt{2}}, \frac{e^{i\varphi}}{\sqrt{2}})$, the columns of the 4×2 matrix with rows v_i . The products $K_1^*K_2 = \text{diag}(0, 0, \frac{1}{2}, \frac{e^{i\varphi}}{2})$ and $K_2^*K_1 = \text{diag}(0, 0, \frac{1}{2}, \frac{e^{-i\varphi}}{2})$ are independent of each other and of $K_1^*K_1$, $K_2^*K_2$ exactly when $\sin \varphi \neq 0$. The complementary channel is $X \mapsto \sum_i X_{ii}|v_i\rangle\langle v_i|$ on $\mathbb{C}^2$, so $H(\Xi_\varphi) = 1$, attained at $\frac{1}{2}(E_{11} + E_{22})$. By Proposition 3.3(c), $\kappa(\Xi_\varphi) = 1$ for $0 < \varphi < \pi$. For $\varphi = 0$ put $u_1 = X$, $u_2 = Z$ and $u_3 = u_4 = (X+Z)/\sqrt{2}$, Hermitian unitaries with $\tau_2(u_j^*u_i) = (G_0)_{ij}$. The unitary $\sum_i E_{ii} \otimes u_i$ on $\mathbb{C}^4 \otimes \mathbb{C}^2$ with the flat bath realizes $X \mapsto \sum_{ij} X_{ij}\tau_2(u_j^*u_i)E_{ij} = \Xi_0(X)$. The Choi state $\frac{1}{4} \sum_{ij} (G_\varphi)_{ij} E_{ij} \otimes E_{ij}$ is continuous in φ . $\square$

7.4 Exact abstract realization

Proof of Corollary 3.5. By [28, Theorem 12.10] the closure of the quantum correlation sets differs from the commuting-operator set, which by [29, 30, 31] gives a negative answer to Connes' embedding problem. Haagerup and Musat [27, Theorem 3.7] showed that a positive answer is equivalent to the following: for every $d \geq 3$, every generally factorizable channel on M_d is a limit, in cb-norm, of channels with exact factorizations through $M_d \otimes M_l$. Their proof of the relevant direction produces, from a failure of the embedding, a Schur multiplier with this property failing, via Dykema and Juschenko [32]. In fixed dimension all norms on maps are equivalent, so this channel Φ is not in $\mathcal{K}_d$, and $\Delta_J(\Phi) > 0$ by Theorem 3.1(a). The remaining claims are Theorem 3.1(a)–(c). $\square$

8 A one-round bound for memory states that are not flat

In this section $\|\cdot\|_F$ is the unnormalized Hilbert–Schmidt norm, ρ is a state on $\mathbb{C}^M$ and $\xi = \sqrt{\rho}$, so $\|\xi\|_F = 1$. For operators W on $\mathbb{C}^M$ put

$$E_\rho(W) = \|(W - I)\xi\|_F, \quad \chi_\rho(W) = \|W\xi - \xi W\|_F.$$

For unitaries u, v one has $E_\rho(uv) \leq E_\rho(u) + E_\rho(v)$, $E_\rho(u^{-1}) = E_\rho(u)$, $E_\rho(uWu^{-1}) \leq E_\rho(W) + 2\chi_\rho(u)$ for $\|W\|_{\text{op}} \leq 1$, $\chi_\rho(uv) \leq \chi_\rho(u) + \chi_\rho(v)$, and $\|Fv\xi\|_F \leq \|F\xi\|_F + \|F\|_{\text{op}}\chi_\rho(v)$.

8.1 A weighted sector bound

Theorem 8.1 (Weighted sector bound). *Let c_i, x_i , $1 \leq i \leq N$, be unitaries on $\mathbb{C}^M$ with $c_i^3 = x_i^2 = I$ and $x_i c_i x_i = c_i^{-1}$, and no condition relating x_i and x_j . Put $f(g) = c_1^{g_1} \cdots c_N^{g_N}$ for $g \in \mathbb{Z}_3^N$, $\Delta_i = \|(c_i - I)\xi\|_F$, $\theta_i = \chi_\rho(c_i)$, $\eta_i = \chi_\rho(x_i)$, and, for uniform g ,*

$$A_i^2 = \mathbb{E}_g \|(f(g + e_i) - f(g)c_i)\xi\|_F^2, \quad B_i^2 = \mathbb{E}_g \|(x_i f(g^{(i)}) - f(g)x_i)\xi\|_F^2.$$

Then

$$S(\rho) \geq \sum_{i=1}^N \left[\frac{1}{3} \left(\Delta_i - \frac{4}{\sqrt{3}} \left(\sum_{j>i} \theta_j^2 \right)^{1/2} \right)_+^2 - \frac{(B_i + \eta_i)^2}{2 \ln 2} - G((A_i + \theta_i)^2) \right],$$

with G as in Theorem 4.9, and

$$A_i^2 \leq \frac{16}{3} \sum_{j>i} \|(c_i c_j - c_j c_i) \xi\|_F^2 + 8 \sum_{j>i} \theta_j^2, \quad B_i^2 \leq \frac{16}{3} \sum_{j \neq i} \|(x_i c_j - c_j x_i) \xi\|_F^2 + 8 \sum_j \theta_j^2.$$

Proof. Row sums. For order-three unitaries L_j on a Hilbert space, the averages $P_j = \frac{1}{3}(I + L_j + L_j^2)$ are orthogonal projections with $\|(I - P_j)w\|^2 = \frac{1}{3}\|L_j w - w\|^2$, so Lemma 4.7 gives $\mathbb{E}\|L_m^{g_m} \cdots L_1^{g_1} w - w\|^2 \leq \frac{4}{3} \sum_j \|L_j w - w\|^2$ for independent uniform exponents. We use it in the ordinary Hilbert–Schmidt space. For A_i , with $B = c_{i+1}^{g_{i+1}} \cdots c_N^{g_N}$,

$$B^* c_i B \xi - c_i \xi = (B^* (c_i \xi) B - c_i \xi) + B^* c_i (B \xi - \xi B).$$

Apply the inequality to the vectors $c_i \xi$ and ξ under conjugation by the c_j , $j > i$; since $\|c_j^{-1} c_i \xi c_j - c_i \xi\|_F \leq \|(c_i c_j - c_j c_i) \xi\|_F + \theta_j$, the bound for A_i^2 follows from $(u+v)^2 \leq 2u^2 + 2v^2$. For B_i use the order-three maps $Z \mapsto c_j^{-1} Z c_j$ for $j \neq i$ and $Z \mapsto c_i^{-1} Z c_i^{-1}$; their ordered product applied to x_i is $f(g)^* x_i f(g^{(i)})$, and the relation $x_i c_i x_i = c_i^{-1}$ makes the local error at $j = i$ at most θ_i .

Twirls. Let $Vw = 3^{-N/2} \sum_g f(g) w \otimes |g\rangle$ and $\sigma = V \rho V^*$, so $S(\sigma) = S(\rho)$, and let S_i shift the i -th coordinate so that $\|(S_i V - V c_i) \xi\|_F = A_i$. The three states $S_i^k \sigma S_i^{-k}$ have purifications $S_i^k V \xi c_i^{-k}$, and the order-three map $Z \mapsto S_i Z c_i^{-1}$ moves the base purification by at most $s_i = A_i + \theta_i$. So the uniform mixture of the three purifications has rank at most three and largest eigenvalue at least $1 - \frac{2}{3} s_i^2$, and entropy at most $G(s_i^2)$. By monotonicity of the Holevo quantity under partial trace, a consequence of strong subadditivity [22], the twirl $\mathcal{T}_i$ by the powers of S_i raises the entropy by at most $G(s_i^2)$; the twirls commute, and by data processing for the Holevo quantity each twirl raises the entropy of the already twirled state by no more than that of σ , so $\bar{\sigma} = \mathcal{T}_1 \cdots \mathcal{T}_N \sigma$ has $S(\bar{\sigma}) \leq S(\rho) + \sum_i G(s_i^2)$.

Sectors. Let p be the distribution of the joint Fourier labels of the group register in $\bar{\sigma}$, with values in $\{0, +, -\}^N$. The joint eigenspaces of the shifts are one-dimensional in the group register, so $S(\bar{\sigma}) \geq H(p)$. For each i , $3p\{s_i \neq 0\} = \mathbb{E}_B \|(c_i - I) B \xi\|_F^2$, and the row-sum inequality applied to ξ gives $p\{s_i \neq 0\} \geq \frac{1}{3} (\Delta_i - \frac{4}{\sqrt{3}} (\sum_{j>i} \theta_j^2)^{1/2})_+^2$. The unitary x_i together with negation of the i -th coordinate moves the purification by at most $B_i + \eta_i$, so $\sum_s (\sqrt{p(s)} - \sqrt{p(s^{(i)})})^2 \leq (B_i + \eta_i)^2$. As in the proof of Theorem 4.9, $H(p) \geq \sum_i H(S_i | S_{-i})$ and $h_2(t) \geq 1 - (1 - 2\sqrt{t(1-t)}) / \ln 2$ give $H(p) \geq \sum_i [p\{s_i \neq 0\} - \frac{1}{2 \ln 2} \sum_s (\sqrt{p(s)} - \sqrt{p(s^{(i)})})^2]$, which completes the proof. $\square$

Only row sums of squared commutators and of squared centrality defects enter, so the exponent $1/(2p+1)$ of the argument of [26, Section 4] survives the weighting.

8.2 Charged transport and rounding

Lemma 8.2 (Charged transport). *Suppose unitaries $\phi(a), \phi(b), \phi(\alpha)$ satisfy $E_\rho(\phi(r_j)) \leq e$ for all j and $\chi_\rho(\phi(s)) \leq e$ for $s \in \{a, b, \alpha\}$. If a null word w has area A_w , then $E_\rho(\phi(w)) \leq (25A_w + 2|w|)e$.*

Proof. Take a van Kampen diagram of area A_w , mark each face at a vertex where its boundary reads a relator, and cut the disk, with each face punctured and its puncture joined to its marked vertex, along a spanning tree of the one-skeleton rooted at the base point. This writes

w freely as a nested product of conjugates of relators in which every tree edge occurs once as a conjugating letter. There are at most $12A_w + |w|$ edges. Each face costs at most e and each tree edge at most $2e$, which gives $A_w e + 2(12A_w + |w|)e$. $\square$

Lemma 8.3 (Weighted S_3 rounding). *If unitaries X, Y satisfy $E_\rho(X^2), E_\rho(Y^2), E_\rho((XY)^3) \leq \delta$ and $\chi_\rho(X), \chi_\rho(Y) \leq \delta$, there are unitaries x, c with $x^2 = c^3 = I$ and $xcx = c^{-1}$ such that $\|(x - X)\xi\|_F \leq \delta$, $\|(c - XY)\xi\|_F \leq 29\delta$, $\chi_\rho(x) \leq 3\delta$ and $\chi_\rho(c) \leq 56\delta$.*

Proof. Round the spectra of X and Y to ± 1 as in Lemma 4.8. Since $|z - \operatorname{sgn} \operatorname{Re} z| \leq |z^2 - 1|$ on the circle, the involutions x, y satisfy $\|(x - X)\xi\|_F, \|(y - Y)\xi\|_F \leq \delta$; the differences are normal, so their left- and right-weighted norms agree, and $\chi_\rho(x), \chi_\rho(y) \leq 3\delta$. Then $w = xy$ has $\|(w - XY)\xi\|_F \leq 4\delta$ and, telescoping the cubes and charging the centrality of XY , $E_\rho(w^3) \leq 25\delta$. Rounding the spectrum of w to the cube roots of unity by a rule that commutes with complex conjugation and sends -1 to 1 moves each point z by at most $|z^3 - 1|$ and preserves $xwx = w^*$; this gives c , and the bounds follow from the triangle inequality. $\square$

8.3 Extraction without tracialization

Lemma 8.4 (Weighted extraction). *Let Ψ , ρ , ℓ and a be as in Theorem 3.28, put $u = d_\diamond(\Psi, \Phi_H) \leq 1/8$, $\nu = \ell + a$ and $e = 2^{17}\sqrt{\nu}$, and suppose $e \leq 1/20$. There are unitaries $\phi(a), \phi(b), \phi(\alpha)$ on $\mathbb{C}^M$ with $E_\rho(\phi(r_j)) \leq e$ for all j , $\chi_\rho(\phi(s)) \leq e$ for $s \in \{a, b, \alpha\}$, and $E_\rho(\phi(\gamma)) \geq 1$.*

Proof. Let $d = 104$.

Entropy production controls centrality. Choose bath eigenbases with $\rho = \operatorname{diag}(\lambda_i)$ and $T(\rho) = \operatorname{diag}(\mu_j)$, both decreasing; changing the output bath basis does not change Ψ . With $t_{ji} = \frac{1}{d} \operatorname{Tr}(U_{ji}^* U_{ji})$ for the $d \times d$ blocks of U , the matrix t is doubly stochastic and $D(p||q) = a$ for $p_{ji} = t_{ji}\lambda_i$ and $q_{ji} = t_{ji}\mu_j$, as in Lemma 6.1. The bound $D_{\text{nat}}(p||q) \geq 2(1 - \sum_{ji} \sqrt{p_{ji}q_{ji}})$ gives $\sum_{ji} t_{ji}(\sqrt{\lambda_i} - \sqrt{\mu_j})^2 \leq a \ln 2$, and the rearrangement inequality gives $\sum_i (\sqrt{\lambda_i} - \sqrt{\mu_i})^2$ at most the same quantity. Hence $\|U(I \otimes \xi) - (I \otimes \xi)U\|_F \leq 2\sqrt{da \ln 2} =: h$.

Column-local residuals. Write $U = \sum_{o,i} |o\rangle\langle i| \otimes A_{oi}$ and put $e_0 = \sqrt{d\ell}$ and $s = e_0 + h$. Projecting the purification $d^{-1/2} \sum_{o,i} |oi\rangle \otimes |A_{oi}\xi\rangle$ of $\mathcal{J}(\Psi)$ onto the span of the K_g gives $A_{oi} = \sum_g (K_g)_{oi} X_g + R_{oi}$ with $\sum_{o,i} \|R_{oi}\xi\|_F^2 = d\ell$. For each singleton diagonal position w , including that of the identity, let V_w be a unitary polar extension of the diagonal block A_w , a contraction on the same bath. Column unitarity gives $\|(V_w - A_w)\xi\|_F \leq e_0$, row unitarity and the centrality bound give $\|\xi(V_w - A_w)\|_F \leq e_0 + h$, and so $\chi_\rho(V_w) \leq 3s$.

Triangles. For a triangle $xy = z$ with system blocks $B_{11}, B_{12}, B_{21}, B_{22}$, comparison with the diagonal anchors gives $\|(B_{11} - V_e/\sqrt{2})\xi\|_F, \|(B_{12} - V_y/\sqrt{2})\xi\|_F, \|(B_{21} - V_x/\sqrt{2})\xi\|_F, \|(B_{22} - V_z/\sqrt{2})\xi\|_F \leq 3e_0$. The two full columns of U are orthogonal, and the rows outside the block contribute at most e_0 ; replacing the four blocks by the anchors costs at most $12s$ in each internal product, so $\|(V_x V_e^* V_y - V_z)\xi\|_F \leq 50s$. With $\phi(w) = V_e^* V_w$ every triangle satisfies $\|(\phi(x)\phi(y) - \phi(z))\xi\|_F \leq 50s$ and $\chi_\rho(\phi(w)) \leq 6s$. Following the prefix and inverse triangles through a relator of length at most 12 gives $E_\rho(\phi(r_j)) \leq 4000s$, and the same bound compares the prefix symbol of γ with its word.

Displacement. The word γ is a prefix of r_2 and has a diagonal anchor. The maximally entangled input on the anchors of the identity and of γ , together with $\tau(\lambda(\gamma)) = 0$, gives $|\operatorname{Re} \operatorname{Tr}(\rho A_e^* A_\gamma)| \leq 2u$, and with the anchor bounds $\|(\phi(\gamma) - I)\xi\|_F^2 \geq 2 - 4u - 4e_0$ for the prefix symbol. Finally $4000s \leq 4000\sqrt{d}(1 + 2\sqrt{\ln 2})\sqrt{\nu} \leq 2^{17}\sqrt{\nu} = e$, and $e \leq 1/20$ with $u \leq 1/8$ gives the claims. $\square$

Proof of Theorem 3.28. Amplification. Let ϕ be as in Lemma 8.4, let $N \geq 2$, put $A = \max\{N, A^*(3N)\}$, and suppose $10^8 \sqrt{N} A e \leq 1$. With $s_j = \phi(a)^j \phi(\alpha) \phi(a)^{-j}$, apply Lemma 8.3

to $X_i = s_{3i}$ and $Y_i = s_{3i+1}$, $0 \leq i < N$; the conjugation estimate shows that its hypotheses hold with $\delta = 16Ne$. The resulting exact pairs satisfy $\chi_\rho(c_i), \chi_\rho(x_i) \leq 20000Ne$ and, since $X_i Y_i$ is a conjugate of $\phi(\gamma)$, $\|(c_i - I)\xi\|_F \geq 1 - 476Ne$. For different copies each raw commutator is a conjugate of some C_k with $k \leq 3N$, so Lemma 8.2 gives $\|(s_j s_k - s_k s_j)\xi\|_F \leq 200Ae$; expanding the commutators of the products and adding the rounding errors gives $\|(c_i c_j - c_j c_i)\xi\|_F, \|(x_i c_j - c_j x_i)\xi\|_F \leq 20000Ae$. Theorem 8.1 then applies with $\Delta_i \geq \frac{1}{2}$, $A_i + \theta_i \leq 10^{-3}$, $B_i + \eta_i \leq 10^{-3}$ and $\frac{4}{\sqrt{3}}(\sum_{j>i} \theta_j^2)^{1/2} \leq 10^{-3}$; since $G(10^{-6}) < 10^{-3}$, every summand exceeds $1/16$, and $S(\rho) \geq N/16$.

Choice of N . With $A^*(3N) \leq 30(3N)^{\log 130}$ (Theorem 4.5) and $e = 2^{17}\sqrt{\nu}$, the condition reads $K_H N^{\log 130 + 1/2} \sqrt{\nu} \leq 1$. Put $X = (K_H^2 \nu)^{-\beta_H}$. If $X \geq 2$, take $N = \lfloor X \rfloor$: the condition holds, it forces $e \leq 1/20$, and $S(\rho) \geq \lfloor X \rfloor / 16 \geq X/16 - 1/8$. If $X < 2$ the bound is trivial. If $\nu = 0$ every N is allowed, which is impossible for a finite bath. $\square$

9 Lower bounds for services

Proof of Theorem 3.29. Apply Theorem 3.18 with $B = P + C$. Since $\epsilon \leq 1/16$, the selected round has a bath of dimension 2^Q , $d_\diamond(\Psi_t, \Phi_H) \leq 1/15 < 1/8$ and $q_t + a_t \leq t_n$. As $\epsilon + h_2(\epsilon) < 0.4$ and $0.4 + \frac{1}{16} \log n \leq \log(n+1)$, $t_n \leq \frac{16}{15}(B + \log(n+1))/n$. Theorem 3.28 gives $Q \geq S(\rho_t) \geq c_* t_n^{-\beta_H} - \frac{1}{8} \geq c_a(n/(B + \log(n+1)))^{\beta_H} - \frac{1}{8}$. $\square$

Proof of Theorem 3.34. Feed fresh halves of maximally entangled pairs in every round, without testing, and let ρ_t be the memory state entering round t and $a_t = S(T_t(\rho_t)) - S(\rho_t)$. If j_t qubits are exported after round t and replaced by a state of entropy $j_t - C_t$, then $S(\rho_{t+1}) \geq S(T_t(\rho_t)) - j_t + (j_t - C_t) = S(\rho_t) + a_t - C_t$. Summing, with $S(\rho_1) = Q - P$ and $S(\rho_{n+1}) \leq Q$, gives $\sum_t a_t \leq P + C \leq n\zeta$, so some round has $a_t \leq \zeta$. Holding the earlier inputs fixed and varying the input and reference of that round shows $d_\diamond(\Psi_t, \Phi_H) \leq \epsilon$, so its support leakage is at most ϵ . Theorem 3.28 gives $Q \geq S(\rho_t) \geq c_*(\epsilon + \zeta)^{-\beta_H} - \frac{1}{8}$. $\square$

9.1 Purity at the rank rate

Lemma 9.1 (Entropy variance). *Let p be the spectrum of a state on $q \geq 1$ qubits with deficit $D = q - S(p)$. Then the surprisal $-\log p_i$, drawn with probability p_i , has variance at most $(q + 2/\ln 2) D \leq 5qD$.*

Proof. Put $\varphi(u) = u \ln u - u + 1 \geq 0$. For $u \geq 0$ we claim $u(\ln u)^2 \leq (2 + \ln_+ u) \varphi(u)$. For $u = e^t$ with $t \geq 0$ the difference is $f(t) = e^t(t-2) + t + 2$, with $f(0) = f'(0) = 0$ and $f''(t) = te^t \geq 0$. For $t \leq 0$ it is at least $g(t) = 2e^t(t-1) + 2 - e^t t^2$, with $g(0) = 0$ and $g'(t) = -t^2 e^t \leq 0$. Now put $m = 2^q$ and $u_i = mp_i \leq m$, so that $\frac{1}{m} \sum_i u_i = 1$ and $D \ln 2 = \frac{1}{m} \sum_i \varphi(u_i)$. The variance is at most $\mathbb{E}[(\log(mp_i))^2] = \frac{1}{m(\ln 2)^2} \sum_i u_i (\ln u_i)^2 \leq \frac{2+q \ln 2}{(\ln 2)^2} D \ln 2 = (q + 2/\ln 2)D$, and $2/\ln 2 \leq 4q$. $\square$

Proof of Theorem 3.32. Put $b = P + C$ and assume $b > 0$. *Preloading.* The imports are fixed independent states and the exports are never used again, so the observer's final state does not change if all imports are prepared at the start in a register of J qubits and all exports are kept in the device. The device then starts in the product state $\sigma_0 = \beta_0 \otimes \bigotimes_t \omega_t$ on $L = Q + J$ qubits, of entropy $S = L - b$, and against a non-adaptive observer the whole experiment is one unitary V on the inputs, the references and these L qubits.

Typical subspace. The surprisal of σ_0 is a sum of independent surprisals of its factors. Each factor has at most Q qubits, because imports replace exported memory qubits, and the deficits of the factors add up to b . By Lemma 9.1 the variance is at most $5Qb$, so by

Chebyshev's inequality with $t = \sqrt{10Qb}$ the eigenvalues of σ_0 above $2^{-(S+t)}$ carry weight at least $\frac{1}{2}$; there are at most 2^{S+t} of them.

Rank. The observer feeds $\psi_*^{\otimes n}$ and keeps the references. With $\sigma_0 = \sum_i p_i |e_i\rangle\langle e_i|$, its final state is $\tau_O = \sum_i p_i \text{Tr}_D[V(\psi_*^{\otimes n} \otimes |e_i\rangle\langle e_i|)V^*]$, where D is the device side, of dimension 2^L . Each summand is a marginal of a pure state and has rank at most 2^L . Let E be the support projection of the sum over the typical eigenvalues: then $\text{rank } E \leq 2^{L+S+t} = 2^{2L-b+t}$ and $\text{Tr}(E\tau_O) \geq \frac{1}{2}$.

Comparison. The error bound gives $\text{Tr}(E\omega^{\otimes n}) \geq \frac{1}{2} - \epsilon \geq \frac{7}{16}$, and $\omega^{\otimes n}$ is flat of rank r^n , so $\text{Tr}(E\omega^{\otimes n}) \leq \text{rank } E/r^n$. Hence $2L - b + t \geq n \log r - \log \frac{16}{7}$. With $L = Q + J$ and $2J \leq n \log r + 2E_x$, where $E_x = (J - \frac{n}{2} \log r)_+$, this is $b \leq 2Q + 2E_x + \sqrt{10Qb} + \log \frac{16}{7}$, and $\sqrt{10Qb} \leq \frac{b}{2} + 5Q$ gives $b \leq 14Q + 4E_x + 2 \log \frac{16}{7}$. $\square$

The count needs no assumption on when imports happen. Replacing the rank count by the entropy count of Proposition 3.6 gives $P + C \leq 2Q + 2n\epsilon \log d + 1$, which is useful only for ϵ of order $1/n$.

10 A sublinear service of the Houghton channel

The device keeps a finite piece of H_3 as bath, a hidden selector seed, a register recording the first failure, a clock, and a flat record buffer. In round t it applies the channel's block unitary with labels twisted by θ^{j_t} , where $j_t \in \{0, 1, 2\}$ is computed from the seed and the clock, and it writes a random phase code of its new state into the buffer. Buffers are exported in blocks. The record makes different Kraus histories almost unable to interfere, at a cost of $\frac{1}{2} \log 26$ exported qubits per call.

10.1 A near-orthogonal record of complete histories

Consider a device with a finite set $\mathcal{X}$ of basis states, $|\mathcal{X}| = A$, whose interaction at time t , for each value of a selector from an alphabet of size k , has the form

$$|\psi\rangle|x\rangle \mapsto \sum_{g \in E_{t,x}} K_{t,x,g} |\psi\rangle |F_{t,x}(g)\rangle, \quad |E_{t,x}| \leq r, \quad \sum_g K_{t,x,g}^* K_{t,x,g} = I, \quad (10)$$

with distinct destinations $F_{t,x}(g)$ for distinct g . A *history* is the sequence of device states after each call; from a common start, distinct branch sequences give distinct histories. Let H_D be the normalized Walsh–Hadamard matrix on $D = 2^q$ levels. A *block* consists of two half-blocks of at most h calls. After each call of the first half the device applies to a flat q -qubit buffer the diagonal sign unitary $Z_{t,x} = \text{diag}_z \xi_{t,x,z}$ of its current state x , and after each call of the second half the rotated sign unitary $X_{t,x} = H_D \text{diag}_z \eta_{t,x,z} H_D$. Recording a register by controlled unitaries on a flat buffer is the dephasing mechanism of [24]; here the unitaries are only nearly orthogonal on complete histories, and the two bases let one buffer record two half-blocks, which halves the exchange.

Lemma 10.1 (History record). *Let $0 < \delta \leq 1$ and $D \geq 3r^h \delta^{-2} \ln(8nA(kr)^h)$. There are fixed sign tables for at most $2n$ half-blocks such that for every block, every starting state and every selector string, the Gram matrix Γ of the buffer tags of its complete histories satisfies $\|\Gamma - I\| \leq 3\delta$. Consequently, for every observer and every input that is classical on the device register, the block differs from the same block followed by dephasing of the device state after every call by at most $\frac{3}{2}\delta$ in half trace norm.*

Proof. One half-block. Fix a start and a selector string and let $\mathcal{P}$ be the $N \leq r^h$ histories of the half-block. With random independent signs, the vector $\zeta_z = (\prod_t \xi_{t,x_t,z})_{p \in \mathcal{P}}$ has

$\mathbb{E}\zeta_z\zeta_z^* = I$: two distinct histories differ at some time, where their product contains two distinct independent signs once each. The matrices $\zeta_z\zeta_z^*$ are independent, positive and of norm N , so the matrix Chernoff inequality [25] gives $\|D^{-1}\sum_z \zeta_z\zeta_z^* - I\| \leq \delta$ except with probability $2N \exp(-D\delta^2/(3N)) \leq 1/(4nAk^h)$. A union bound over the Ak^h starts and selector strings of each half-block and over the $2n$ half-blocks leaves positive probability, so fixed tables exist. Writing $v_p = D^{-1/2}(\prod_t \xi_{t,x_t,z})_z$, the matrix V with columns v_p satisfies $(1-\delta)I \preceq V^*V \preceq (1+\delta)I$, and likewise for the second-half vectors w .

Two halves. A complete history (a, c) acts on the buffer by

$$H_D \text{diag}(\sqrt{D} w_{a,c}) H_D \text{diag}(\sqrt{D} v_a).$$

Purifying the flat buffer, and using $H_D^2 = I$ and $|(H_D)_{zy}|^2 = 1/D$, the tags satisfy $\langle T_{a,c}, T_{a',c'} \rangle = \langle v_a, v_{a'} \rangle \langle w_{a,c}, w_{a',c'} \rangle$. The map $|a, c\rangle \mapsto v_a \otimes w_{a,c}$ factors as $(V \otimes I)(\bigoplus_a W_a)$, where W_a has columns $w_{a,c}$ for the continuations of a , which start from its middle state. Hence $(1-\delta)^2 I \preceq \Gamma \preceq (1+\delta)^2 I$ and $\|\Gamma - I\| \leq 3\delta$.

Observers. Purify the observer. For a fixed start the final vector is $\sum_p |\psi_p\rangle |x_p\rangle |T_p\rangle$ over complete histories p , where $|\psi_p\rangle$ collects all outputs and references. With M the matrix of columns $|\psi_p\rangle |x_p\rangle$, the actual state is $\rho = M\Gamma M^*$ and the dephased one is $\sigma = MM^*$, so $-3\delta\sigma \preceq \rho - \sigma \preceq 3\delta\sigma$ and $\frac{1}{2}\|\rho - \sigma\|_1 \leq \frac{3}{2}\delta$. Convexity handles a classical mixture of starts correlated with the observer. $\square$

10.2 Hidden selectors

Lemma 10.2 (A t -wise independent seed). *Let $t \geq 4$ be even, let $h_S = \max\{1, \lceil \log_3 n \rceil\}$, let $x_1, \dots, x_n \in \mathbb{F}_{3^{h_S}}$ be distinct, let $\lambda : \mathbb{F}_{3^{h_S}} \rightarrow \mathbb{F}_3$ be a nonzero linear functional, and put $j_s = \lambda(\sum_{i < t} A_i x_s^i)$ with $A_0, \dots, A_{t-1}$ uniform and independent. The j_s are uniform and t -wise independent. If $X_s = h_s(j_s)$ with deterministic functions h_s of mean zero over $\mathbb{F}_3$ and $|h_s| \leq B$, then with $N = 2^{\lceil \log n \rceil}$ and $c = 1 - 2^{-1/4}$,*

$$\Pr \left\{ \max_{m \leq n} \left| \sum_{s \leq m} X_s \right| > r \right\} \leq \left(\frac{B\sqrt{2tN}}{cr} \right)^t.$$

Proof. Evaluation of a polynomial of degree below t at any $\min\{t, n\}$ distinct points is onto (Vandermonde), and λ is onto, which gives uniformity and t -wise independence. For an interval I of indices, extend $X_s = 0$ beyond n and expand $\mathbb{E}(\sum_I X_s)^t$ over t -tuples of indices. By t -wise independence the expectation of a product factorizes over the blocks of equal indices, and it vanishes if some block has size one. A partition of the t positions into m blocks of size at least two has $m \leq t/2$, there are at most $\binom{t}{m} m^{t-m}$ of them (choose the least element of each block, then place the rest), and each contributes at most $|I|^m B^t$. If $|I| \geq t/2$, using $m^{t-m} \leq (t/2)^{t-m}$ and $(2|I|/t)^m \leq (2|I|/t)^{t/2}$,

$$\mathbb{E} \left(\sum_I X_s \right)^t \leq B^t \left(\frac{t}{2} \right)^t \sum_m \binom{t}{m} \left(\frac{2|I|}{t} \right)^m \leq B^t \left(\frac{t}{2} \right)^t 2^t \left(\frac{2|I|}{t} \right)^{t/2} = (2tB^2|I|)^{t/2},$$

and if $|I| < t/2$ the trivial bound $(B|I|)^t$ is smaller. At dyadic level ℓ the 2^ℓ aligned blocks of length $N2^{-\ell}$ have maximal sum A_ℓ with $\|A_\ell\|_t \leq 2^{\ell/t} (2tB^2N2^{-\ell})^{1/2} \leq B\sqrt{2tN} 2^{-\ell/4}$. Every prefix is a disjoint union of at most one aligned block per level, so $\|\max_m |\sum_{s \leq m} X_s|\|_t \leq B\sqrt{2tN}/c$, and Markov's inequality applies. $\square$

Lemma 10.3 (A hidden coin walk). *Let $j_1, \dots, j_n$ be the stationary chain on $\mathbb{Z}/3$ that moves by ± 1 with probability $\frac{1}{2}$ each, and let $X_t = h_t(j_t)$ as in Lemma 10.2. Then*

$$\Pr \left\{ \max_{m \leq n} \left| \sum_{t \leq m} X_t \right| \geq B + r \right\} \leq 2 \exp \left(- \frac{r^2}{18nB^2} \right).$$

Proof. The transition operator P acts as $-\frac{1}{2}$ on mean-zero functions. Put $u_t = \sum_{s=0}^{n-t} (-\frac{1}{2})^s h_{t+s}$, so $u_t = h_t + Pu_{t+1}$, $\|u_t\|_\infty \leq 2B$ and $\|Pu_t\|_\infty \leq B$. Then $D_1 = u_1(j_1)$ and $D_t = u_t(j_t) - (Pu_t)(j_{t-1})$ are martingale differences bounded by $3B$, and $\sum_{t \leq m} X_t = \sum_{t \leq m} D_t - (Pu_{m+1})(j_m)$. The exponential supermartingale $\exp(\lambda \sum_{t \leq m} D_t - m\lambda^2(3B)^2/2)$, stopped at the first crossing, gives the bound for each sign. $\square$

10.3 The service

Registers and initial state. Let c, N, t, R and M be as in Theorem 3.30. The bath register holds F_M in $q_F = \lceil \log(M^2(2M)!) \rceil$ qubits and starts uniform on the fibre over $(R+61, R+61)$, of entropy $\log((2M)!)$. The seed register holds $(A_0, \dots, A_{t-1})$ of Lemma 10.2 in $\lceil t h_S \log 3 \rceil$ qubits, uniform on 3^{ths} states. A failure register with values $0, 1, \dots, n$ and a clock, each of $\lceil \log(n+1) \rceil$ qubits, start in $|0\rangle$. The record buffer consists of flat qubits, sized below.

Safe positions and the round. Call $f \in F_M$ *safe* if both coordinates of $\pi(f)$ lie in $[60, M-61]$. By Lemma 4.3 and the word lengths of Section 4.1, for safe f the map $|\psi\rangle|f\rangle \mapsto \sum_g K_g |\psi\rangle |f\theta^j(g)^{-1}\rangle$ is an exact isometry into $\mathbb{C}^{104} \otimes \ell^2(F_M)$ with 26 distinct destinations; extend it to a unitary W_j . At the start of round t , for every unsafe f , swap $|f, 0\rangle$ with $|f, t\rangle$ in the bath and failure registers; no reachable state carries flag t earlier, so this marks the first failure reversibly. If the flag is 0, apply W_{j_t} to the input and bath; otherwise apply the identity. Then record the device state (bath and flag) in the buffer as in Lemma 10.1, and release the output. Each round has either 26 branches with distinct destinations or one branch, so the lemma applies with $r = 26$, $k = 3$ and $A = (n+1)|F_M|$.

Blocks. Put $\delta = \epsilon/(3n)$, $q(u) = \lceil u \log 26 + \log(27n^2/\epsilon^2) + \log \ln(8nA78^u) \rceil$ and $\Lambda = 1 + \log(27n^2/\epsilon^2) + \log \ln(8nA78^n)$, so that $q(u) \leq u \log 26 + \Lambda$ for $u \leq n$. Let $b = \min\{\lceil n/2 \rceil, \lceil \sqrt{n\Lambda/(2 \log 26)} \rceil\}$ and $B = \max\{1, \lfloor n/(2b) \rfloor\}$. The first $B-1$ blocks have $2b$ rounds each; the last has the remaining $\ell_{\text{last}} = n - 2b(B-1) < 4b$ rounds, with half-length $b_* = \lceil \ell_{\text{last}}/2 \rceil \leq 2b$. (If $B = 1$ the last block is the whole run, and $\ell_{\text{last}} = n$ may be $2b-1$.) The buffer has $q(b_*)$ flat qubits; ordinary blocks use a fixed subset of $q(b)$ of them, which is exported and replaced by fresh flat qubits after the block. The last block uses all $q(b_*)$ and is not exported.

Lemma 10.4 (The reference process). *Replace the bath by $\ell^2(H_3)$, the finite unitaries by U_j^∞ , and dephase the bath after every call. For every adaptive observer this reproduces the ideal experiment exactly, and the whole sequence of Kraus labels $(g_1, \dots, g_n)$ is independent of the seed.*

Proof. From a bath basis state f , a call maps $\rho \otimes |f\rangle\langle f|$ to $\sum_g K_g \rho K_g^* \otimes |f\theta^j(g)^{-1}\rangle\langle f\theta^j(g)^{-1}|$, with distinct endpoints for distinct g . Tracing the bath gives $\Phi_H \otimes \text{id}$ for every f and j ; since the bath is dephased, different Kraus histories never interfere, and the observer's state is the incoherent sum over histories, as in the ideal experiment with independent environments. The observer-side operators K_g depend neither on j nor on the bath, so after summing the bath the weight of a label sequence is $\text{Pr}(\text{seed}) \text{Tr} \omega_{\mathbf{g}}$. $\square$

Proof of Theorem 3.30. Records. Compare the service with the same process in which the device state is dephased after every call. Replace the blocks one at a time, starting with the first: after the replaced blocks the device register is classical (the initial bath, seed and registers are diagonal), so Lemma 10.1 bounds each replacement by $\frac{3}{2}\delta$, and the later blocks are channels. With at most n blocks the total is at most $\frac{3}{2}n\delta = \epsilon/2$.

Window. In the dephased finite process, histories that never raise the failure flag start every call at a safe position, where W_j acts as U_j^∞ ; they coincide with the corresponding histories of the reference process. By Lemma 10.4 we may condition on the whole Kraus history; the quotient increments $-\pi(\theta^{j_t}(g_t))$ have mean zero over j_t by (6) and are bounded by 2, so Lemma 10.2 with $B = 2$, for both coordinates, bounds the probability of leaving the square of radius R around the start by $2(2\sqrt{2tN}/(cR))^t \leq 2^{1-t} \leq \epsilon/2$. The never-failing histories give a common positive part of trace at least $1 - \epsilon/2$ in the dephased finite process and in the ideal experiment, so these two differ by at most $\epsilon/2$. Together the error is at most ϵ .

Resources. $J = (B - 1)q(b)$ and $C = 0$. If $B \geq 2$ then $b = \lceil \sqrt{n\Lambda/(2\log 26)} \rceil$ and

$$J \leq \left(\frac{n}{2b} - 1\right)(b \log 26 + \Lambda) = \frac{n}{2} \log 26 + \frac{n\Lambda}{2b} - b \log 26 - \Lambda \leq \frac{n}{2} \log 26,$$

since $b^2 \geq n\Lambda/(2\log 26)$. The memory is $q_F + \lceil th_S \log 3 \rceil + 2\lceil \log(n+1) \rceil + q(b_*)$ with $q_F = O(R \log R) = O(\sqrt{n \log(2/\epsilon)} \log(n/\epsilon))$, $th_S = O(\log(2/\epsilon) \log(n+1))$, $\log \ln A = O(\log M)$, $\Lambda = O(\log(n/\epsilon))$ and $q(b_*) \leq 2b \log 26 + \Lambda = O(\sqrt{n \log(n/\epsilon)})$. The deficits are the bath's $q_F - \log((2M)!) < 2 \log M + 1$, the seed's below 1, and the two pure registers. $\square$

Remark 10.5. Replacing the seed by the coin walk of Lemma 10.3, driven by one fresh flat qubit per call, gives exponential tails: with $R = 2 + \lceil 12\sqrt{n \ln(8/\epsilon)} \rceil$, Lemma 10.3 with $B = 2$ and $r = R - 2$, applied to both coordinates, bounds the exit probability by $4 \exp(-(R - 2)^2/(72n)) \leq \epsilon/2$, so $Q = O(\sqrt{n \log(2/\epsilon)} \log(n/\epsilon))$ at exchange $J \leq n(\frac{1}{2} \log 26 + 1)$, with the state of the walk held in a two-qubit register in place of the seed; the record lemma applies unchanged because the coins act as a hidden selector string. The seed of Lemma 10.2 gives the same radius at the optimal exchange.

Proofs of Corollaries 3.31 and 3.35. For exact factors use Theorem 3.8; there $x = \log(128n/\epsilon)$ is $O(\log n)$. For Φ_H at constant error combine Theorems 3.29 and 3.30, and Theorem 3.7 for the optimality of the rate. At $\epsilon_n = n^{-a}$, Theorem 3.29 applies because $\epsilon_n \leq 1/16$ eventually, Theorem 3.34 gives the second lower bound with $\zeta = O(\log n/n)$, and Theorem 3.30 with $\log(2/\epsilon_n) = O_a(\log n)$ gives the upper bound. $\square$

11 Memory for exact factors

11.1 A concentration inequality for rectangular frames

Lemma 11.1. *Let $F : \mathbb{C}^M \rightarrow \mathbb{C}^D \otimes \mathbb{C}^S$, $R = FF^*$, $g = \|F^*F\|$ and $\alpha = \|\text{Tr}_{\mathbb{C}^D} R\|$. For Hermitian A on $\mathbb{C}^D$ and Haar-random $V \in U(D)$ put $Z = F^*((V^*AV) \otimes I)F - \tau_D(A)F^*F$. Then*

$$\Pr\{\|Z\| \geq t\} \leq 2M \exp\left[-\frac{Dt^2}{4\alpha g\|A\|^2}\right].$$

Proof. Let (T_a) be a Hermitian orthonormal basis of M_D for the Hilbert–Schmidt inner product, so that $\sum_a T_a X T_a = \text{Tr}(X)I$ and $\sum_a T_a^2 = DI$. Differentiate along $V \mapsto V e^{itT_a}$. For $X = V^*AV$, $\sum_a [T_a, [T_a, X]] = 2DX - 2\text{Tr}(X)I$, so the Laplacian satisfies $\mathcal{L}Z = -2DZ$. With $Y_a = i[T_a, X] \otimes I$, the inequality $(P - P')R(P - P')^* \leq 2PRP^* + 2P'RP'^*$, the identity $\sum_a (T_a \otimes I)R(T_a \otimes I) = I \otimes \text{Tr}_D R$ and the cyclicity of the partial trace for operators $X \otimes I$ give $\sum_a Y_a R Y_a^* \leq 4\alpha\|A\|^2 I$, hence $\sum_a (\partial_a Z)^2 \leq 4\alpha\|A\|^2 F^*F$. Let $f(\theta) = \mathbb{E} \text{Tr} e^{\theta Z}$. Integration by parts with $\mathcal{L}Z = -2DZ$ and the formula $\partial_a e^{\theta Z} = \theta \int_0^1 e^{u\theta Z} (\partial_a Z) e^{(1-u)\theta Z} du$ give

$$f'(\theta) = \frac{\theta}{2D} \mathbb{E} \sum_a \int_0^1 \text{Tr} [(\partial_a Z) e^{u\theta Z} (\partial_a Z) e^{(1-u)\theta Z}] du \leq \frac{\theta}{2D} \mathbb{E} \text{Tr} \left[\sum_a (\partial_a Z)^2 e^{\theta Z} \right],$$

using $\int_0^1 e^{ua+(1-u)b} du \leq (e^a + e^b)/2$ in an eigenbasis of Z . Hence $f' \leq (2\theta\alpha g\|A\|^2/D)f$, $f(\theta) \leq Me^{\theta^2\alpha g\|A\|^2/D}$, and the exponential Markov inequality for $\pm Z$ gives the bound. $\square$

11.2 A Rényi basis

Fix the unitary U on $\mathbb{C}^d \otimes \mathbb{C}^m$ of Theorem 3.8 and let r be the Choi rank of Φ . If $r = 1$ the channel is unitary and needs no bath, so assume $r \geq 2$.

Lemma 11.2. *For every $0 < a \leq 1/2$ there are minimal Kraus operators $K_1, \dots, K_r$ of Φ , matrices $B_1, \dots, B_r \in M_m$ with $U = \sum_i K_i \otimes B_i$ and $\tau_m(B_i^* B_j) = \delta_{ij}$, and a probability vector q with all $q_i > 0$, such that*

$$\sum_i q_i B_i^* B_i = I_m, \quad \sum_i q_i^{-a} K_i^* K_i \leq 2^{aH(\Phi)+2ra^2} I_d. \quad (11)$$

Proof. Expand $U = \sum_p E_p \otimes C_p$ over a τ_m -orthonormal basis (C_p) of M_m ; then (E_p) is a Kraus family of Φ , so $E_p = \sum_i u_{pi} K_i$ for minimal Kraus operators K_i and a matrix u with $u^* u = I_r$. Hence $U = \sum_i K_i \otimes B_i$ with $B_i = \sum_p u_{pi} C_p$ and $\tau_m(B_i^* B_j) = \delta_{ij}$. The concave function $\omega \mapsto \text{Tr} \omega^{1-a}$ attains its maximum over the compact convex set of complementary outputs $\Phi^c(\rho)$, whose entries are $\text{Tr}(\rho K_j^* K_i)$, at some ρ_a ; rotating the K_i by a unitary, and the B_i by its conjugate, makes $\Phi^c(\rho_a) = \text{diag}(q)$. Every q_i is positive: $\Phi^c(I_d/d)$ is faithful because the K_i are linearly independent, and mixing a weight t of it into ρ_a would gain order t^{1-a} in a missing direction while losing only order t elsewhere. Applying $\text{Tr}(\rho_a \cdot)$ on the first factor of $U^* U = \sum_{ij} K_i^* K_j \otimes B_i^* B_j = I$ gives the first identity. For any input σ , optimality gives $\text{Tr}(q^{-a}(\Phi^c(\sigma) - \text{diag}(q))) \leq 0$, that is $\sum_i q_i^{-a} \text{Tr}(\sigma K_i^* K_i) \leq \sum_i q_i^{1-a}$. Finally $g(t) = \log \sum_i q_i^{1-t}$ has $g(0) = 0$ and $g'(0) = H(q) \leq H(\Phi)$, and for $0 \leq t \leq \frac{1}{2}$, since $\sum_i q_i^{1-t} \geq 1$ and $q^{1-t} \leq \sqrt{q}$, $g''(t) \leq (\ln 2)^{-1} \sum_i \sqrt{q_i} (\ln q_i)^2 \leq 16r/(e^2 \ln 2) < 4r$. Hence $\sum_i q_i^{1-a} \leq 2^{aH(\Phi)+2ra^2}$. $\square$

11.3 Cost of histories

Fix an observer and consider the ideal experiment with the Stinespring isometry $\xi \mapsto \sum_i K_i \xi \otimes |i\rangle$. Measuring every environment in the basis $(|i\rangle)$ at the end yields a random history $y = (i_1, \dots, i_n)$ with $\text{Pr}(y) = \|\psi_y\|^2$, where $\sum_y \psi_y \otimes |y\rangle$ is the observer's final vector. Since the environments are never touched, earlier measurement gives the same law, and conditionally on the past $\text{Pr}(i_t = i \mid \text{past}) = \text{Tr}(\rho_t K_i^* K_i)$ for the conditional input ρ_t . With $\ell_i = -\log q_i$ and $H = H(\Phi)$, (11) gives

$$\mathbb{E}[2^{a(\ell_{i_t} - H)} \mid \text{past}] = 2^{-aH} \sum_i q_i^{-a} \text{Tr}(\rho_t K_i^* K_i) \leq 2^{2ra^2}. \quad (12)$$

Put $\nu = H/2 + s$, $J_t = \lfloor \nu t \rfloor$, $j_t = J_t - J_{t-1}$, $x = \log(128n/\epsilon)$, $a_0 = \min\{\frac{1}{2}, \sqrt{3x/(2rn)}\}$ and $b = 2 + \min\{2rx/s, 3x/a_0 + 2rna_0\}$ (only the second term when $s = 0$). Use the basis of Lemma 11.2 with $a = s/r$ if the first term attains the minimum and with $a = a_0$ otherwise; since $r \geq 2$ and $s \leq 1$, $a \leq \frac{1}{2}$. Note $b = O(d^2 \min\{\sqrt{nx}, x/s\})$ when $x < n$. Define the reflected cost $z_0 = 0$, $z_t = \max\{0, z_{t-1} + \ell_{i_t} - 2j_t\}$. A history *exits* at the first t with $z_t > b$.

Lemma 11.3. *For every observer, $p_{\text{exit}} := \text{Pr}(\text{the history exits}) \leq \epsilon^2/(128^2 n)$.*

Proof. $z_t > b$ forces an interval $(u, t]$ with $\sum_{u < k \leq t} (\ell_{i_k} - 2j_k) > b$, and $2 \sum_{u < k \leq t} j_k \geq 2\nu(t - u) - 2$, so $\sum_{u < k \leq t} (\ell_{i_k} - H - 2s) > b - 2$. If $b - 2 = 3x/a_0 + 2rna_0$, iterating (12) and Markov's inequality bound the probability for one interval by $2^{-a_0(b-2)+2ra_0^2(t-u)} \leq 2^{-3x}$, and a union over at most n^2 intervals gives $p_{\text{exit}} \leq n^2 2^{-3x} \leq \epsilon^2/(128^2 n)$. If $b - 2 = 2rx/s$, then

with $a = s/r$ we have $2ra^2 - 2as = 0$, so for each start u the process $2^a \sum_{u < k \leq t} (\ell_{i_k} - H - 2s)$ is a supermartingale by (12); Ville's maximal inequality and a union over n starts give $p_{\text{exit}} \leq n2^{-a(b-2)} = n2^{-2x} = \epsilon^2/(128^2n)$. $\square$

11.4 Frames, mixing and export

A *frame* is a linear map $F : \mathbb{C}^M \rightarrow \mathbb{C}^D \otimes \mathbb{C}^S$; its columns will be the bath-and-spectator vectors attached to histories, $\mathbb{C}^D$ the active bath and $\mathbb{C}^S$ the spectators (the purifier of the initial bath, exported registers and purifiers of imports). Write $R = FF^*$, $g(F) = \|F^*F\|$ and $\alpha(F) = \|\text{Tr}_{\mathbb{C}^D} R\|$.

Lemma 11.4 (Export). *Let $\mathbb{C}^D = \mathbb{C}^{D/e} \otimes \mathbb{C}^e$. There is a unitary W on $\mathbb{C}^D$ such that the frame F' obtained by applying W , exporting the factor $\mathbb{C}^e$ and importing a flat e -dimensional register purified by a spectator satisfies $F'^*F' = F^*F$ and, for every $\eta > 0$,*

$$\alpha(F') \leq \frac{\alpha}{e^2} + 4\sqrt{\frac{\alpha g \ln(8Se^2)}{D}} \leq (1 + \eta) \frac{\alpha}{e^2} + \frac{4e^2 g \ln(8Se^2)}{\eta D}.$$

Moreover a Haar-random W has this property with probability at least $1/2$.

Proof. Gram preservation is clear. Tracing the retained bath gives $Y_W \otimes I/e$ with $Y_W = \text{Tr}_{\mathbb{C}^{D/e}}[(W \otimes I)R(W^* \otimes I)]$, so $\alpha(F') = \|Y_W\|/e$, and $\mathbb{E}Y_W = (I_e/e) \otimes \text{Tr}_{\mathbb{C}^D} R$. Define $\hat{F} : \mathbb{C}^S \rightarrow \mathbb{C}^D \otimes \mathbb{C}^M$ by $\hat{F}_{(q,y),\sigma} = \overline{F}_{(q,\sigma),y}$. A direct index computation gives $\text{Tr}_{\mathbb{C}^D}[(X \otimes I)R] = \hat{F}^*(X^T \otimes I)\hat{F}$ for every X on $\mathbb{C}^D$, together with $\hat{F}^*\hat{F} = \text{Tr}_{\mathbb{C}^D} R$ and $\text{Tr}_{\mathbb{C}^D}(\hat{F}\hat{F}^*) = F^*F$, so the product αg is the same for $\hat{F}$. The (k, l) block of Y_W is $\text{Tr}_{\mathbb{C}^D}[(X_{kl} \otimes I)R]$ with $X_{kl}^T = \overline{W}^*(I \otimes |k\rangle\langle l|)\overline{W}$, and $\overline{W}$ is Haar distributed. Apply Lemma 11.1 to $\hat{F}$ and to the Hermitian parts of $I \otimes |k\rangle\langle l|$ (norm at most one) at level $z = 2\sqrt{\alpha g \ln(8Se^2)/D}$: the $2e^2$ failure probabilities add up to at most $1/2$. Then each block deviates from its mean by at most $2z$, the block matrix by at most $2ez$, and $\|Y_W\| \leq \alpha/e + 2ez$. The last inequality is $2\sqrt{uv} \leq u + v$. $\square$

11.5 The construction

Put $\delta = \epsilon/8$, $h = \lceil H/2 + 1 \rceil$, $E_{\max} = 2^h$ and $e_t = 2^{j_t} \leq E_{\max}$. The active bath has dimension D , divisible by $mE_{\max}$, and contains $\mathbb{C}^m$ as a tensor factor; it starts in the flat state. Call t consists of a bath-only unitary $W_t^{(1)}$, the unitary U on the input and the factor $\mathbb{C}^m$, release of the output, a bath-only unitary $W_t^{(2)}$, and export of j_t qubits of the bath replaced by flat qubits. The unitaries are chosen once, before any observer, as follows.

For a history y let v_y be its bath-and-spectator vector, $v_\emptyset$ the purified flat bath, and $v_{(y,i)} = (B_i W_t^{(1)} \otimes I)v_y$ followed by the fixed operations $W_t^{(2)}$ and the exchange. A history *survives* to time t if it has not exited; survival depends only on the labels. Let F_t be the frame of surviving histories at time t and $F_t^w = F_t \text{diag}(w_y^{1/2})$ its weighted version, $w_y = 2^{-z_t(y)} \in [2^{-b}, 1]$. Let

$$L_D = \ln(8DE_{\max}^{2(n+1)}), \quad A_D = 3(1 + 8E_{\max}^2 n^2 L_D), \quad L_c = \ln(8r^{n+2}),$$

and assume

$$D^2 \geq 32 m^2 r^2 n^2 2^b A_D L_c / \delta^2. \quad (13)$$

Lemma 11.5 (Induction). *The unitaries can be chosen so that for every $t \leq n$: the Gram matrix of F_t is within $t\delta/n$ of the identity in operator norm, and $\alpha(F_t^w) \leq A_D/D$. The Gram matrix of the children of surviving histories at time t , including those that exit at t , is also within $t\delta/n$ of the identity.*

Proof. At $t = 0$ the frame is one unit vector and $\alpha = 1/D$. Suppose the claims at $t - 1$. Since $w \geq 2^{-b}$, $F_{t-1}F_{t-1}^* \leq 2^b F_{t-1}^w F_{t-1}^{w*}$, so $\alpha(F_{t-1}) \leq 2^b A_D/D$, and $g(F_{t-1}) \leq 2$.

Children. The (i, j) block of the children's Gram matrix is $F_{t-1}^*((W^*(B_i^* B_j \otimes I)W) \otimes I)F_{t-1}$ with $W = W_t^{(1)}$, whose Haar mean is $\tau_m(B_i^* B_j)F_{t-1}^* F_{t-1} = \delta_{ij}F_{t-1}^* F_{t-1}$. The Hermitian parts of $B_i^* B_j$ have norm at most $\max_i \|B_i\|^2 \leq m$. Lemma 11.1 at level $\delta/(2rn)$, with at most r^n columns, and a union over $2r^2$ events fail with probability at most $4r^{n+2} \exp[-D^2 \delta^2 / (32m^2 r^2 n^2 2^b A_D)] \leq 1/2$ by (13). Then each block deviates by at most $\delta/(rn)$ and the $r \times r$ block matrix by at most δ/n . Restriction to survivors does not increase this.

Weighted marginal. The weights satisfy $w_{(y,i)} = \min\{1, e_t^2 q_i w_y\}$. Hence, using (11) and cyclicity of the partial trace over the bath, the weighted marginal right after the collision is at most $e_t^2 \sum_i q_i \text{Tr}_{\mathbb{C}^D}[(W^* B_i^* B_i W) \otimes I] R_{t-1}^w = e_t^2 \text{Tr}_{\mathbb{C}^D} R_{t-1}^w$; bath-only unitaries do not change it and omitting exited histories only decreases it. Lemma 11.4 with $W = W_t^{(2)}$, $\eta = 1/n$, $g \leq 2$ and $Sc_t^2 \leq DE_{\max}^{2(n+1)}$ gives $\alpha(F_t^w) \leq (1 + 1/n)\alpha(F_{t-1}^w) + 8E_{\max}^2 n L_D / D$. Unrolling from $1/D$ with $(1 + 1/n)^n < 3$ gives A_D/D . Export and import preserve all Gram matrices. $\square$

Proof of Theorem 3.8. Error. Purify the observer. By induction over rounds the actual final vector is $\sum_y \psi_y \otimes v_y$ and the ideal one is $\sum_y \psi_y \otimes |y\rangle$ with the same observer vectors ψ_y , because the device acts only on bath and spectators and the observer only on its own registers. At a first exit at time t remove the vector $\sum_c \psi_c \otimes v_c$ over the exiting children; its squared norm is $\text{Tr}(\Psi G^T) \leq \|G\| \sum_c \|\psi_c\|^2 \leq (1 + \delta)p_t$, where Ψ and G are the Gram matrices of the ψ_c and of the v_c , and p_t is the ideal exit probability at t . All later operations are isometries in the proof dilation, so by Cauchy–Schwarz the removals cost at most $\sqrt{(1 + \delta)np_{\text{exit}}}$. On the final survivors the Gram matrix G is within δ of I ; writing $v = uG^{1/2}$ with orthonormal u_y , replacing v_y by u_y costs at most $\|G^{1/2} - I\| \leq \delta$. Map $|y\rangle \mapsto u_y$ by an isometry, extended on omitted labels into a larger comparison space; the omitted ideal vector has norm $\sqrt{p_{\text{exit}}}$. Hence the vector error is at most $\delta + 3\sqrt{np_{\text{exit}}} \leq 19\epsilon/128$ by Lemma 11.3, and tracing the inaccessible registers does not increase the trace distance.

Registers. If $x < n$, take $q \geq h + \lceil \log(4m/\epsilon) \rceil$ qubits, $L = \lfloor 2^{q-h}/m \rfloor$ and $D = mL E_{\max}$, increasing q until (13) holds; the active bath is an mL -dimensional subspace of the first $q - h$ qubits tensored with the last h qubits, which serve as the export port. All operations preserve this sector, and the flat state of q qubits has weight less than $mE_{\max}/2^q \leq \epsilon/4$ outside it, which costs at most $\epsilon/4$ once. Taking logarithms in (13), where $\log A_D$ and $\log L_c$ are $O(\log n + \log d + \log \log D)$, one can take $q \leq \log m + \log(2 + \log m) + b/2 + O_d(x)$; moreover $b = O(d^2 \min\{\sqrt{nx}, x/s\})$ and $x \leq \min\{\sqrt{nx}, x/s\}$ in this branch, and h , $E_{\max}$ and r are bounded in terms of d . A clock of $\lceil \log(n+1) \rceil$ pure qubits selects the round's unitaries. The bills are $Q = q + \lceil \log(n+1) \rceil$, $P = \lceil \log(n+1) \rceil$, $C = 0$, $J = J_n \leq n(H/2 + s)$, and the error is below $51\epsilon/128$. If $x \geq n$, use n untouched m -dimensional baths, one per round, encoded in $\lceil n \log m \rceil$ qubits prepared uniformly on an m^n -dimensional subspace; this is exact with $J = C = 0$, $Q = O_\Phi(n)$ and an extra initial deficit below one. $\square$

11.6 Block tests and a logarithmic lower bound

Lemma 11.6 (Block rank). *In a causal service of Φ , consider k consecutive rounds during which j qubits are exported, and let an observer feed the input halves of fresh copies of a fixed pure state ψ in these rounds, keeping the references. The block's outputs and references have rank at most $2^{2(Q+j)}$. The measurement onto their support is passed by the service with certainty, leaves the joint state of the actual experiment unchanged, and is passed by k ideal uses with probability at most the sum of the $2^{2(Q+j)}$ largest eigenvalues of $\omega^{\otimes k}$, $\omega = (\Phi \otimes \text{id})(\psi)$. For maximally entangled ψ the ideal failure probability is therefore at least $(r^k - 2^{2(Q+j)})_+ \lambda^k$,*

with r and λ the rank and least nonzero eigenvalue of $\mathcal{J}(\Phi)$.

Proof. Purify the memory state at the start of the block, of rank at most 2^Q , and the j qubits imported during the block. The block's outputs and references, the final memory, the j exported qubits and the purifiers are then in a pure state, so the outputs and references have rank at most $2^Q 2^j 2^Q 2^j$. Let S be their support. The actual state passes $\{S, I - S\}$ with certainty, and since a state is supported on the support of a marginal tensored with the rest, the measurement does not disturb it. The ideal state is $\omega^{\otimes k}$, and a projection of rank D captures at most its D largest eigenvalues; for maximally entangled ψ these are the r^k nonzero eigenvalues of $\mathcal{J}(\Phi)^{\otimes k}$, each at least λ^k . $\square$

Proof of Theorem 3.10. Feed fresh maximally entangled inputs in every round, and partition the first $k\lfloor n/k \rfloor$ rounds into blocks of $k = \lceil 2(Q+1)/g \rceil$ rounds. Since the measurements of Lemma 11.6 do not disturb the actual experiment, the observer can compute each block's support from the public protocol and test every block. Call a block cheap if $2(Q+j) \leq k \log r - 2$; its ideal failure probability is at least $\frac{3}{4}(r\lambda)^k = \frac{3}{4}2^{-\mu k}$. A block that is not cheap exports more than $\frac{k}{2} \log r - Q - 1 \geq k(\frac{1}{2} \log r - \frac{g}{2})$ qubits, so with $\eta = g/(\log r - g)$ at most $(1 - \eta)n/k$ blocks are not cheap, and at least $L \geq \eta n/k - 1$ are cheap. The ideal tests are independent and the actual ones all pass, so the error is at least $1 - \exp(-\frac{3}{4}2^{-\mu k}L)$. Error ϵ therefore forces

$$n \leq \frac{k}{\eta} \left(1 + \frac{4}{3} \ln \frac{1}{1 - \epsilon} 2^{\mu k} \right),$$

and taking logarithms with $k \leq 2(Q+1)/g + 1$ gives the bound. For the second statement, $H(\Phi) < \log r$ forces $\lambda < 1/r$, since the complementary output of the maximally mixed input has the nonzero spectrum of $\mathcal{J}(\Phi)$ and hence entropy below $\log r$; at a rate R with $H/2 < R < \frac{1}{2} \log r$ apply the bound with $g = \frac{1}{2} \log r - R$, and Theorem 3.8 with $s = R - H/2$ gives $O(\log n)$. $\square$

For biased dephasing (Remark 3.14) at $J \leq 7n/16$, blocks of $16Q + 16$ rounds with cheap blocks those exporting at most $7Q + 7$ qubits give $n \leq (16Q + 16)(7Q + 8)(1 + \frac{4}{3} \ln \frac{1}{1 - \epsilon} 2^{16Q + 16})$, so $Q \geq \frac{1}{16} \log n - O_\epsilon(\log \log n)$, while Theorem 3.8 at slack $\frac{7}{16} - \frac{1}{2}h_2(\frac{1}{4}) \approx 0.032$ gives $O_\epsilon(\log n)$.

11.7 Windows and a slack law

Proof of Theorem 3.9. The spectrum. The entropy-maximizing complementary output is faithful: $\Phi^c(I_d/d)$ is faithful, and mixing it into a singular maximizer would raise the entropy at first order through a term $-t \log t$. So ω has r positive eigenvalues, equal to those of that output, and $H < \log r$ means $V, R > 0$. For $\delta \geq 0$ let $R_\delta(\rho)$ be the least rank of a projection P with $\text{Tr}(P\rho) \geq 1 - \delta$; a projection of rank D captures at most the D largest eigenvalues, even when it is not diagonal.

A spectral excess. For $0 \leq u \leq 1/(2R)$ let $q_u(i) \propto \lambda_i^{1-u}$, $\mu(u) = \mathbb{E}_{q_u} Z$ and $D(u) = \sum_i q_u(i) \log(q_u(i)/\lambda_i)$. Then $\mu' = (\ln 2) \text{Var}_{q_u} Z$ and $D' = u\mu'$. Since $q_u(i)/\lambda_i \geq 2^{-uR}$, the pairwise form of the variance gives $\text{Var}_{q_u} Z \geq 2^{-2uR}V \geq V/2$, and $\text{Var}_{q_u} Z \leq R^2/4$; so $a \leq \mu' \leq A$, $\mu(u) - H \geq au$ and $D(u) \leq Au^2/2$. Let $b \geq x \geq x_0$ be integers and $u = \gamma\sqrt{x/b}$. Under $q_u^{\otimes b}$ the sum of b copies of Z lies within $R\sqrt{b}$ of $b\mu(u)$ with probability at least $3/4$ by Chebyshev's inequality, and on this band $\lambda^{\otimes b} \geq q_u^{\otimes b} 2^{-bD(u) - uR\sqrt{b}}$. So the band has $\lambda^{\otimes b}$ -mass $w \geq \frac{3}{4}2^{-x/4} \geq 2 \cdot 2^{-x}$, using $A\gamma^2 \leq \frac{1}{4}$ and $x \geq 64R^2\gamma^2$, and its eigenvalues are at most $2^{-b\mu(u) + R\sqrt{b}}$. A projection capturing $1 - 2^{-x}$ of $\omega^{\otimes b}$ captures at least $w/2$ of the band, so $\log R_{2^{-x}}(\omega^{\otimes b}) \geq b\mu(u) - R\sqrt{b} + \log(w/2)$. By the choice of γ and x_0 each of the losses $R\sqrt{b}$, $(A\gamma^2/2)x$, $R\gamma\sqrt{x}$ and $\log(8/3)$ is at most $(a\gamma/8)\sqrt{bx}$, whence

$$\log R_{2^{-x}}(\omega^{\otimes b}) \geq bH + c\sqrt{bx} \quad (b \geq x \geq x_0). \quad (14)$$

(a) Put $B = \min\{n, c^2x/(64\sigma^2)\}$, or $B = n$ if $\sigma = 0$; the hypotheses give $B \geq 4x$. Split the rounds into $N = \lceil n/B \rceil$ windows of nearly equal lengths b_i ; then $b_i \geq B/4 \geq x$ and $n/N \leq B$. The observer feeds the pure maximizer in every round. By Lemma 11.6 the record of window i has rank at most $2^{2(Q+J_i)}$, where J_i counts the exports inside the window and $\sum_i J_i \leq J$. Testing that window alone, the observer sees error at most $\epsilon \leq 2^{-x}$, so the support has rank at least $R_{2^{-x}}(\omega^{\otimes b_i})$, and (14) gives $2(Q + J_i) \geq b_i H + c\sqrt{b_i x}$. Summing, $2NQ + 2J \geq nH + \frac{c}{2}N\sqrt{Bx}$, and $J \leq n(H/2 + \sigma)$ gives $Q \geq \frac{c}{4}\sqrt{Bx} - \sigma B \geq \frac{c}{8}\sqrt{Bx}$, since $\sigma\sqrt{B} \leq \frac{c}{8}\sqrt{x}$. The two values of B give the two terms.

(b) Split the rounds into N windows of lengths $b_i \in \{b, b+1\}$ and let Π_i be the support of the actual record of window i . A state is supported in the product of the supports of its marginals, so the actual experiment passes $\bigotimes_i \Pi_i$ with certainty, while n ideal uses, independent across windows, pass with probability $\prod_i (1 - m_i)$, $m_i = 1 - \text{Tr}(\Pi_i \omega^{\otimes b_i})$. Hence $\sum_i m_i \leq -\ln(1 - \epsilon) \leq 2\epsilon \leq \frac{1}{2}$, and at most $\delta_0 N$ windows have $m_i > 2^{-y}$. On the others (14) gives $\log \text{rank } \Pi_i \geq b_i H + c\sqrt{b_i y}$. On the rest $m_i \leq \frac{1}{2}$; by Chebyshev three quarters of the mass of $\omega^{\otimes b_i}$ sits on eigenvalues at most $2^{-b_i H + 2\sqrt{b_i y}}$, so $\log \text{rank } \Pi_i \geq b_i H - 2\sqrt{2b_i y} - 2$. With $\log \text{rank } \Pi_i \leq 2(Q + J_i)$, $n \leq 2Nb$ and $\delta_0(2\sqrt{2b_i y} + 2) \leq \frac{c}{8}\sqrt{b_i y}$, summing gives $2NQ \geq \frac{3cN}{8}\sqrt{by} - 2sn \geq \frac{3cN}{8}\sqrt{by} - 4sNb$.

For the consequence put $K = (3c/64)^2$, $y_1 = \log(1/\epsilon) + \frac{\eta}{2} \log n$, $b_0 = \lfloor Ky_1/(4s^2) \rfloor$ and $N = \lfloor n/b_0 \rfloor$. Under the stated conditions, for n large and s_1 small, $b_0 \leq b \leq 2b_0$, $\delta_0 N/2 \geq n^{\eta/2}$ so that $y \geq y_1$, and $b \geq y \geq x_0$. Then $b \leq Ky/(2s^2)$, so $2sb \leq \frac{3c}{32}\sqrt{by}$, and $\sqrt{by} \geq \sqrt{K/8} y_1/s$ with $y_1 \geq \frac{\eta}{2} \log(n/\epsilon)$; the bound of (b) becomes $Q \geq \frac{3c}{32}\sqrt{K/8} \frac{\eta}{2} \log(n/\epsilon)/s \geq (\eta c^2/1300) \log(n/\epsilon)/s$. $\square$

11.8 Constant memory at the rank rate

Proof of Proposition 3.12. The memory is one qubit F , initially maximally mixed. In the first round of each pair apply $V_Z = \sqrt{p}I + i\sqrt{1-p}R \otimes Z$ to the input and F , in the second $V_X = \sqrt{p}I + i\sqrt{1-p}R \otimes X$; both are unitary because $R \otimes Z$ and $R \otimes X$ are Hermitian unitaries. After every completed pair except the last pair or single round, export F and import a maximally mixed qubit. Purify the observer and put $c_0 = \sqrt{p}$, $c_1 = i\sqrt{1-p}$. For a pair, the actual final vector is $\sum_{a,b \in \{0,1\}} c_a c_b \psi_{ab} \otimes (X^b Z^a \otimes I)\xi$, where ψ_{ab} is the observer vector obtained by applying R^a in the first round and R^b in the second and ξ purifies the flat state of F ; the ideal experiment is $\sum_{a,b} |c_a c_b|^2 |\psi_{ab}\rangle\langle\psi_{ab}|$. The four operators $X^b Z^a$ are orthonormal for the normalized trace, so tracing out F and its purifier keeps exactly the diagonal terms. A single final round uses V_Z alone, and fresh imports make pairs independent. The count is $Q = 1$, $P = C = 0$ and $J = \lfloor (n-1)/2 \rfloor$. The protocol uses round-dependent unitaries, as Definition 2.3 allows; a time-independent protocol needs one more pure qubit as a clock. Recording histories by orthonormal unitaries on a flat register is the device of [24]; here one flat qubit serves two rounds.

For $R = Z$ the Choi rank is 2. At zero error Theorem 3.7(c) gives $J \geq n/2 - Q$. At a fixed error below one and a rate below $\frac{1}{2}$, Theorem 3.10 forces memory $\Omega(\log n)$ when $p \neq \frac{1}{2}$, since then the least Choi eigenvalue is below $\frac{1}{2}$; for $p = \frac{1}{2}$ the spectrum ω is flat on two levels and Theorem 3.7 gives $2(Q + J) \geq n + \log(1 - \epsilon)$. $\square$

Proof of Proposition 3.13. The vectors $(U_a \otimes I)|\Omega\rangle$ are orthonormal, so $\mathcal{J}(\Phi)$ is flat of rank r ; the complementary output of the maximally mixed input has this spectrum, and r bounds the entropy of a complement of dimension r , so $H(\Phi) = \log r = q$. Hold a flat bath of q qubits, with computational projections P_a and Fourier projections Q_b . Apply $V_1 = \sum_a U_a \otimes P_a$ in the first round of each pair and $V_2 = \sum_b U_b \otimes Q_b$ in the second, and refresh the bath after every completed pair except the last. The bath operator of the history (a, b) is $Q_b P_a$, and

$\tau((Q_{b'}P_{a'})^*Q_bP_a) = \delta_{aa'}\delta_{bb'} |\langle a|\hat{b}\rangle|^2/r = \delta_{aa'}\delta_{bb'}/r^2$, which is the weight of that history in two ideal uses; as in the previous proof, tracing out the bath keeps exactly the diagonal terms. The count is $Q = q$, $P = C = 0$ and $J = q\lfloor(n-1)/2\rfloor$, and $\frac{q}{2} = \frac{H}{2}$ is optimal by Theorem 3.7. $\square$

11.9 Closure channels

Proof of Theorem 3.19. Let Ψ have a single finite tracial factorization of bath dimension $m = D_\Phi^J(\epsilon/(8dn))$ with $d_J(\Psi, \Phi) \leq \epsilon/(8dn)$; by (1), $d_\diamond(\Psi, \Phi) \leq u := \epsilon/(8n)$. Apply Theorem 3.8 to Ψ with error $\epsilon/2$ and slack s ; its parameter $\log(128n/(\epsilon/2))$ is the present $x < n$, and its constants depend only on d . Replacing the n ideal uses of Ψ by uses of Φ changes every observer's final state by at most $nu \leq \epsilon/8$. The service exports at most $n(H(\Psi)/2 + s)$ qubits. For a common purified input, the complementary entropies of Ψ and Φ are the entropies of their output-and-reference states, of dimension d^2 and at trace distance at most u , so by the Fannes–Audenaert inequality $H(\Psi) \leq H(\Phi) + f(u)$ with $f(u) = h_2(u) + u \log(d^2 - 1)$, and $nf(u) \leq x + 2 \log d$. Replace the last $\lceil nf(u)/2 \rceil + 1$ exports by swaps into initially flat spare slots of the memory that are never touched again; seen from outside, this is identical to exporting the qubit and importing a flat one. The slots add $O_d(x)$ to Q and nothing to P or C , and the error is at most $\epsilon/2 + \epsilon/8 < \epsilon$. $\square$

11.10 Service at the rank rate

Let Φ on M_d have Choi rank $r \geq 2$, and let Ψ have a factorization with unitary U on $\mathbb{C}^d \otimes \mathbb{C}^m$, Kraus rank $R \leq d^2$ and $\delta_J = d_J(\Psi, \Phi)$. Put $t = d\delta_J$. The construction of Theorem 3.8 used H in exactly one place, the exponential moment (12) of the cost of a history; the following three lemmas replace it by a bound that refers only to r .

Lemma 11.7 (A rank- r tail). *There is a projection P_E of rank at most r on the minimal environment $\mathbb{C}^R$ of Ψ with $\text{Tr}[(I - P_E)\Psi^c(\rho)] \leq t$ for every state ρ .*

Proof. Let $L_1, \dots, L_R$ be Kraus operators of Ψ with $\text{Tr}(L_i^* L_j) = 0$ for $i \neq j$, ordered so that the eigenvalues $\lambda_i = \text{Tr}(L_i^* L_i)/d$ of $\mathcal{J}(\Psi)$ decrease. By Ky Fan's principle a projection of rank r captures at most $\sum_{i \leq r} \lambda_i$, so $\sum_{i > r} \lambda_i \leq \text{Tr}[(I - \Pi_\Phi)\mathcal{J}(\Psi)] = \text{Tr}[(I - \Pi_\Phi)(\mathcal{J}(\Psi) - \mathcal{J}(\Phi))] \leq \delta_J$. Hence $A = \sum_{i > r} L_i^* L_i$ has trace at most t and $A \leq tI$. With P_E the projection onto the first r basis vectors, $\text{Tr}[(I - P_E)\Psi^c(\rho)] = \text{Tr}(\rho A) \leq t$. $\square$

Lemma 11.8 (Pinching). *For every state ρ and $0 < a < 1$, $\text{Tr}[\Psi^c(\rho)]^{1-a} \leq r^a + (d^2)^a t^{1-a}$.*

Proof. Pinching $\omega = \Psi^c(\rho)$ by P_E and $I - P_E$ is unital, so for the concave $f(s) = s^{1-a}$, $\text{Tr} f(\omega)$ does not decrease. A positive block of dimension k and trace w has $\text{Tr} X^{1-a} \leq k^a w^{1-a}$ by Jensen's inequality. The blocks have dimensions at most r and $R \leq d^2$ and traces at most 1 and t . $\square$

Lemma 11.9 (A basis for the rank rate). *For $0 < a \leq \frac{1}{2}$ there are minimal Kraus operators $K_1, \dots, K_R$ of Ψ , matrices $B_i \in M_m$ with $U = \sum_i K_i \otimes B_i$ and $\tau_m(B_i^* B_j) = \delta_{ij}$, and a probability vector q with all $q_i > 0$, such that $\sum_i q_i B_i^* B_i = I_m$ and $\sum_i q_i^{-a} K_i^* K_i \leq (r^a + (d^2)^a t^{1-a}) I_d$. Consequently, in the notation of (12) for the ideal experiment with Ψ ,*

$$\mathbb{E}[2^{a(\ell_{it} - \log r)} \mid \text{past}] \leq 1 + \left(\frac{d^2}{r}\right)^a t^{1-a} =: 2^\zeta. \quad (15)$$

Proof. Follow the proof of Lemma 11.2 for Ψ , with R Kraus operators, up to the optimality inequality $\sum_i q_i^{-a} \text{Tr}(\sigma K_i^* K_i) \leq \sum_i q_i^{1-a} = \text{Tr}[\Psi^c(\rho_a)]^{1-a}$ for every state σ . Bound the right side by Lemma 11.8 instead of by 2^{aH+2Ra^2} . $\square$

Proof of Theorem 3.21. Put $\eta = \epsilon/2$, $x = \log(128n/\eta)$, $L = \log(8n/\epsilon) \geq 3$, $a = 1/L$, and take $t = \epsilon/(8n)$, which is allowed since $d\delta_J \leq \epsilon/(8n)$. Then $t^{1-a} = 2t$ exactly and $(d^2/r)^a \leq d$, so $\zeta \leq 2dt/\ln 2$ and $n\zeta \leq d\epsilon/(4\ln 2)$.

Schedule and exits. Run the construction of Section 11 for Ψ at tolerance η , with three changes: the basis of Lemma 11.9 replaces that of Lemma 11.2; the schedule is $\nu = \frac{1}{2} \log r$; and the threshold is $b = 2 + L(3x + n\zeta) \leq 2 + 3x^2 + 0.361 dx$. The export port has $h = \lceil \frac{1}{2} \log r + 1 \rceil$ qubits, since $j_t \leq \nu + 1$. As in Lemma 11.3, an exit forces an interval $(u, t]$ with $\sum_{u < k \leq t} (\ell_{i_k} - \log r) > b - 2$. By (15) and Markov's inequality this has probability at most $2^{-a(b-2)+n\zeta} = 2^{-3x}$, and a union over n^2 intervals gives $p_{\text{exit}} \leq n^2 2^{-3x} \leq \eta^2/(128^2 n)$, the bound of Lemma 11.3.

The rest. Lemmas 11.4 and 11.5 use only the first identity of (11), the weights $w_y \in [2^{-b}, 1]$, $E_{\max}$ and the number of Kraus operators, which is now R ; they hold with R in place of r in L_c and (13). The error paragraph of the proof of Theorem 3.8 uses only p_{exit} , so the service of Ψ has error below $51\eta/128$. Taking logarithms in (13), with q qubits, $L = \lfloor 2^{q-h}/m \rfloor$ and $D = mLE_{\max}$, the condition holds once $q \geq \log m + b/2 + \frac{1}{2} \log(2+q) + C_d x$, and the least such q is at most $\log m + \log(2 + \log m) + b/2 + \log(2+b) + O_d(x)$. Initialize all q qubits maximally mixed, including the padding sector, exactly as in the register argument of Theorem 3.8. No clock is needed: Definition 2.3 permits a prescribed unitary for each round, so the selected mixers and replacements can be specified directly as that sequence. The protocol uses no random controller or additional retained register.

From Ψ to Φ . By (1), $d_\diamond(\Psi, \Phi) \leq \epsilon/(8n)$, so replacing the ideal uses of Ψ by uses of Φ changes every observer's final state by at most $\epsilon/8$, and the total error is below ϵ . The exchange is $J = \lfloor n\nu \rfloor$ and $P = C = 0$. Unlike in the proof of Theorem 3.19, no exchange is spent on $H(\Psi) - H(\Phi)$: the excess cost of the approximant is paid by the term $n\zeta$ in b . $\square$

Proof of Theorem 3.15. Run the construction for $\Psi = \Phi$ at tolerance ϵ . Now $t = 0$, so Lemma 11.9 with $a = \frac{1}{2}$ gives $\zeta = 0$, and $b = 2 + 6x$ gives $p_{\text{exit}} \leq n^2 2^{-3x} \leq \epsilon^2/(128^2 n)$; the register count gives $b/2 + \log(2+b) = O(x)$. If $H(\Phi) = \log r$, an input ρ_* with $\Phi^c(\rho_*) = I_r/r$ maximizes $\text{Tr}[\Phi^c(\rho)]^{1-a}$ for every a , so the basis may be taken with $q_i = 1/r$. Then every $\ell_i = \log r$, the cost is deterministic, and $\sum_{u < k \leq t} (\log r - 2j_k) \leq 2$ for every interval, so $z_t \leq 2$ always: with $b = 2$ no history exits. $\square$

Proof of Corollary 3.16. The spectrum of ω is that of the maximizing complementary output, which is faithful on r levels, so ω is flat exactly when $H(\Phi) = \log r$. Part (a) is Theorem 3.15. Part (b) is Theorem 3.9(a) with $\sigma = 0$ and Theorem 3.8 with $s = 0$. $\square$

Proof of Corollary 3.37. Φ_H has $d = 104$, Choi rank $r = 26 = 2^{H(\Phi_H)}$, and lies in $\mathcal{K}_{104}$. Apply Theorem 3.21 with $m = D_{\Phi_H}^J(\epsilon/(832n))$. By Theorem 3.36, $\log m = O((n/\epsilon)^{1/3} \log^{4/3}(n/\epsilon))$, which dominates the term $O(\log^2(n/\epsilon))$. The window at fixed error is Theorem 3.29 with $B = P + C = O(\log n)$. $\square$

Proof of Corollary 3.38. The channel Ξ . The 32 unitaries Z_{208}^a are orthonormal for the normalized trace, since $\tau(Z_{208}^b) = 0$ for $0 < |b| < 208$, and $32 = 2^5$, so Proposition 3.13 gives Choi rank 32, $H = 5$ with flat complementary output, and (a). The unitary $\sum_a Z_{208}^a \otimes |a\rangle\langle a|$ with a flat 32-dimensional bath is an exact factorization, so $\Xi \in \mathcal{K}_{208}$.

Invariants of $\hat{\Phi}$. Its Kraus operators are $|0\rangle\langle 0| \otimes K_g$ for the 26 labels of Proposition 4.2 and $|1\rangle\langle 1| \otimes 6^{-1/2} Z_{104}^a$ for $a < 6$. They are linearly independent, so the Choi rank is 32. The two branches have orthogonal environments, so $\hat{\Phi}^c(\rho) = \Phi_H^c(\rho_{00}) \oplus \Theta_6^c(\rho_{11})$. For $\rho = \frac{26}{32}|0\rangle\langle 0| \otimes \rho_* + \frac{6}{32}|1\rangle\langle 1| \otimes \tau_{104}$, with ρ_* as in Proposition 4.2, this is $\frac{26}{32} \frac{I_{26}}{26} \oplus \frac{6}{32} \frac{I_6}{6} = \frac{I_{32}}{32}$, since $\text{Tr}(Z_{104}^a \tau_{104} Z_{104}^{-b}) = \delta_{ab}$ for $a, b < 6$. So $H(\hat{\Phi}) = 5$, the probe spectrum is flat of rank 32, and $\frac{n}{2} \log 32 = \frac{5}{2}n$.

Approximants. Let Ψ have a factorization with unitary U and bath dimension m , and let $V = \sum_{a \leq 6} Z_{104}^a |a\rangle\langle a|$ on $\mathbb{C}^{104} \otimes \mathbb{C}^6$. On the tag, the system and a flat bath $\mathbb{C}^m \otimes \mathbb{C}^6 \otimes \mathbb{C}^2$, apply $|0\rangle\langle 0| \otimes U \otimes I_6 + |1\rangle\langle 1| \otimes V \otimes I_m$ (each acting on its own bath factor), followed by the controlled phase between the tag and the last bath qubit. The last step dephases the tag, and tracing the flat bath factors that a branch does not use leaves $X \mapsto |0\rangle\langle 0| \otimes \Psi(X_{00}) + |1\rangle\langle 1| \otimes \Theta_6(X_{11})$, a factorization of bath dimension $12m$. The maximally entangled input puts weight $\frac{1}{2}$ on each tag and the tag-coherent part of the Choi state vanishes for both channels, so its Choi distance to $\hat{\Phi}$ is $\frac{1}{2}d_J(\Psi, \Phi_H)$. Hence $D_{\hat{\Phi}}^J(\delta/2) \leq 12 D_{\Phi_H}^J(\delta)$ and $\hat{\Phi} \in \mathcal{K}_{208}$, so $\kappa(\hat{\Phi}) = 0$ by Theorem 3.1, and the optimal rate is $\frac{5}{2}$ by Corollary 3.2 and Proposition 3.6.

Upper bound. Theorem 3.21 for $\hat{\Phi}$, with $d = 208$ and $r = 32$, and $\log D_{\hat{\Phi}}^J(\epsilon/(1664n)) \leq \log 12 + \log D_{\Phi_H}^J(\epsilon/(832n))$, together with Theorem 3.36, gives (c).

Lower bound. Let S be a causal service of $\hat{\Phi}$ with memory Q , exchange $J \leq \frac{5}{2}n$, deficits P, C and error ϵ . Build a service S' of Φ_H with one extra memory qubit T , initially $|0\rangle$. In round t it feeds T as the tag, together with the input, into the round of S , releases the 104-dimensional output and keeps the tag output in T ; its exports and imports are those of S . So $Q' = Q + 1$, $P' = P + 1$, $C' = C$ and $J' = J$. Given an observer O of Φ_H , let O' be the observer of $\hat{\Phi}$ that acts as O , supplies $|0\rangle$ as the first tag, supplies each later tag as the tag output it received in the previous round, and discards the last one. The experiment of O with S' is that of O' with S , restricted to the registers of O . In the ideal experiment with O' every tag output is $|0\rangle$ and every other output is that of Φ_H , so the restriction is the ideal experiment of O . Partial traces contract the trace distance, so S' has error at most ϵ . Since the probe of $\hat{\Phi}$ is flat of rank 32 and $J \leq \frac{n}{2} \log 32$, Theorem 3.32 gives $P + C \leq 14Q + 3$, hence $P' + C' \leq 14Q'$. Theorem 3.29 applied to S' , as in the proof of Corollary 3.33(c), gives $Q' = \Omega(n^{\beta_H/(1+\beta_H)})$, and with $P + C \leq A \log(n + 1)$ it gives the second bound. Above the exchange $\frac{5}{2}n$, Theorem 3.32 gives $P + C \leq 14Q + 4E_x + 3$, hence $P' + C' \leq 14Q' + 4E_x$, and the argument of Corollary 3.33(d) applies to S' , whose exchange is that of S . $\square$

12 Group channels

Throughout, $\mathcal{P} = \langle S \mid R \rangle$ and Φ_G are as in Section 3.4, and $(\mathcal{A}, \tau)$ is a finite-dimensional tracial pair: M_M with τ_M , or $\bigoplus_j M_{m_j}$ with $\tau = \sum_j p_j \tau_{m_j}$, $p_j > 0$. On $M_d \otimes \mathcal{A}$ put $\|X\|_{2,d,\tau} = ((\tau_d \otimes \tau)(X^* X))^{1/2}$. For an assignment $\phi : S \rightarrow U(\mathcal{A})$ the gadget is $U_\phi = \sum_{a \in \mathcal{T}} c_a E_a \otimes \phi(w_a)$, with E_a the matrix unit of slot a , and $\Psi_\phi(\rho) = (\text{id} \otimes \tau)[U_\phi(\rho \otimes 1)U_\phi^*]$ is in $\mathcal{F}_M$ when every block of $\mathcal{A}$ has size at most M . Write $[w]$ for the element of G represented by a word. The labels $[w_a]$ are e and the labels of the symbols, since the product slot of a triangle carries a prefix or e . Every label g has a *singleton anchor*: a diagonal slot i_g with coefficient 1 whose row and column contain no other slot. Put $A_g = \sum_{[w_a]=g} c_a E_a$ and $\mu_g = \|A_g\|_F^2 \geq 1$; the A_g have disjoint supports, $\sum_g \mu_g = d$, and $\mathcal{J}(\Phi_G) = \frac{1}{d} \sum_g |A_g\rangle\langle A_g|$, with support projection $\Pi = \sum_g \mu_g^{-1} |A_g\rangle\langle A_g|$.

12.1 The correspondence

Lemma 12.1 (Gadget Choi states). *For every assignment ϕ into $(\mathcal{A}, \tau)$, with $|a\rangle = |i_a j_a\rangle$,*

$$\mathcal{J}(\Psi_\phi) = \frac{1}{d} \sum_{a,b \in \mathcal{T}} c_a \bar{c}_b \tau(\phi(w_a) \phi(w_b)^*) |a\rangle\langle b|.$$

More generally, let $V = \sum_{ij} E_{ij} \otimes V_{ij}$ be a unitary on $\mathbb{C}^d \otimes \mathcal{A}$ with channel Ψ , and let $|V\rangle = d^{-1/2} \sum_{ij} |ij\rangle \otimes V_{ij}$, a unit vector in $\mathbb{C}^{d^2} \otimes L^2(\mathcal{A}, \tau)$. Then $\mathcal{J}(\Psi)$ is the partial trace

of its pure state, and

$$\ell_{\Phi_G}(\Psi) = \|V - W\|_{2,d,\tau}^2, \quad W = (\Pi \otimes 1)V = \sum_g A_g \otimes B_g, \quad B_g = \mu_g^{-1} \sum_{[w_a]=g} \bar{c}_a V_{i_a j_a}.$$

Proof. $\Psi_\phi(|i\rangle\langle j|) = \sum_{a,b} c_a \bar{c}_b \tau(\phi(w_a)\phi(w_b)^*) E_a |i\rangle\langle j| E_b^*$, and $\mathcal{J} = \frac{1}{d} \sum_{ij} \Psi(|i\rangle\langle j|) \otimes |i\rangle\langle j|$. The second formula is the first with all d^2 positions as slots, since $\tau(V_{ij} V_{kl}^*)$ is the inner product in $L^2(\mathcal{A}, \tau)$. Hence $\ell = \text{Tr}[(I - \Pi)\mathcal{J}(\Psi)] = \|((I - \Pi) \otimes 1)|V\rangle\|^2$, and the projection onto the range of $\Pi \otimes 1$ is W . $\square$

Proof of Theorem 3.23(a). By Lemma 12.1, the Choi matrices differ by

$$\mathcal{J}(\Psi_\phi) - \mathcal{J}(\Phi_G) = \frac{1}{d} \sum_{a,b} c_a \bar{c}_b (\tau(\phi(w_a)\phi(w_b)^*) - \mathbf{1}[w_a = w_b]) |a\rangle\langle b|,$$

so $d_J(\Psi_\phi, \Phi_G) \leq \frac{z}{2d} (\sum_a |c_a|)^2 \leq \frac{z}{2d} |\mathcal{T}| d \leq dz$, and $d_\diamond \leq d d_J \leq d^2 z$. For the leakage, if $[w_a] = [w_b]$ then $w_a w_b^{-1}$ is a product of at most $A_{\mathcal{P}}$ conjugates of relators, so $\|\phi(w_a) - \phi(w_b)\|_2 \leq A_{\mathcal{P}} e$. Fixing a slot $a(g)$ for every label, $W' = \sum_g A_g \otimes \phi(w_{a(g)})$ lies in the range of $\Pi \otimes 1$, so by Lemma 12.1 $\ell \leq \|U_\phi - W'\|_{2,d,\tau}^2 = \frac{1}{d} \sum_a |c_a|^2 \|\phi(w_a) - \phi(w_{a(g)})\|_2^2 \leq A_{\mathcal{P}}^2 e^2$. Minimizing assignments give the two bounds. $\square$

Proof of Theorem 3.23(b). Write $\Psi = \sum_j p_j \Psi_j$ with factorization unitaries U_j of bath dimensions $m_j \leq M$, and put $\mathcal{A} = \bigoplus_j M_{m_j}$, $\tau = \sum_j p_j \tau_{m_j}$ and $V = \bigoplus_j U_j$; then Ψ is the channel of V .

Anchors. By Lemma 12.1, $\|V - W\|_{2,d,\tau}^2 = \ell \leq v$. Fix a label g with anchor i . The positions (j, i) with $j \neq i$ are not slots, so column unitarity gives $\tau(I - V_{ii}^* V_{ii}) = \sum_{j \neq i} \|V_{ji}\|_2^2 \leq dv$. Extend the polar part of V_{ii} to a unitary $S_g \in \mathcal{A}$; since $(1 - x)^2 \leq 1 - x^2$ on $[0, 1]$, $\|V_{ii} - S_g\|_2^2 \leq dv$. The anchor component of $V - W$ is $E_{ii} \otimes (V_{ii} - B_g)$, so $\|V_{ii} - B_g\|_2^2 \leq dv$ and $\|B_g - S_g\|_2 \leq 2\sqrt{dv}$. With $Z = \sum_g A_g \otimes S_g$ this gives $\|W - Z\|_{2,d,\tau} \leq 2\sqrt{dv}$ and $\|V - Z\|_{2,d,\tau} \leq h_d \sqrt{v}$.

Multiplicativity. Z has the block structure of the gadget, with diagonal entries S_g and triangle blocks $2^{-1/2} \begin{pmatrix} S_e & S_{[y]} \\ S_{[x]} & -S_{[z]} \end{pmatrix}$, so $\|Z\|_\infty \leq \sqrt{2}$ and $\|Z^* Z - I\|_{2,d,\tau} \leq \|Z^*(Z - V)\| + \|(Z - V)^* V\| \leq (1 + \sqrt{2}) h_d \sqrt{v}$. The off-diagonal entry of a triangle block of $Z^* Z - I$ is $\frac{1}{2}(S_e^* S_{[y]} - S_{[x]}^* S_{[z]})$, and each entry has $\|\cdot\|_2$ at most $\sqrt{d} \|Z^* Z - I\|_{2,d,\tau}$. With $T_g = S_e^* S_g$ and unitary invariance, $\|T_{[x]} T_{[y]} - T_{[z]}\|_2 = \|S_e^* S_{[y]} - S_{[x]}^* S_{[z]}\|_2 \leq b_d \sqrt{v}$ for every triangle.

The assignment. Put $\phi(s) = T_{[s]}$ and $\beta = b_d \sqrt{v}$. The inverse triangles give $\|\phi(s^{-1}) - T_{[s^{-1}]}\|_2 \leq \beta$, and induction along the prefix triangles gives $\|\phi(p_{i,l}) - T_{[p_{i,l}]}\|_2 \leq (2l - 1)\beta$. At $l = \ell_i$, $[r_i] = e$ and $T_e = I$, so $e(\phi) \leq (2L - 1)\beta \leq 2L b_d \sqrt{v}$.

Moments. Every slot word satisfies $\|\phi(w_a) - T_{[w_a]}\|_2 \leq 2L\beta$, and $(I \otimes S_e^*)Z = \sum_a c_a E_a \otimes T_{[w_a]}$, so $\|(I \otimes S_e^*)V - U_\phi\|_{2,d,\tau} \leq (h_d + 2L b_d) \sqrt{v} = c_{\mathcal{P}} \sqrt{v}$. Left multiplication by $I \otimes S_e^*$ does not change the channel, and $\frac{1}{2} \| |\psi\rangle\langle\psi| - |\psi'\rangle\langle\psi'| \|_1 \leq \|\psi - \psi'\|$ for unit vectors, so $d_J(\Psi, \Psi_\phi) \leq c_{\mathcal{P}} \sqrt{v}$ and $d_\diamond(\Psi_\phi, \Phi_G) \leq q := u + d c_{\mathcal{P}} \sqrt{v}$. For slots $a = (i, j)$ and $b = (k, l)$, $\langle i | \Psi_\phi(E_{jl}) | k \rangle = c_a \bar{c}_b \tau(\phi(w_a)\phi(w_b)^*)$ and the same entry of $\Phi_G(E_{jl})$ is $c_a \bar{c}_b \mathbf{1}[w_a = w_b]$. Since $\|E_{jl}\|_1 = 1$ and $|c_a c_b| \geq \frac{1}{2}$, $z(\phi) \leq 4q$.

For a single factorization with $d_J \leq \delta$: the leakage is at most δ , $d_\diamond \leq d\delta$, and $\mathcal{A} = M_M$. $\square$

Proof of Theorem 3.23(c)–(f). (c) By Remark 3.20, $\mathfrak{D}_{\Phi_G}(u + d\sqrt{2 \ln 2 v}, v + \sqrt{2 \ln 2 v}) \leq D_{\Phi_G}^{\text{ent}}(u, v)$. Apply (b) with $v' = v + \sqrt{2 \ln 2 v} \leq (1 + \sqrt{2 \ln 2}) \sqrt{v}$ and use $v \leq \sqrt{v} \leq v^{1/4}$ for $v \leq 1$ and $2L b_d \leq c_{\mathcal{P}}$.

(d) Let ϕ on $\bigoplus_j M_{m_j}$ with weights p attain M . The moments and the squared relator defects are linear in p , so by Carathéodory some p' supported on at most $N_{\mathcal{P}}$ blocks gives

the same values. Put $K_0 = \lceil 2N_{\mathcal{P}}M/\eta \rceil + N_{\mathcal{P}}M$, $k_j = \lfloor p'_j K_0/m_j \rfloor$ and $K = \sum_j k_j m_j \in [K_0 - N_{\mathcal{P}}M, K_0]$, and let $\phi''(s) = \bigoplus_j \phi_j(s) \otimes I_{k_j}$ on $\mathbb{C}^K$. Its normalized trace is the weighted trace with weights $q_j = k_j m_j/K$, and $\sum_j |q_j - p'_j| \leq (\sum_j m_j + N_{\mathcal{P}}M)/K \leq \eta$. So every moment moves by at most η , and every squared defect, the trace of $(\phi(r) - I)^*(\phi(r) - I)$ of norm at most 4, by at most 4η . Finally $K \leq K_0 \leq \lceil 3N_{\mathcal{P}}M/\eta \rceil$.

(e) Let ϕ attain $\mathfrak{h}_{\mathcal{P}}(e, z_0)$ and put $\hat{\phi}(s) = (\phi(s) \otimes \phi(s))^{\otimes k}$. Then $\tau(\hat{\phi}(w_a)\hat{\phi}(w_b)^*) = |\tau(\phi(w_a)\phi(w_b)^*)|^{2k}$. For $[w_a] \neq [w_b]$ this is at most $z_0^{2k} \leq \delta/(2d)$. For $[w_a] = [w_b]$, $\operatorname{Re} \tau(\phi(w_a)\phi(w_b)^*) = 1 - \frac{1}{2}\|\phi(w_a) - \phi(w_b)\|_2^2 \geq 1 - \frac{1}{2}A_{\mathcal{P}}^2 e^2$, so the amplified moment is at least $1 - kA_{\mathcal{P}}^2 e^2 = 1 - \delta/(2d)$. By (a), $d_J(\Psi_{\hat{\phi}}, \Phi_G) \leq \delta/2$ with bath dimension $\mathfrak{h}_{\mathcal{P}}(e, z_0)^{2k}$.

(f) By (a) and (b), $\Phi_G \in \mathcal{K}_d$ if and only if $\mathfrak{h}_{\mathcal{P}}$ is finite everywhere, and by (d) this is equivalent to the same for $\mathfrak{h}_{\mathcal{P}}^{\oplus}$. If G is hyperlinear, then for every finite $F \subset G$ and $\eta > 0$ there is $\theta : G \rightarrow U(M)$ with $\|\theta(g)\theta(h) - \theta(gh)\|_2 \leq \eta$ for $g, h \in F$ and $|\tau_M(\theta(g)) - \mathbf{1}[g=e]| \leq \eta$ for $g \in F$ [35]. Take F to contain the labels of all prefixes of the relators and of the words $w_a w_b^{-1}$, and put $\phi(s) = \theta([s])$. Telescoping along a word of length m gives $\|\phi(u) - \theta([u])\|_2 \leq 3m\eta$, so the defects and the moment errors are $O_{\mathcal{P}}(\eta)$, and $\mathfrak{h}_{\mathcal{P}}$ is finite everywhere. $\square$

Proof of Theorem 3.24. (a) The last clause of Theorem 3.18 gives $D_{\Phi_G}^{\text{ent}}(\epsilon/(1-\epsilon), t_n) \leq 2^Q$ for every channel. By Theorem 3.23(c), $\mathfrak{h}_{\mathcal{P}}^{\oplus}(C_{\mathcal{P}}t_n^{1/4}, 4\epsilon/(1-\epsilon) + C_{\mathcal{P}}t_n^{1/4}) \leq 2^Q$, and $4\epsilon/(1-\epsilon) \leq \frac{4}{15} = \frac{1}{2} - \frac{7}{30}$. For the second bound apply Theorem 3.23(d) with $\eta = C_{\mathcal{P}}^2 \sqrt{t_n}/4$.

(b) Theorem 3.23(e) with $\delta = \epsilon/(8dn)$ gives $\log D_{\Phi_G}^J(\epsilon/(8dn)) \leq 2k_n \log \mathfrak{h}_{\mathcal{P}}(e_n, z_0)$. Insert this into Theorem 3.19, using $\min\{\sqrt{nx}, x/s\} \leq x/s$ and that $y \mapsto y + \log(2+y)$ increases.

(c) Insert the same bound into Theorem 3.21. Its exchange $\frac{n}{2} \log r$ is $nH(\Phi_G)/2$ by Proposition 3.22.

(d) By Proposition 3.22 the probe spectrum of Φ_G is flat of rank r and $\frac{n}{2} \log r = nH(\Phi_G)/2$, so Theorem 3.32 gives $P+C \leq 14Q + 2 \log \frac{16}{7} < 14Q + 3$, and (a) applies with this budget. $\square$

12.2 The Slofstra channel

Slofstra's proof of [4, Theorem 1.2] uses the binary representatives $z(k) = \prod_{i=0}^{\ell} t^i z^{k_i} t^{-i}$, where $k = \sum_i k_i 2^i$. Free cancellation gives

$$z(k) = z^{k_0} t z^{k_1} t \dots t z^{k_{\ell}} t^{-\ell}, \quad |z(k)| \leq 3\ell + 1.$$

Thus $x(k) = z(k)x_0 z(k)^{-1}$ has length $O(\log k)$, rather than the $O((\log k)^2)$ estimate used there. The embedding in [4, Corollary 2.4] has isoperimetric function Cn^{26} . Each defining relation of $\mathcal{C}(N)$ therefore costs at most $C_3(\log N)^{26}$ cells after substitution, with a fixed C_3 . Lemma 3.4 there consequently gives a $C_3(N+1)^2(\log N)^{26}\epsilon$ -homomorphism of $\mathcal{C}(N)$ from every ϵ -representation of Slofstra's group, whose value at the central element J is $\phi(w)$. Lemma 3.1 there forces dimension at least $2^{\lfloor N/2 \rfloor - 1}$ whenever J moves by more than 42 times that error. This elementary shortening improves the logarithmic factor, not the polynomial exponent.

Lemma 12.2 (Slofstra's bound with the logarithm). *Let ϕ be an ϵ -representation of Slofstra's group of dimension m with $\|\phi(w) - I\|_2 \geq \delta_0 > 0$, and put $\varrho = \delta_0/\epsilon$ and $c_1 = \min\{1, (2^{25}/(168C_3))^{1/2}\}$. If $\varrho \geq 2$ and $N = \lfloor c_1 \varrho^{1/2} (\log \varrho)^{-13} \rfloor \geq 2$, then $\log m \geq \lfloor c_1 \varrho^{1/2} / (2(\log \varrho)^{13}) \rfloor - 1$.*

Proof. Since $\varrho \geq 2$, $\log \varrho \geq 1$, and with $c_1 \leq 1$ this gives $N \leq \varrho^{1/2}$, hence $\log N \leq \frac{1}{2} \log \varrho$. Also $(N+1)^2 \leq 4N^2 \leq 4c_1^2 \varrho (\log \varrho)^{-26}$. So $42C_3(N+1)^2(\log N)^{26}\epsilon \leq 168C_3c_1^2 2^{-26}\delta_0 \leq \delta_0/2$, and the dimension bound of [4] applies with this N . $\square$

Lemma 12.3 (Component selection). *Let ϕ be a direct-sum assignment for $\mathcal{P}_S$ with components ϕ_a of weights p_a , $e(\phi) \leq e$ and $z(\phi) \leq \frac{1}{2}$. Then some component has all relator defects at most $\epsilon' = 4\sqrt{|R|+1}e$ and $\|\phi_a(j) - I\|_2 \geq 2^{-1/2}$. If $\epsilon' \leq 2^{-1/2} - \frac{1}{2}$, its restriction to S is an ϵ' -representation of Slofstra's group with $\|\phi_a(w) - I\|_2 \geq \frac{1}{2}$.*

Proof. Since $[j] \neq e$, $|\tau(\phi(j))| \leq z(\phi) \leq \frac{1}{2}$, so $\sum_a p_a \|\phi_a(j) - I\|_2^2 \geq 1$; each term is at most 4, so the components with $\|\phi_a(j) - I\|_2^2 \geq \frac{1}{2}$ have weight at least $\frac{1}{7}$. By Markov's inequality the components whose squared defects sum to more than $16(|R|+1)e^2$ have weight at most $\frac{1}{16}$. Pick a component in the first set and not in the second. Finally $\|\phi_a(w) - I\|_2 \geq \|\phi_a(j) - I\|_2 - \|\phi_a(j^{-1}w) - I\|_2 \geq 2^{-1/2} - \epsilon'$. $\square$

Proof of Theorem 3.25. By Theorem 3.24(a), for n large every such service yields a direct-sum assignment with blocks of dimension at most 2^Q , $e(\phi) \leq C_{\mathcal{P}_S} t_n^{1/4}$ and $z(\phi) \leq \frac{1}{2}$. Lemma 12.3 gives an ϵ' -representation of dimension at most 2^Q with $\epsilon' = 4\sqrt{|R|+1}C_{\mathcal{P}_S} t_n^{1/4}$ that moves w by $\frac{1}{2}$, and Lemma 12.2 with $\varrho = 1/(2\epsilon') = \Theta(t_n^{-1/4})$, which is at least 2 for n large, gives $Q \geq c\varrho^{1/2}(\log \varrho)^{-13}$. Since $t_n = \Theta(\log n/n)$, this is the claimed bound. The proof uses no assumption on $\mathcal{K}_d$.

At the rank rate, Theorem 3.24(d) allows the budget $B = 14Q+3$, so $t_n = O((Q+\log n)/n)$. Let $t_* > 0$ be such that for $t \leq t_*$ all conditions of Theorem 3.24(a) and Lemmas 12.2 and 12.3 hold at $\epsilon' = 4\sqrt{|R|+1}C_{\mathcal{P}_S} t^{1/4}$; we may assume $Q \leq t_* n/16$, since otherwise $Q \geq n^{1/9}$ for large n . If $Q \leq \log n$ that argument gives a contradiction for large n . Otherwise $t_n = O(Q/n)$, and the same argument gives $Q \geq c'(n/Q)^{1/8}(\log n)^{-13-1/8}$, that is $Q^{9/8} \geq c'n^{1/8}(\log n)^{-13-1/8}$ and $Q \geq n^{1/9-o(1)}$.

In branch (i), Theorem 3.24(c) provides services with $P = C = 0$, and the rank rate is optimal by Corollary 3.2. In branch (ii), $\Delta_J(\Phi_S) > 0$ and the purity bound are Theorem 3.1; Φ_S is generally factorizable through the group von Neumann algebra; and if the group were hyperlinear, Theorem 3.23(f) would put Φ_S in $\mathcal{K}_d$. At the rank rate with $\epsilon < \min\{\Delta, 1/16\}$, Theorem 3.24(d) gives $P + C \leq 14Q + 3$, and with $P + C \geq n(\Delta - \epsilon)^2/(2\ln 2)$ this is the linear bound. $\square$

The same argument applied to the mixture profile, with Theorem 3.23(b) in place of (c), gives $\log D_{\Phi_S}^{\text{mix}}(\delta) \geq c\delta^{-1/4}(\log(1/\delta))^{-13}(1+o(1))$ as $\delta \rightarrow 0$.

13 Accuracy scales and open problems

The accuracy at which repeated service reads a profile depends on how per-round deviations accumulate. Two facts frame the question.

Proposition 13.1 (Support-preserving approximants). *Suppose $\text{supp } \mathcal{J}(\Psi) \subseteq \text{supp } \mathcal{J}(\Phi)$, let $\lambda > 0$ be the least nonzero eigenvalue of $\mathcal{J}(\Phi)$, and let $\delta = d_J(\Psi, \Phi) \leq \lambda$. Then every adaptive observer distinguishes n independent uses of Ψ from n independent uses of Φ with advantage at most $(\delta/\lambda)\sqrt{n/2}$. In particular this holds when $\mathcal{J}(\Phi) \geq \lambda I$.*

Proof. The case $\delta = 0$ is immediate. Otherwise put $t = \delta/\lambda$ and let Π be the support projection of $\mathcal{J}(\Phi)$. The Choi difference is traceless, supported on Π , and of operator norm at most δ , while $\mathcal{J}(\Phi) \geq \lambda\Pi$; so $\mathcal{J}(\Phi) \pm (\mathcal{J}(\Psi) - \mathcal{J}(\Phi))/t \geq 0$, with reference marginal τ_d , and $\Gamma_{\pm} = \Phi \pm (\Psi - \Phi)/t$ are channels, $\Phi = \frac{1}{2}(\Gamma_+ + \Gamma_-)$ and $\Psi = \frac{1+t}{2}\Gamma_+ + \frac{1-t}{2}\Gamma_-$. Both experiments are generated by the same sequence of conditional channels from n independent coins, fair for Φ and of bias t for Ψ . One biased coin has relative entropy at most t^2 from a fair one, and Pinsker's inequality bounds the distance of the coin strings by $t\sqrt{n/2}$. $\square$

Remark 13.2. No square-root bound in single-use trace distance holds for arbitrary independent approximants. For $\Phi = \text{id}$ and $\Psi = \text{Ad}_{\text{diag}(1, e^{i\vartheta})}$, $d_J(\Psi, \Phi) = |\sin(\vartheta/2)|$, whereas an n -qubit GHZ input gives distance $|\sin(n\vartheta/2)|$. Rare errors also add linearly: $(1-p)\text{id} + p \text{Ad}_Z$ has n -use distance $1 - (1-p)^n$ from the identity. Support preservation is sufficient for Proposition 13.1, but not necessary, as the following example shows.

Proposition 13.3 (Leaking approximants with square-root accumulation). *Let $\Phi = \Delta_Z$ be qubit dephasing, let $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, and put $\Psi_\theta = \text{Ad}_{R_\theta} \circ \Delta_Z$, with $0 \leq \theta \leq \pi/2$ and $\delta = \sin \theta$. Both channels are exact finite tracial factors, and*

$$d_J(\Psi_\theta, \Phi) = d_\diamond(\Psi_\theta, \Phi) = \delta, \quad \ell_\Phi(\Psi_\theta) = \delta^2.$$

Their complete adaptive n -use distance is exactly $\sqrt{1 - (1 - \delta^2)^n}$.

Proof. Each channel is a mixture of two unitary channels: $\Delta_Z = (\text{id} + \text{Ad}_Z)/2$ and $\Psi_\theta = (\text{Ad}_{R_\theta} + \text{Ad}_{R_\theta Z})/2$. Their Choi states are equally weighted blocks with classical reference label x and output $|x\rangle$ or $R_\theta|x\rangle$; these pure states have distance δ , and the latter leaves the former's support with weight δ^2 . For the adaptive upper bound use a common processor that measures the input in the Z basis, obtains x , and applies $(XZ)^x$ to a supplied program state. Programs $|0\rangle$ and $\cos \theta|0\rangle + \sin \theta|1\rangle$ implement Φ and Ψ_θ respectively. This holds also for reference-entangled inputs. Every adaptive experiment is therefore a common channel applied to the corresponding n -fold program states, whose distance is $\sqrt{1 - \cos^{2n} \theta}$. All-zero inputs attain that distance, proving equality and the diamond claim at $n = 1$. The processor is a comparison argument, not an uncharged physical resource in the service model. $\square$

For independently repeated approximants, a Bell-input support test gives the lower bound $1 - (1 - \ell_\Phi(\Psi))^n$. Proposition 13.3 shows why this does not imply linear accumulation in d_J : leakage may be quadratic in d_J . For Φ_H , no finite tracial mixture of zero leakage lies within diamond distance $1/8$ (Theorem 3.27); determining the relation between leakage and useful approximation accuracy remains essential.

Open problems.

- (1) *The exponent for Φ_H .* At fixed error and logarithmic purity the least memory lies between $c(n/\log n)^{\beta_H}$, at every exchange, and $O(n^{1/3} \log^{4/3} n)$, at the optimal exchange (Corollary 3.37). The lower bound is read from the one-round profile, and by Proposition 4.18 that profile cannot certify more than order $n^{1/3}$ at a fixed error; so a one-round bound with the static exponent $\frac{1}{3}$, that is $\log D_{\Phi_H}^{\text{ent}}(u_0, v) \geq v^{-1/3+o(1)}$ for some fixed $u_0 > 0$, would give memory $n^{1/3 \pm o(1)}$. Arguments that use more of the extracted round, or several rounds, are not limited in this way. The present exponent β_H comes from the uniform-area transport and sector argument used here, whose exponent is capped at $\frac{1}{5}$ in $1/v$ (that is, $\frac{2}{5}$ in the inverse relator defect) because the area is at least quadratic [26]; the smoothed models spread a defect in a way that has no analogue in our permutation models, and a lower bound of exponent $\frac{1}{3}$ has to see this spreading.
- (2) *The scale mismatch.* For a gadget channel, Theorem 3.24 reads the group at relator defect of order $(\log n/n)^{1/4}$ below and $(n \log n)^{-1/2}$ above. For presentations whose profile $\log \mathbf{hp}(e, z)$ is a power of $1/e$ this fixes the memory exponent up to a factor of 2. The loss is partly on the channel side (two square roots in the extraction) and partly in the service (the accuracy the compiler needs). Is there a lower bound that reads the profile at defect of order $n^{-1/2}$, or a service that reads it at defect of order $n^{-1/4}$? More generally: does one profile, at one accuracy scale, determine the memory of every channel in $\mathcal{K}_d$ at logarithmic purity?

- (3) *A modulus for κ .* On $\mathcal{K}_d$, κ vanishes, and near interior points it is Lipschitz (Proposition 3.3(d)). At the boundary it jumps, but the examples of Proposition 3.4 are not generally factorizable. Is there, on the generally factorizable channels on M_d , a function f_d with $f_d(\delta) \rightarrow 0$ as $\delta \rightarrow 0$ and $\kappa \leq f_d(\Delta_J)$? And outside $\mathcal{K}_d$, how do memory and purity trade?
- (4) *An explicit Slofstra channel.* Theorem 3.25 applies to a channel whose dimension is not known, because Slofstra’s presentation is produced by an embedding theorem. An explicit presentation with a steep profile would give an explicit channel. Which branch of Theorem 3.25 holds is open; a proof of branch (ii) would show that Slofstra’s group is not hyperlinear. For any finitely presented nonhyperlinear group, a further question would be whether a finite gadget channel, after suitable Tietze moves, must lie outside $\mathcal{K}_d$; nonhyperlinearity alone does not identify such a gadget.
- (5) *The logarithmic end.* Flat exact factors need only $O(\log(n/\epsilon))$ memory at their optimal rate (Theorem 3.15), and some need $O(1)$ (Proposition 3.13). Does some flat exact factor need $\Omega(\log n)$ at its optimal rate? Is $\frac{1}{2} \log r$ the least rate with bounded memory for every exact factor?
- (6) *Exact service.* At zero error Theorem 3.7(c) requires exchange rate $\frac{1}{2} \log r$ with sublinear memory, and Propositions 3.12 and 3.13 attain it with constant memory. Is $\frac{1}{2} \log r$ attained at zero error by every exact factor, and with how much memory?

Use of AI assistance

Generative AI tools assisted with developing arguments and constructions, checking proofs and numerical identities, searching the literature, writing verification code, and drafting and editing the manuscript. Automated review is not a substitute for mathematical proof. The accompanying verification notes distinguish finite executable checks and kernel-checked arithmetic from the analytical and operational arguments proved in the text. They do not claim a formal verification of the full paper. The authors take responsibility for the content.

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